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Chapter 1

Summary and Display of Univariate Data

1.1 Frequency Table and Histogram

Engineers and applied scientists are often involved with the generation and collection of data and the retrieval of information contained in data sets. They must also communicate to different audiences the results of complex numerical studies including one or more data sets.

Experience shows that data sets are often messy, difficult to grasp and hard to analyze. In this chapter we introduce some statistical techniques and ideas which can be used to **summarize and display** data.

Table 1.1: Live Load Data

Bay	1st	2d	3d	4th	5th	6th	7th	8th	9th	10th
A	44.4	130.4	127.6	127.7	108.4	184.0	139.1	120.6	174.1	187.9
B	138.4	236.4	202.5	128.7	154.3	117.0	125.9	127.2	175.6	114.1
D	164.7	110.4	185.7	185.0	150.0	198.7	144.5	121.5	93.2	202.2
E	98.3	154.5	171.9	104.8	230.1	102.8	156.6	136.1	93.8	197.8
F	178.0	108.1	197.9	112.0	66.6	160.9	106.8	123.2	162.5	118.3
G	123.7	185.4	130.3	169.2	91.8	134.5	153.5	131.4	254.0	194.6
H	157.5	62.3	65.2	94.4	156.1	133.6	101.9	117.6	87.6	142.4
I	119.4	74.1	118.2	144.4	212.0	132.3	136.1	184.3	177.2	151.8
J	150.4	137.8	105.5	55.2	122.9	127.8	180.6	53.0	150.1	138.4
K	92.2	54.0	139.2	116.7	32.1	184.8	127.1	171.8	159.6	123.8
L	169.8	168.4	169.9	159.6	179.6	33.5	193.3	99.5	124.3	208.6
M	181.5	147.5	104.1	167.4	172.4	128.8	138.6	110.1	141.1	189.3
N	105.4	133.1	62.0	144.9	129.1	94.9	147.6	167.9	136.7	173.2
O	157.6	164.6	195.0	136.3	136.6	223.7	134.0	179.1	85.7	122.3
P	168.4	173.5	150.4	116.4	143.7	179.5	84.5	161.5	140.5	94.1
Q	161.0	132.8	161.0	147.1	199.8	141.4	178.1	145.7	124.8	179.8
R	156.3	128.6	111.8	157.6	129.3	115.2	73.3	94.3	161.9	154.7
S	152.3	169.5	162.1	106.6	112.0	141.0	110.7	145.8	206.1	88.8
T	138.9	101.1	127.9	178.3	127.5	145.1	53.5	182.4	147.9	138.0
U	112.3	135.1	123.9	258.9	192.1	155.0	122.3	86.1	147.0	118.0

Frequency Table

Consider the 200 measurements of the live load distribution (pounds per square foot) on ten floors and twenty bays of a large warehouse (Table 1.1). The live load is the load supported by the structure excluding the weight of the structure itself. Notice how hard it is to understand data presented in this raw form. They must clearly be organized and summarized in some fashion before their analysis can be attempted.

One way to summarize a large data set is to condense it into a *frequency table* (see Table 1.2). The **first step** to construct a frequency table is to determine an appropriate *data range*, that is, an interval that contains all the observations and that has end points close (but not necessarily equal) to the smallest and largest data values. The **second step** is to determine the number k of *bins*. The data range is divided into k smaller subintervals, the bins, usually taken of the same size. Normally, the number of bins k is chosen between 7 and 15, depending on the size of the data set with fewer bins producing simpler but less detailed tables. For example, in the case of the live load data, the smallest and largest observations are 32.1 and 258.9, the data range is $[20, 260]$ and there are 12 bins of size 20. The **third step** is to calculate the *bin mark*, c_i, which represents that bin. The bin mark is the center of the bin interval (that is, one half of the sum of the bin's end points). For example, $30 = (20 + 40)/2$ for the first bin in Table 1.2. The **fourth step** is to calculate the *bin frequencies*, n_i. The bin frequency is equal to the number of data points lying in that bin. Each data point must be counted once; if a data point is equal to the end points of two successive bins, then it is included (only) in the second. For example, a live load of 60 is included in the third bin (see Table 1.2). The **fourth step** is to calculate the *relative frequencies*

$$f_i = \frac{n_i}{n_1 + n_2 + \ldots + n_k}$$

and the *cumulative relative frequencies*

$$F_i = \frac{n_1 + \ldots + n_i}{n_1 + n_2 + \ldots + n_k}.$$

Notice that $f_i 100\%$ gives the percentage of observations in the ith bin and $F_i 100\%$ gives the percentage of observations below the end point of the ith bin. For example, from Table 1.2, 18% of the live loads are between 140 and 160 psf, and 95% of the live loads are below 200 psf.

Table 1.2: Frequency Table

Class	c_i	n_i	f_i	F_i
20–40	30	2	0.010	0.010
40–60	50	5	0.025	0.035
60–80	70	6	0.030	0.065
80–100	90	15	0.075	0.140
100–120	110	28	0.14	0.280
120–140	130	47	0.235	0.515
140–160	150	36	0.180	0.695
160–180	170	32	0.160	0.855
180–200	190	19	0.095	0.950
200–220	210	5	0.025	0.975
220–240	230	3	0.015	0.990
240–260	250	2	0.010	1.000

At this point it is worth comparing Table 1.1 and Table 1.2. We can quickly learn, for instance, from Table 1.2 that only 2 live loads lie between 20 and 40, but we cannot say which they are. On

the other hand, with considerably effort, we can find out from Table 1.1 that these live loads are 32.1 and 33.5. Table 1.2 looses some information in exchange for clarity. The loss of information and gain in clarity are proportional to the number of bins.

Histogram:

The information contained in a frequency table can be graphically displayed in a picture called **histogram** (see Figure 1.1). Bars with areas proportional to the bin frequencies are drawn over each bin. Notice that in the case of bins of equal size the bar areas are proportional to the bar heights. The histogram shows the shape or distribution of the data and permits a direct visualization of its general characteristics including typical values, spread, shape, etc. The histogram also helps to detect unusual observations called **outliers**. From Figure 1.1 we notice that the distribution of the live load is approximately symmetric: the central bin $120 - 140$ is the most frequent and the frequency of the other bins decrease as we move away from this central bin.

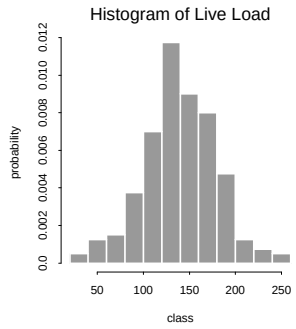


Figure 1.1: Histogram of the Live Load

Many data sets encountered in practice are not symmetric. For example the histogram of Tobin's Q-ratios (market value to replacement cost, out of 250) for 50 firms in Figure 1.2 (a) shows high positive skewness. There are a few firms which are highly over rated. The age of officers attaining the rank of colonel in the Royal Netherlands Air Force (Figure 1.2 (b)) exhibit a pattern of negative skewness. There appear to be more "whizzes" than "laggards" in the Netherlands Air Force. Figure 1.2 (c) displays Simon Newcomb's measurements of the speed of light. Newcomb measured the time required for light to travel from his laboratory on the Potomac River to a mirror at the base of the Washington Monument and back, a total distance of about 7400 meters. These measurements were used to estimate the speed of light. The histogram of Newcomb's data (Figure 1.2 (c)) shows a symmetric distribution except for two outliers. Deleting these outliers gives the symmetric histogram on Figure 1.2 (d).

Data sets can be further summarized in terms of just two numbers, one giving their location and the other their dispersion. These summaries are very convenient and perhaps unavoidable when we

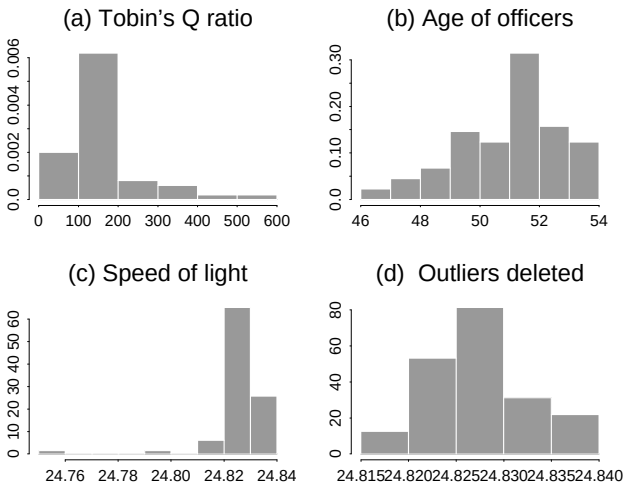


Figure 1.2: Some Non-Symmetric Histograms

must compare several data sets (e.g. the production figures from several plants and shifts). The loss of information is not severe in the case of data sets with approximately symmetric histograms, but may be very severe in other cases.

Two commonly used measures of location and dispersion are the *sample mean* and the *sample standard deviation*. They are studied in the next two sections.

1.2 Sample Mean

Quantitative variables such as the live load are usually denoted by upper case letters X, Y, etc. The particular measurements for these variables are denoted by the corresponding lower case letters, x_i, y_i, etc. The subscripts give the order in which the measurements have been taken. For example, the variable "live load" can be represented by X and, if the measurements were made floor by floor from the first to the tenth, from bay A to bay U, then

$$x_1 = 44.4, \quad x_2 = 138.4, \quad \ldots \quad x_{10} = 92.2, \quad \ldots \quad x_{200} = 118.0.$$

The sample mean $\overline{x}$ (also called *sample average*) of a data set or "sample" is defined as

$$\overline{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n},$$

where n represents the number of data points (observations). For the live load data (see Table 1.1)

$$\overline{x} = 140.156 \text{ pounds per ft}^2.$$

The sample average can also be approximately calculated from a frequency table using the formula

$$\overline{x} \approx \frac{\sum_{i=1}^{k} c_i n_i}{\sum_{i=1}^{k} n_i} = \sum^{k} c_i f_i.$$

The approximation is better when the measurements are symmetrically distributed over each bin. For the live load data (see Table 1.2) we have

$$\begin{aligned} \overline{x} &\approx \frac{(30 \times 2) + (50 \times 5) + \ldots + (250 \times 2)}{2 + 5 + \ldots + 2} \\ &= (30 \times 0.01) + (50 \times 0.035) + \ldots + (250 \times 0.01) = 139.8 \text{ pounds per ft}^2, \end{aligned}$$

which is close to the exact value, 140.156.

Properties of the Sample Mean

Linear Transformations: If the original measurements, x_i are linearly transformed to obtain new measurements

$$y_i = a + bx_i,$$

for some constants a and b, then

$$\overline{y} = a + b\overline{x}.$$

In fact,

$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{\sum_{i=1}^{n}(a + bx_i)}{n} = \frac{na + b\sum_{i=1}^{n} x_i}{n} = a + b\frac{\sum_{i=1}^{n} x_i}{n} = a + b\overline{x}.$$

Example 1.1 Suppose that each live load from Table 1.1 is increased by 5 kilograms and converted to *kilograms per square foot*. Since one pound equals 0.4535 kilograms, the revised measurements are $y_i = 5 + 0.4535x_i$ and $\overline{y} = 5 + 0.4535\overline{x} = 5 + 0.4535 \times 140.2 = 68.58kg$.

Sum of Variables: If new measurements z_i are obtained by adding old measurements x_i and y_i then

$$\overline{z} = \overline{x} + \overline{y}.$$

In fact,

$$\overline{z} = \frac{\sum_{i=1}^{n} z_i}{n} = \frac{\sum_{i=1}^{n}(x_i + y_i)}{n} = \frac{\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i}{n} = \overline{x} + \overline{y}.$$

Example 1.2 Let u_i and v_i $(i = 1, \ldots, 10)$ represent the live loads on bays A and B. The mean load across floors for these two bays are (see Table 1.1)

$$\begin{aligned} \overline{u} &= (44.4 + 130.4 + \ldots + 187.9)/10 = 134.42 \qquad \text{(Bay A)} \\ \overline{v} &= (138.4 + 236.4 + \ldots + 114.1)/10 = 152.01 \qquad \text{(Bay B)}. \end{aligned}$$

If w_i represent the combined live loads on bays A and B (i.e. $w_i = u_i + v_i$) then the combined mean load across floors for these two bays is

$$\overline{w} = \overline{u} + \overline{v} = 134.42 + 152.01 = 286.43.$$

Least Squares: The sample mean has a nice *geometric interpretation*. If we represent each observation x_i as a point on the real line, then the sample mean is the point which is "closest" to entire collection of measurements. More precisely, let $S(t)$ be the sum of the squared distances from each observation x_i to the point t:

$$S(t) = \sum^{n}(x_i - t)^2.$$

Then $S(t) \geq S(\overline{x})$ for all t. To prove this write

$$\begin{aligned}
S(t) &= \sum^{n}[(x_i - \overline{x}) + (\overline{x} - t)]^2 \\
&= \sum^{n}[(x_i - \overline{x})^2 + (\overline{x} - t)^2 + 2(x_i - \overline{x})(\overline{x} - t)] \\
&= \sum^{n}(x_i - \overline{x})^2 + n(\overline{x} - t)^2 + 2(\overline{x} - t)\sum^{n}(x_i - \overline{x}) \\
&= S(\overline{x}) + n(\overline{x} - t)^2, \qquad \text{since} \quad \sum^{n}(x_i - \overline{x}) = n\overline{x} - n\overline{x} = 0 \\
&\geq S(\overline{x}), \qquad \text{since} \quad n(\overline{x} - t)^2 \geq 0, \quad \text{for all} \;\; t.
\end{aligned}$$

Moreover, equality holds only if all the measurements are equal.

Center of Gravity: The sample mean has also a nice *physical interpretation*. If we think of the observations x_i as points on a uniform beam where vertical equal forces, $\mathbf{F_i}$, are applied (see Figure 1.3), then the sample mean is the center of gravity of this system. To see this consider the magnitude and the placement of the opposite force $\mathbf{F}$ needed to achieve static equilibrium. Since all the forces are vertical, the horizontal component of $\mathbf{F}$ must be equal to zero. To achieve translation equilibrium the sum of the vertical components of all the forces must also be equal to zero. If we denote the vertical components of $\mathbf{F_i}$ by F_i, and the vertical component of $\mathbf{F}$ by F, then

$$F + (F_1 + F_2 + \ldots + F_n) = 0 \qquad \text{(Static Equilibrium)}.$$

Since the $F_i's$ are all equal ($F_i = -w$, say) we have $F - nw = 0$ and so $F = nw$. To achieve torque equilibrium, the placement d of F must satisfy

$$dF + (x_1F_1) + (x_2F_2) + \ldots + (x_nF_n) = 0 \qquad \text{(Torque Equilibrium)}.$$

Replacing F_i by $-w$ and F by nw we have

$$dnw - w(x_1 + x_2 + \ldots + x_n) = 0.$$

Therefore,

$$d = \frac{x_1 + x_2 + \ldots + x_n}{n} = \overline{x}.$$

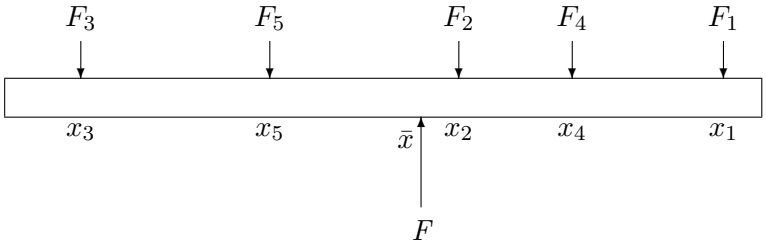


Figure 1.3: The Sample Mean As Center of Gravity

1.3 Sample Standard Deviation, Variance and Covariance

Given the measurements (or sample) $x_1, x_2, \ldots, x_n$, their sample standard deviation $\mathrm{SD}(x)$ is defined as

$$\mathrm{SD}(x) = +\sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}.$$

The expression inside the square root is called the *sample variance*, and denoted $\mathrm{Var}(x)$. In the case of the live load data (Table 1.1)

$$\mathrm{Var}(x) = 1583.892 \text{ square pounds per ft}^4 \quad \text{and} \quad \mathrm{SD}(x) = 39.798 \text{ pounds per ft}^2.$$

The standard deviation can be approximately calculated from a frequency table using the formula

$$\mathrm{SD}(x) \approx +\sqrt{\frac{\sum_{i=1}^{k}(c_i - \overline{x})^2 n_i}{n-1}}.$$

The approximation is better when the observations are symmetrically distributed on each bin. For the live load (Table 1.2) we have

$$\mathrm{SD}(x) \approx \sqrt{\frac{(30-139.8)^2 \times 2 + (50-139.8)^2 \times 5 + \cdots + (250-139.8)^2 \times 2}{199}} = 37.75 \text{ pounds per ft}^2,$$

which is close to the exact value, 39.798.

Properties of the Sample Variance

Linear Transformations: If the original measurements, x_i are linearly transformed to obtain new measurements

$$y_i = a + bx_i,$$

for some constants a and b, then

$$\mathrm{Var}(y) = b^2\mathrm{Var}(x).$$

In fact, since $\overline{y} = a + b\overline{x}$,

$$\begin{aligned}\mathrm{Var}(y) &= \frac{\sum(y_i - \overline{y})^2}{(n-1)} = \frac{\sum(a + bx_i - a - b\overline{x})^2}{(n-1)} \\ &= \frac{\sum[b(x_i - \overline{x})]^2}{(n-1)} = b^2\frac{\sum(x_i - \overline{x})^2}{(n-1)} = b^2\mathrm{Var}(x).\end{aligned}$$

Example 1.3 As in Example 1.1, each live load in Table 1.1 is increased by 5 kilograms per square foot and converted to *kilograms per square foot*. Since one pound equals 0.4535 kilograms, the revised measurements are $y_i = 5 + 0.4535x_i$ kilograms per square foot and so $\text{Var}(y) = 0.4535^2 \times \text{Var}(x) = 0.2056623 \times 1583.892 = 325.747kg^2$ square kilograms per ft^4. The corresponding standard deviation is $\text{SD}(y) = \sqrt{325.747} = 18.048kg$ kilograms per square foot.

Sum of Variables: If new measurements z_i are obtained by adding old measurements x_i and y_i then

$$\text{Var}(z) = \text{Var}(x) + \text{Var}(y) + 2\text{Cov}(x, y), \tag{1.1}$$

where

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{n-1},$$

is the *covariance* between x_i and y_i. The covariance will be further discussed in the next Chapter. The important point here is to notice that the variances of x_i and y_i cannot simply be added to obtain the variance of z_i.

To prove (1.1) write

$$\begin{aligned}
\text{Var}(z) &= \frac{\sum_{i=1}^n (z_i - \overline{z})^2}{n-1} = \frac{\sum_{i=1}^n (x_i + y_i - \overline{x} - \overline{y})^2}{n-1} = \frac{\sum_{i=1}^n [(x_i - \overline{x}) + (y_i - \overline{y})]^2}{n-1} \\
&= \frac{\sum_{i=1}^n [(x_i - \overline{x})^2 + (y_i - \overline{y})^2 + 2(x_i - \overline{x})(y_i - \overline{y})]}{n-1} \\
&= \frac{\sum_{i=1}^n (x_i - \overline{x})^2 + \sum_{i=1}^n (y_i - \overline{y})^2 + 2\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{n-1}.
\end{aligned}$$

Example 1.4 As in Example 1.2 let u_i and v_i be the live loads on bays A and B. The variances and covariance for these loads are (see Table 1.1 and Example 1.2)

$$\text{Var}(u) = \frac{(44.4 - 134.42)^2 + (130.4 - 134.42)^2 + \cdots + (187.9 - 134.42)^2}{9} = 1777.128 \qquad \text{(Bay A)}$$

$$\text{Var}(v) = \frac{(138.4 - 152.01)^2 + (236.4 - 152.01)^2 + \cdots + (114.1 - 152.01)^2}{9} = 1657.93 \qquad \text{(Bay B)}$$

$$\text{Cov}(u, v) = \frac{(44.4 - 134.42)(138.4 - 152.01) + \cdots + (187.9 - 134.42)(114.1 - 152.01)}{9} = -218.650.$$

If w_i represents the combined live loads on bays A and B (i.e. $w_i = u_i + v_i$) then

$$\text{Var}(w) = \text{Var}(u) + \text{Var}(v) + 2\text{Cov}(u, v) = 1777.128 + 1657.93 + 2 \times (-218.6502) = 2997.758$$

Two Simple Identities: the following identities are very useful for handling calculations of variances and covariances:

$$\sum_{i=1}^n (x_i - \overline{x})^2 = \sum_{i=1}^n x_i^2 - n\overline{x}^2 = \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2/n \tag{1.2}$$

and

$$\sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y})=\sum_{i=1}^{n}x_iy_i-n\overline{x}\ \overline{y}=\sum_{i=1}^{n}x_iy_i-(\sum_{i=1}^{n}x_i)(\sum_{i=1}^{n}y_i)/n. \tag{1.3}$$

To prove (1.2) write

$$\sum_{i=1}^{n}(x_i-\overline{x})^2 \;=\; \sum_{i=1}^{n}(x_i^2+\overline{x}^2-2x_i\overline{x})=\sum_{i=1}^{n}x_i^2+n\overline{x}^2-2\overline{x}\sum_{i=1}^{n}x_i.$$

The identities in (1.2) follow now because $\sum_{i=1}^{n}x_i=n\overline{x}$ and so

$$n\overline{x}^2-2\overline{x}\sum_{i=1}^{n}x_i=n\overline{x}^2-2n\overline{x}^2=-n\overline{x}^2=-(\sum_{i=1}^{n}x_i)^2/n.$$

The proof of (1.3) is similar and is left as an exercise.

Table 1.3: Variance and Covariance Calculations

Floor (i)	Bay A (u_i)	Bay B (v_i)	u_i^2	v_i^2	u_iv_i
1	44.4	138.4	1971.36	19154.56	6144.96
2	130.4	236.4	17004.16	55884.96	30826.56
3	127.6	202.5	16281.76	41006.25	25839.00
4	127.7	128.7	16307.29	16563.69	16434.99
5	108.4	154.3	11750.56	23808.49	16726.12
6	184.0	117.0	33856.00	13689.00	21528.00
7	139.1	125.9	19348.81	15850.81	17512.69
8	120.6	127.2	14544.36	16179.84	15340.32
9	174.1	175.6	30310.81	30835.36	30571.96
10	187.9	114.1	35306.41	13018.81	21439.39
Total	1344.2	1520.1	196681.5	245991.8	202364.0

Example 1.5 To illustrate the use of (1.2) and (1.3), let's calculate again $\mathrm{Var}(u)$, $\mathrm{Var}(v)$ and $\mathrm{Cov}(u,v)$ where u_i and v_i are as in Example 1.4. Using (1.2) and the totals from Table 1.3 we have

$$\mathrm{Var}(u)=\frac{196681.5-\frac{(1344.2)^2}{10}}{9}=1777.128 \quad \text{and} \quad \mathrm{Var}(v)=\frac{245991.8-\frac{(1520.1)^2}{10}}{9}=1657.93.$$

Using (1.3) and the totals from Table 1.3 we have

$$\mathrm{Cov}(u,v)=\frac{202364.0-\frac{(1344.2)(1520.1)}{10}}{9}=-218.650.$$

1.4 Sample Quantiles, Median and Interquartile Range

The location of non-symmetric data sets may be poorly represented by the sample mean because the sample mean is very sensitive to the presence of outliers in the data. Notice that observations far from the center have high "torque" or "leverage" and attract the sample mean (center of gravity) toward them. The dispersion of non-symmetric data sets may also be poorly represented by the sample standard deviation.

Example 1.6 A student with an average of 94.7% (SD=2.8%) on the first 10 assignments had a personal problem and did very poorly on the eleventh where he got zero. Calculate his current average and standard deviation.

Solution The mean drops from 95 to

$$\overline{x} = \frac{(10 \times 95) + 0}{11} = 86.09.$$

To calculate the new standard deviation notice that $\sum_{i=1}^{10}(x_i - 95)^2 = 9 \times 2.8^2 = 70.56$ and by (1.2)

$$\sum_{i=1}^{10} x_i^2 = \sum_{i=1}^{10}(x_i - 95)^2 + 10 \times 95^2 = 70.56 + 90250 = 90320.56.$$

Therefore,

$$\text{Var}(x) = \frac{90320.56 + 0^2 - (11 \times 86.09^2)}{10} = 879.4191,$$

and the standard deviation, then, increases from 2.8 to $\sqrt{879.4191} = 29.66$. □

We will see that data sets which are asymmetric or include outliers may be better summarized using the sample quantiles defined below.

Sample Quantiles

Let $0 < p < 1$ be fixed. The sample quantile of order p, $Q(p)$, is a number with the property that approximately $p100\%$ of the data points are smaller than it. For example, if the 0.95 quantile for the class final grades is $Q(0.95) = 85$ then 95% of the students got 85 or less. If your grade is 87 then you are in the the top 5% of the class. On the other hand, if your mark were smaller than $Q(0.10)$ than you would be in the lowest 10% of the class.

To compute $Q(p)$ we must follow the following steps

1 Sort the data from smallest data point, $x_{(1)}$, to largest data point, $x_{(n)}$, to obtain

$$x_{(1)} \leq x_{(2)} \leq \ldots \leq x_{(n)}.$$

The ith largest data point is denoted $x_{(i)}$.

2 Compute the number $np + 0.5$. If this number is an integer, m, then

$$Q(p) = x_{(m)}.$$

If $np + 0.5$ is not an integer and $m < np + 0.5 < m + 1$ for some integer m then

$$Q(p) = \frac{x_{(m)} + x_{(m+1)}}{2}.$$

Example 1.7 Let u_i and v_i be the live loads on the first two floors (see Table 1.4). Calculate the quantiles of order 0.25, 0.50 and 0.75 for the live load on floors 1 and 2 and for the differences $w_i = u_i - v_i$ between the live loads on these two floors.

Solution

To calculate the quantile of order 0.25 for the live load on floor 1, $Q_u(0.25)$, observe that $n = 20$, $p = .25$ and so $np + .5 = 20 \times .25 + .5 = 5.5$ is between 5 and 6. Using the column $u_{(i)}$ from Table 1.4 we obtain

$$Q_u(0.25) = \frac{u_{(5)} + u_{(6)}}{2} = \frac{112.3 + 119.4}{2} = 115.85.$$

Similar calculations give $Q_v(0.25) = 109.25$ and $Q_w(0.25) = -25.25$. To calculate $Q_u(0.50)$ notice that $np + .5 = 20 \times .50 + .5 = 10.5$ is between 10 and 11. Again, using the column $u_{(i)}$ from Table 1.4 we obtain

$$Q_u(0.50) = \frac{u_{(10)} + u_{(11)}}{2} = \frac{150.4 + 152.3}{2} = 151.35.$$

The reader can check using similar calculations that $Q_v(0.50) = 134.1$, $Q_w(0.50) = 7$, $Q_u(0.75) = 162.85$, $Q_v(0.75) = 166.5$ and $Q_w(0.75) = 38$.

Unfortunately, the sample quantiles do not have the same nice properties as the the sample mean in relation with sums and differences of variables. For example

$$Q_u(0.50) - Q_v(0.50) = 151.35 - 134.1 = 17.25$$

is quite different from $Q_{u-v}(0.50) = Q_w(0.50) = 7$. Also

$$Q_u(0.25) - Q_v(0.25) = 115.85 - 109.25 = 6.6 \neq -25.25 = Q_{u-v}(0.50)$$

and

$$Q_u(0.75) - Q_v(0.75) = 151.35 - 134.1 = 17.25 \neq 38 = Q_{u-v}(0.75).$$

Median and Interquartile Range

The quantiles $Q(0.25)$, $Q(0.5)$ and $Q(0.75)$ are particularly useful and given special names: *lower quartile*, *median* and *upper quartile*. Notice that the lowest 25% of the data is below $Q(0.25)$ and the lowest 75% of the data is below $Q(0.75)$. Because of that, $Q(0.25)$ and $Q(0.75)$ are also called *first* and *third qartiles*.

The lowest 50% of the data is below $Q(0.5)$ and the other half is above it. Therefore the median divides the data into two equal pieces, regardless the shape of the histogram. Because of this property and the fact that the median is not much affected by outliers, it is often used as a measure of location (instead of the mean).

The mean and the median are equal in the case of perfectly symmetric data sets. They are also close in the presence of mild asymmetry. But very asymmetric data sets can produce very different means and medians. When the mean and the median roughly agree we will normally prefer the mean because of its nicer numerical properties (see the comments at the end of Problem 1.7). When they do not, however, we will normally prefer the median because of its resistance to outliers. The difference between the mean and the median is a strong indication of the presence outliers in the data which are severe enough to upset the sample mean.

Table 1.4: Live Load on the First and Second Floors

i	u_i	$u_{(i)}$	v_i	$v_{(i)}$	w_i	$w_{(i)}$
1	44.4	44.4	130.4	54.0	-86.0	-98.0
2	138.4	92.2	236.4	62.3	-98.0	-86.0
3	164.7	98.3	110.4	74.1	54.3	-61.7
4	98.3	105.4	154.5	101.1	-56.2	-56.2
5	178.0	112.3	108.1	108.1	69.9	-27.7
6	123.7	119.4	185.4	110.4	-61.7	-22.8
7	157.5	123.7	62.3	128.6	95.2	-17.2
8	119.4	138.4	74.1	130.4	45.3	-7.0
9	150.4	138.9	137.8	132.8	12.6	-5.1
10	92.2	150.4	54.0	133.1	38.2	1.4
11	169.8	152.3	168.4	135.1	1.4	12.6
12	181.5	156.3	147.5	137.8	34.0	27.7
13	105.4	157.5	133.1	147.5	-27.7	28.2
14	157.6	157.6	164.6	154.5	-7.0	34.0
15	168.4	161.0	173.5	164.6	-5.1	37.8
16	161.0	164.7	132.8	168.4	28.2	38.2
17	156.3	168.4	128.6	169.5	27.7	45.3
18	152.3	169.8	169.5	173.5	-17.2	54.3
19	138.9	178.0	101.1	185.4	37.8	69.9
20	112.3	181.5	135.1	236.4	-22.8	95.2
Mean	138.53		135.38		3.145	
SD	34.66		43.61		51.37	

As a **rule of thumb** we will calculate both the mean and the median and use the mean if they are similar. Otherwise we will use the median. To guide our choice we can calculate the *discrepancy index*

$$d = \sqrt{n}\ \frac{|\text{Mean} - \text{Median}|}{2\ IQR}$$

and choose the mean when d is smaller than 1. The *interquartile range* (IQR), used in the denominator of d above, is defined as

$$IQR = Q(0.75) - Q(0.25),$$

The IQR is recommended as a measure of dispersion in the presence of outliers and lack of symmetry. Notice that IQR is proportional to the length of the central half of the data, regardless the shape of the histogram, and it is not much affected by outliers.

Example 1.8 Refer to Example 1.6. Calculate the median, the interquatile range and the discrepancy index d for the student's marks before and after the eleventh assignment (The marks are 94, 93, 95, 91, 96, 91, 98, 93, 99, 97 and 0). just one

Solution Since the sorted marks (before the eleventh assignment) are 91, 91, 93, 93, 94, 95, 96, 97, 98, 99, $Q(0.25) = x_{(3)} = 93$, $Q(0.5) = (x_{(5)} + x_{(6)})/2 = (94 + 95)/2 = 94.5$ and $Q(0.75) = x_{(8)} = 97$. Therefore, Median$(x) = 94.5$, $IQR(x) = 97 - 93 = 4$ and $d = \sqrt{10}(94.7 - 94.5)/(2 \times 4) = 0.07905694$.

Including the eleventh assignment we have $Q(0.25) = (x_{(3)} + x_{(4)})/2 = (91 + 93)/2 = 92$, $Q(0.5) = x_{(6)} = 94$ and $Q(0.75) = (x_{(8)} + x_{(9)})/2 = (96 + 97)/2 = 96.5$. Therefore, the new median and IQR are: Median$(x) = 94$ and $IQR(x) = 96.5 - 92 = 4.5$. Unlike the mean, the median is very little affected by the single poor performance. This is also reflected by the large discrepancy index $d = \sqrt{11}(86.09 - 94)/(2 \times 9) = 2.915$.

□

Example 1.9 Table 1.5 gives the mean, median, standard deviation and IQR for the data sets on Figure 1.2. The mean and median of Tobin's Q ratios show appreciable differences ($d = 2.98$). In addition, their standard deviation is more than twice their IQR. Clearly, the mean and standard deviation are upset by a few heavily over–rated firms. Tobin's Q ratios are then better represented by their median and IQR. The effect of outliers and lack of symmetry is moderate in the case of the "Age of Officers" data. Although $d = 1.07$ the mean and standard deviation still summarize these data well. Finally, for the "Speed of Light" data the two clear (lower) outliers do not seem to have much affect on the sample mean ($d = 0.64$).

Table 1.5: Summary figures for the data sets displayed on Figure 1.2

Data Set	Mean	Median	Discrepancy	S. Deviation	IQR
Tobin's Q ratio	158.6	118.5	2.98	97.749	47.593
Age of officers	51.494	52	1.07	1.739	2.222
Speed of light	24.826	24.827	0.64	0.011	0.005

1.5 Box Plot

The *box plot* is a powerful tool to display and compare data sets. It is just a box with "whiskers" which helps to visualize the main quantiles ($Q(0.25)$, $Q(0.50)$ and $Q(0.75)$) and the extreme data points (maximum and minimum).

For the following discussion refer to Figure 1.4 (b) and (d). The lower and upper ends of the box are determined by the lower and upper quartiles ($Q(0.25)$ and $Q(0.75)$); a line sectioning the box displays the sample median and its relative position within the interquartile range. The median then divides the main box into two smaller sub–boxes which represent the lower and upper central quarters of the data. Symmetric data sets have upper and lower sub–boxes of equal size. Asymmetric data sets have sub–boxes of different sizes, the larger one indicating the direction of the asymmetry. The data on Figure 1.4 (b) is mildly asymmetric with a longer lower tail: the lower sub–box is larger than the upper one and the lower whisker is longer than the upper one. The data on Figure 1.4 (d) is symmetric. The location and dispersion of a data set are also clearly conveyed by the box plot: the position of the box (and the median line) give the location; the size (length) of the box (proportional to the IQR) gives the dispersion. Larger boxes indicate larger dispersion. Finally, the whiskers at either end extend to the extreme values (maximum and minimum).

Points which are above $Q(0.75) + 1.5IQR$ or below $Q(0.25) - 1.5IQR$ are considered *outliers*. The following rule is used to help visualizing outliers in the data: the length of the whiskers should not exceed $1.5IQR$ and points outside this range are displayed as unconnected horizontal lines. This is illustrated by Figure 1.4 (a) and (c) where the presence of outliers is flagged by the existence of unconnected horizontal lines above the upper whisker (Figure 1.4 (a)) or below the lower whisker (Figure 1.4 (c)).



Figure 1.4: Box plots for the data sets displayed on Figure 1.2

Example 1.10 Table 2.3 gives the monthly average flow (cubic meters per second) for the Fraser River at Hope, BC, for the period 1971–1990. Figure 1.5 gives the boxplots for each month, from January to December (from left to right). The year to year distributions of the monthly flows are mildly asymmetric, with longer upper tails, and there are some outliers. However, the location and dispersion summaries (see Table 1.10) are roughly consistent for most months and point to the same conclusion: the river flow, and its variability as well, are much larger in the summer.

Table 1.6: Fraser River Monthly Flow (cms)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Mean	957.4	894.8	993.1	1941.0	4994.5	6973.0	5505.0	3548.0	2340.0	1816.0	1588.9	1092.4
Median	868.0	849.5	926.5	2010.0	5000.0	6365.0	5120.0	3380.0	2245.0	1910.0	1525.0	1005.0
SD	274.4	202.8	233.5	477.8	976.4	1434.2	1212.2	886.4	685.6	401.7	366.1	282.2
IQR	174.6	163.0	257.0	427.8	613.0	1325.9	1277.8	505.6	446.3	424.1	377.8	181.1

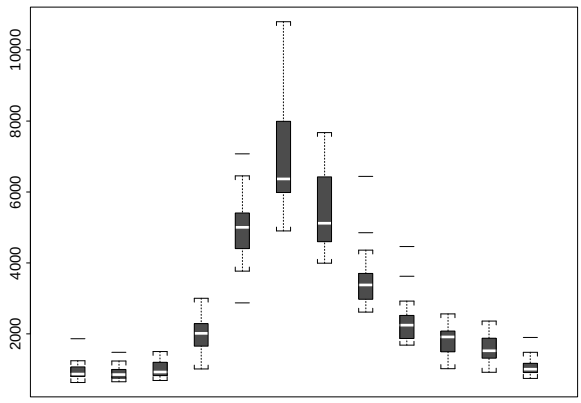


Figure 1.5: Fraser River monthly flow (cms) from January (left) to December (right)

1.6 Exercises

Problem 1.1 The records of a department store show the following total monthly finance charges (in dollars) for 240 customers which accounts included finance charges (see Table 1.7). From a department store's records for a particular month, the total monthly finance charges in dollars were obtained from 240 customers accounts that included finance charges. See the table shown below:
(a) Complete the frequency table. What percentage of customers were charged less than $20?

Table 1.7: Finance Charges from 240 Accounts

Class Limits	Numbers of Customers
0 – 5	65
5 – 10	88
10 – 15	42
15 – 20	27
20 – 25	18

(b) Construct a histogram using the four classes given above.
(c) Calculate the mean, variance and standard deviation.

Problem 1.2 Before microwave ovens are sold, the manufacturer must check to ensure that the radiation coming through the door is below a specified safe limit. The amounts of radiation leakage (mw/cm^2) from 25 ovens, with the door closed, are:

15	9	18	10	5
12	8	5	8	10
7	2	1	5	3
5	15	10	15	9
8	18	1	2	11

(a) Calculate the mean, variance and standard deviation.
(b) What are the median, quartiles and interquartile range?
(c) Compare the results of (a) and (b).
(d) Draw the box plot.

Problem 1.3 The following data are the waiting times (in minutes) between eruptions of Old Faithful geyser between August 6 and 10, 1985.

816	611	796	573	809
778	599	774	748	723
796	1051	820	748	
682	781	772	797	
711	578	696	851	

(a) Calculate the mean, variance and standard deviation.
(b) What are the median, quartiles and interquartile range?
(c) Compare the results of (a) and (b).
(d) Draw the box plot.

Problem 1.4 The following numbers are the final marks of 16 students in a previous STAT 251 class.

64 86 77 68 95 91 58 91 83 97 96 14 32 68 89 75

(a) Calculate the mean, variance and standard deviation.
(b) What are the median, quartiles and interquartile range?
(c) Compare the results of (a) and (b).
(d) Draw the box plot.

Problem 1.5 In 1798, Henry Cavendish estimated the density of the earth (as a multiple of the density of water) by using a torsion balance. The dataset below contains his 29 measurements.

Table 1.8: Cavendish Measurements of the Density of the Earth

5.50	5.47	5.29	5.55	5.75	5.27
5.57	4.88	5.34	5.34	5.29	5.85
5.42	5.62	5.26	5.30	5.10	5.65
5.61	5.63	5.44	5.36	5.86	5.39
5.53	4.07	5.46	5.79	5.58	

(a) Calculate the mean, variance and standard deviation.
(b) What are the median, quartiles and interquartile range?
(c) Compare the results of (a) and (b). I particular calculate the discrepancy index between the mean and median.
(d) Briefly state your conclusions.

Problem 1.6 The mean size of twenty five recent projects at a construction company (in square meters) is 25,689 m^2. The standard deviation is 2,542 m^2.
(a) Calculate the mean, variance and standard deviation in square feet [Hint: 1 foot = 0.3048 m].
(b) A new project of 226050 ft^2 has been just completed. Update the mean, variance and standard deviation.

Problem 1.7 The daily sales in April, 1994 for two departments of a large department store (in thousands of USA dollars) are summarized below.

Table 1.9: Daily Sales, April 1994

	Department A	Department B
Mean	24.3	32.4
Standard Deviation	12.4	10.3
Covariance	96.1	

(a) Convert the figures above to hundreds of Canadian dollars (CN \$1 = US \$0.7)
(b) Calculate the mean and standard deviation for the total daily sales for the two departments. Why do you think the combined daily sales are more variable than the individual ones?
(c) Calculate the mean and standard deviation for the difference in daily sales between the two departments. Comment your results.
(d) Under what conditions would the variance of the sums be smaller than the variance of the differences?

Problem 1.8 A manufacturer of automotive accessories provides bolts to fasten the accessory to the car. Bolts are counted and packaged automatically by a machine. There are several adjustments that affect the machine operation. An experiment to find out how several variables affect the speed of the packaging process was carried out. In particular, the total number of bolts to be counted (10 and 30) and the sensitivity of the electronic eye (6 and 10) have been considered. The observed times (in seconds per bolt) are given in Table 1.10.
(a) Summarize and describe the data.
(b) What adjustments have the greatest effect?
(c) How would you adjust the machine to shorten the packaging time?

Problem 1.9 Find the average, variance and standard deviation for the following sets of numbers.
a) $1,\ 2,\ 3,\ \ 4,\ 5,\ \ \ldots,\ \ 300$
b) $4,\ 8,\ 12,\ 16,\ 20,\ \ \ldots,\ 1200$
c) $1,\ 2,2,\ 3,3,3,\ 4,4,4,4,\ \ \ldots,\ 9,9,9,9,9,9,9,9,9$
Hint: $\sum^n i = n(n+1)/2, \quad \sum^n i^2 = n(n+1)(2n+1)/6, \quad \sum^n i^3 = n^2(n+1)^2/4$ and $\sum^n i^4 = n(n+1)(6n^3+9n^2+n-1)/30$

Table 1.10: Time for Counting and Packaging Bolts

10 Bolts	30 Bolts	Low Sens (6)	High Sens (10)
0.57	0.90	0.57	1.76
1.76	0.65	1.13	0.84
1.13	0.62	1.67	1.20
0.84	0.86	0.92	0.39
1.67	0.63	0.90	0.65
1.20	0.75	0.62	0.86
0.92	0.80	0.63	0.75
0.39	1.00	0.80	1.00
1.34	4.31	1.34	3.43
3.43	3.58	3.97	1.06
3.97	3.72	2.89	3.56
1.06	3.64	1.72	0.60
2.89	3.35	4.31	3.58
3.56	3.64	3.72	3.64
1.72	3.55	3.35	3.64
0.60	4.47	3.55	4.47

Table 1.11: Earthquakes in 1993

Magnitude	frequency
0.1–1.0	9
1.0–2.0	1177
2.0–3.0	5390
3.0–4.0	4263
4.0–5.0	5034
5.0–6.0	1449
6.0–7.0	141
7.0–8.0	15
8.0–9.0	1

Problem 1.10 The number of worldwide earthquakes in 1993 is shown in the following table

(a) Complete the frequency table. What percentage of earthquakes were below 5.0? Above 6.0?
(b) Draw a histogram and comment on it.
(c) Calculate the mean and standard deviation for the earthquake magnitude in 1993.

Problem 1.11 The daily number of customers served by a fast food restaurant were recorded for 30 days including 9 weekends and 21 weekdays. The average and standard deviations are as follows:
Weekends: $\overline{x}_1 = 389.56, \quad \text{SD}_1 = 27.4$
Weekdays: $\overline{x}_2 = 402.19, \quad \text{SD}_2 = 26.2$
Calculate the average and standard deviation for the 30 days.

Problem 1.12 The average and the standard deviation for the weights of 200 small concrete–mix bags (nominal weight = 50 pounds) are 51.2 pounds and 1.5 pounds, respectively. A new sample of 200 large concrete–mix bags (nominal weight = 100 pounds) have just been weighed. Do you expect that the standard deviation for the last sample will be closer to 1.5 pounds or to 3.0 pounds? Justify your answer.

Problem 1.13 Given the data set $x_1 = 1, \quad x_2 = 3, \quad x_3 = 8, \quad x_4 = 12, \quad x_5 = 20$ calculate the function

$$D(t) = \sum^{5} |x_i - t|,$$

for several values of t between 1 and 20, and plot $D(t)$ versus t. Where is the minimum achieved? Do the same "experiment" for the data set $x_1 = 1, \quad x_2 = 3, \quad x_3 = 8, \quad x_4 = 12$. Do you notice any pattern? If so, repeat this experiment for several additional sets of numbers, to investigate the persistence of this pattern. What is your conclusion? Can you prove it mathematically?

Problem 1.14 Each pair (x_i, w_i), $i = 1, \cdots, n$, represents the placement and magnitude of a vertical force acting on a uniform beam. Find the center of gravity of this system. [Hint: see the discussion under "The Sample Mean as Center of Gravity" and notice that in the present case the vertical forces are not equal].

Problem 1.15 Calculate the center of gravity of the system when the placements (x_i) and weights (w_i) are given by Table 1.12.

Table 1.12: Placements of Vertical Forces on a Uniform Beam

x_i	w_i	x_i	w_i
1.8	2.1	1.2	1.5
1.4	1.6	1.3	4.7
1.3	1.4	1.2	2.3
3.8	6.4	1.2	2.3
1.2	1.3	1.4	3.1
1.9	1.2	1.3	1.9
1.2	1.2	1.6	2.4
1.1	3.1	1.1	3.7
1.1	1.1	1.2	1.2

Problem 1.16 Each pair (x_i, w_i), $i = 1, \cdots, n$, represents the placement and magnitude of a vertical force acting on a uniform beam. What values of w_i would make the sample median the center of gravity? Consider the cases when n is even and n odd separately.

Problem 1.17 The maximum annual flood flows for a certain river, for the period 1941-1990, are given in Table 1.6.
(i) Summarize and display these data.
(ii) Compute the mean, median, standard deviation and interquartile range.
(iii) If a one–year construction project is being planned and a flow of 150000 *cfs* or greater will halt construction, what is the "probability" (based on past relative frequencies) that the construction will be halted before the end of the project? What if it is a two-year construction project?

Problem 1.18 The planned and the actual times (in days) needed for the completion of 20 job orders are given in Table 1.14.
(a) Calculate the average and the median planned time per order. Same for the actual time.
(b) Calculate the corresponding standard deviations and interquartile ranges.

Table 1.13: Maximum annual flood flows

Year	Flood, *cfs*	Year	Flood, *cfs*
1941	153000	1966	159000
1942	184000	1967	75000
1943	66000	1968	102000
1944	103000	1969	55000
1945	123000	1970	86000
1946	143000	1971	39000
1947	131000	1972	131000
1948	99000	1973	111000
1949	137000	1974	108000
1950	81000	1975	49000
1951	144000	1976	198000
1952	116000	1977	101000
1953	11000	1978	253000
1954	262000	1979	239000
1955	44000	1980	217000
1956	8000	1981	103000
1957	199000	1982	86000
1958	6000	1983	187000
1959	166000	1984	57000
1960	115000	1985	102000
1961	88000	1986	82000
1962	29000	1987	58000
1963	66000	1988	34000
1964	72000	1989	183000
1965	37000	1990	22000

(c) If there is a delay penalty of \$5000 per day and a before–schedule bonus of \$2500 per day, what is the average net loss (negative loss = gain) due to differences between planned and actual times? What is the standard deviation?
(d) Study the relationship between the planned and actual times.

(e) What would be your advice to the company based on the analysis of these data?

Problem 1.19 Show that (a) $\mathrm{Cov}(x, y) = [(\sum x_i y_i) - n\overline{x}\ \overline{y}]/(n-1)$.
(b) If $u_i = a + b\ x_i \quad$ and $\quad v_i = c + d\ y_i$, then $\mathrm{Cov}(u, v) = bd\mathrm{Cov}(x, y)$.

Table 1.14: The planned and the actual times

Order	Planned Time	Actual Time	Order	Planned Time	Actual Time
1	22	22	11	17	18
2	11	8	12	27	34
3	11	8	13	16	14
4	16	14	14	30	35
5	21	20	15	22	18
6	12	16	16	17	16
7	25	29	17	13	12
8	20	20	18	18	14
9	13	10	19	21	19
10	34	39	20	18	17

Problem 1.20 The total paved area, X (in km^2), and the time, Y (in days), needed to complete the project was recorded for 25 different jobs. The data is summarized as follows:

$$\overline{x} = 12.5 \text{ km}^2 \quad , \quad \text{SD}(x) = 1.2 \text{ km}^2$$
$$\overline{y} = 30.8 \text{ days} \quad , \quad \text{SD}(y) = 3.7 \text{ days}$$
$$\text{Cov}(x, y) = 3.4$$

Give the corresponding summaries when the area is measured in ft^2 and the time is measure in hours.
Hint: 1 foot = 0.3048 m, and 1 km = 1000 m.

Chapter 2

Summary and Display of Multivariate Data

In practice, we usually consider several variables simultaneously. In addition to describing each variable as in Chapter 1, we may wish to investigate their possible relationships. Some examples are provided by the first–crack and failure load data on Table 2.1, the Fraser River flow data in Table 2.3 and the yield data in Table 2.2. Are the first–crack and failure load of concrete beams related? Is it possible to use the first–crack load to predict the failure load? Are the Fraser River mean monthly flows related? Is it possible to use the average flows from previous months to predict the current and future months flows? How does the temperature affect the yield of the chemical process? Is there a simple equation relating the yield response to changes on the temperature?

As explained in the previous chapter, raw data must be summarized and/or graphically displayed to facilitate their analysis. We will now learn some simple techniques which can be used to summarize multivariate data and describe their relationships. In the next sections we will introduce scatter plots, correlation coefficients, multiple correlation coefficients, simple linear regression and multiple linear regression.

2.1 Scatter Plot

Simultaneous observations on a pair of variables (x_i, y_i), $i = 1, \ldots, n$, can be graphically displayed on a *scatter plot*. Each observation is represented as a point with x–coordinate x_i and y–coordinate y_i. Scatter plots help in visualizing statistical relationships between variables (or the lack of them).

Linear Association and Causality

Some examples of scatter plots are presented on Figure 2.1. The dotted lines represent the means for the "x" and "y" variables. For example the mean flows for January, February and June are 957.4, 894.8 and 6973, respectively. Figure 2.1 (a) shows a *positive linear association* between January and February flows: years with higher than average flows in January tend to have also higher than average flows in February and vice versa for lower than average flows. Figure 2.1 (b), on the other hand, shows a lack of linear association: years with higher than average flows in January come together with higher than average and lower than average flows in June with approximately

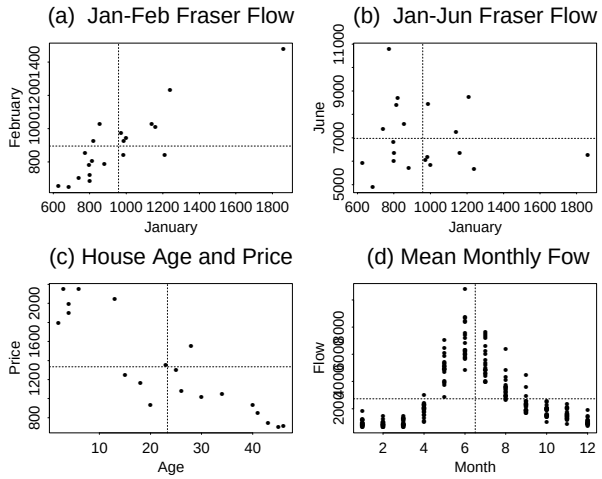


Figure 2.1: Some Examples of Scatter Plots

the same frequency. Similarly for lower than average January flows. Figure 2.1 (c) shows a *negative linear association* between the age and price of twenty randomly selected houses: older than average houses tend to have lower than average prices and vice versa for newer houses; Figure 2.1 (d) shows a *non–linear association* between time of the year and river flow: the monthly mean flows first increase (until June) and then decrease.

A common mistake is to confuse the concepts of linear association and *causality*. If we find a positive linear association between two variables we can say that they tend to take values above and below their means simultaneously. The observed linear association may be the result of a causal relation between the variables – an increase in one of them *causes* an increase in the other. In many occasions, however, observed linear associations are the result of the action of a third variable (called *lurking variable*) which drives the other two. For instance, the linear association between January and February Fraser flows might be due to the effect of a lurking variable, namely the weather. If in a given year we artificially increase the Fraser January flow we cannot expect a naturally occurring higher flow in February.

Several Pairs of Variables

We often wish to investigate the pairwise relations between several pairs of variables. This can be accomplished by several ways. One way is to use different symbols (dots, stars, letters, numbers, etc.) to represent the points and overlay the scatter plots on a single picture, facilitating their comparison. For instance, the weights and heights of men and women could be plotted on a single scatter plot using the letter "w" for women and "m" for male.

Another technique for dealing with several variables is to display the scatter plots in a "matrix" layout. *Scatter plot matrices* are useful for uncovering possible patterns in the pairwise association structure. An example is given by Figure 2.2. Notice that the strength of association decreases as months get further apart. Moreover, while January, February and March show some association, April and May seem to have less (if any) association with other months.

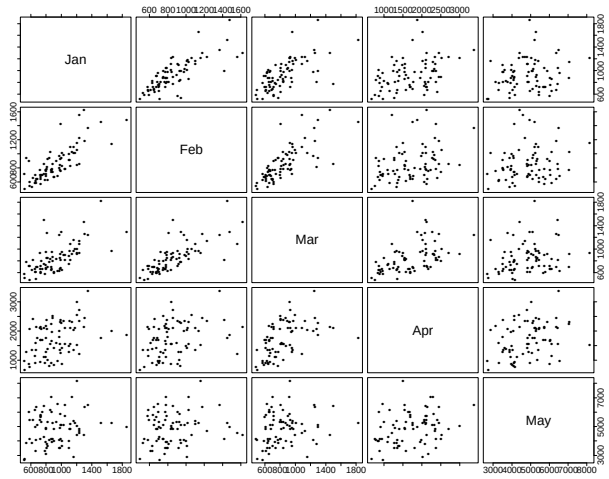


Figure 2.2: Fraser River Monthly Average Flow (1914-1990)

2.2 Covariance and Correlation Coefficient

The *Covariance* and the *correlation coefficient* are used to quantify the degree of linear association between pairs of variables. If two variables, x_i and y_i, are positively associated then when one of them is above (below) its mean the other will also tend to be above (below) its mean. Therefore, the products $(x_i - \overline{x})(y_i - \overline{y})$ will be mostly positive and the *sample covariance*,

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}) \tag{2.1}$$

will be large and positive. On the other hand, if the variables are negatively associated, when one of them is above (below) its mean the other will tend to be below (above) its mean and so the products $(x_i - \overline{x})(y_i - \overline{y})$ will be mostly negative. In this case the sample covariance (2.1) will be large and negative. Finally, if the variables are not positively nor negatively associated the products $(x_i - \overline{x})(y_i - \overline{y})$ will be positive and negative with approximately the same frequency (there will be a fair degree of cancellation) and the sample covariance will be small.

The following formula provides a simple procedure for the hand calculation of the covariance:

$$\begin{aligned} \text{Cov}(x, y) &= \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}) \\ &= \frac{n}{n-1} \left[\overline{xy} - \overline{x}\ \overline{y} \right], \qquad \text{where} \quad \overline{xy} = \frac{1}{n} \sum_{i=1}^{n} x_i y_i \end{aligned} \tag{2.2}$$

Some problems with the interpretation of the covariance and its direct use as a measure of linear association are illustrated in Example 2.1.

Example 2.1 Consider the measurements (x_i, y_i) of the first–crack and failure load (in pounds per square foot) on Table 2.1. Figure 2.3 suggests that there little association between these measurements. Since $\overline{x} = 8396.6$ pounds per square foot, $\overline{y} = 16,064.4$ pounds per square foot, and

$\overline{xy} = 134875,645$ square pounds per ft^4, from (2.2)

$$\text{Cov}(x,y) = (20/19)\,[(134875645) - (8396.6)(16064.4)] = -11,258.99 \text{ square pounds per ft}^4$$

If the loads are given in thousand of pounds per square foot instead of pounds per square foot, then $u_i = x_i/1000$,$v_i = y_i/1000$ and, from Problem 1.19,

$$\text{Cov}(u,v) = \frac{\text{Cov}(x,y)}{1000 \times 1000} = -0.011259 \text{ million square pounds per ft}^4.$$

Table 2.1: Strength of concrete beams

Unit	First–Crack Load (X)	Failure Load (Y)
1	7610	18103
2	9528	15283
3	7071	19171
4	7463	16014
5	4440	12840
6	10929	19606
7	12385	14570
8	5734	16755
9	6342	15713
10	6772	17094
11	7519	13808
12	8511	16480
13	9087	16131
14	9072	15315
15	12157	12683
16	6504	14625
17	6654	16615
18	8700	15643
19	11613	15480
20	9841	19359

Correlation Coefficient

Problem 2.1 illustrates the strong dependency of $\text{Cov}(x,y)$ on the scale of the variables. A measure of linear association which is independent from the variables scale (see 2.5) is provided the *sample correlation coefficient*,

$$\text{r}(x,y) = \frac{\text{Cov}(x,y)}{\sqrt{\text{Var}(x)\text{Var}(y)}} = \frac{\text{Cov}(x,y)}{\text{SD}(x)\ \text{SD}(y)}.$$

More precisely, if $u_i = a + bx_i$ and $v_i = c + dy_i$ then $\text{r}(u,v) = \text{sign}(bd)\text{r}(x,y)$.

Another advantage of $\text{r}(x,y)$ is that it takes values between -1 and 1 (see 2.7). Therefore, values of $\text{r}(x,y)$ close to 1 indicate positive linear association, values of $\text{r}(x,y)$ close to -1 indicate negative linear association. Values of $\text{r}(x,y)$ close to 0 indicate lack of linear association.

For the data in Example 2.1, $\text{Cov}(x,y) = -11258.99$, $\text{SD}(x) = 2193.17$, $\text{SD}(y) = 1949.36$ and

$$\text{r}(x,y) = \frac{-11258.99}{(2193.17)(1949.36)} = -0.0026.$$

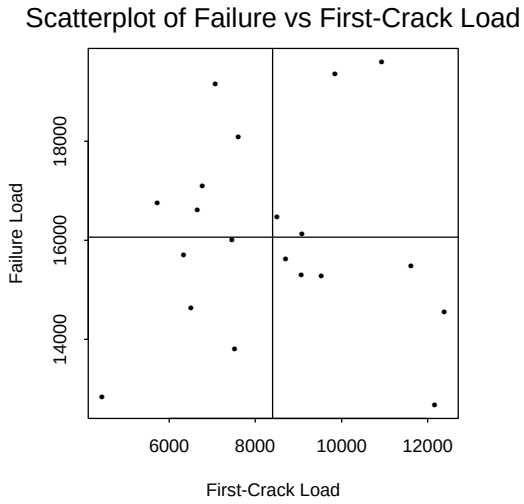


Figure 2.3: First–Crack Load vs Failure Load

The small value of $\mathrm{r}(x, y)$ confirms the qualitative impression from Figure 2.3 that the first crack and the failure loads (in the case of these concrete beams) are not related. The main implication from a practical point of view is that the first crack of a given beam cannot be used to predict its ultimate failure load.

Example 2.2 Table 2.2 gives the results of an experiment to study the relation between temperature (in units of 10° Fahrenheit) and yield of a certain chemical process (percentage). The reader can verify that in this case $\overline{x} = 34.5$, $\overline{y} = 43.07$, $\mathrm{Var}(x) = 77.50$, $\mathrm{Var}(y) = 128.06$ and $\mathrm{Cov}(x, y) = 96.2759$. Therefore, the correlation coefficient,

$$\mathrm{r}(x, y) = \frac{96.2759}{\sqrt{77.50 \times 128.06}} = 0.9664 = 0.97,$$

indicates a strong positive linear association between temperature and yield. This is also clearly suggested by the scatter plot in Figure 2.4. Notice that the relation between yield and tempreature is likely to be causal, that is, the increase in yield may be actually *caused* by the increase in temperature.

Several Pairs of Variables

When we have several variables their covariances and correlation coefficients can be arranged in matrix layouts called *covariance matrix* and *correlation matrix*. Although the covariance matrix is difficult to interpret due to its dependence on the scale of the variables, it is nevertheless routinely computed for future usage.

The correlation matrix is the numerical counterpart of the scatter plot matrix discussed before. For the River Fraser Data (see Figure 2.2) we have

Table 2.2: Yield of a chemical process

Unit	Temp. (X)	Yield (Y)	Unit	Temp. (X)	Yield (Y)
1	20	28	16	35	41
2	21	26	17	36	45
3	22	22	18	37	53
4	23	25	19	38	46
5	24	27	20	39	44
6	25	32	21	40	49
7	26	31	22	41	53
8	27	33	23	42	49
9	28	38	24	43	51
10	29	41	25	44	55
11	30	41	26	45	56
12	31	38	27	46	58
13	32	41	28	47	58
14	33	46	29	48	58
15	34	44	30	49	63

	Jan	Feb	Mar	Apr	May
Jan	1.00	0.78	0.65	0.40	0.18
Feb	0.78	1.00	0.75	0.34	0.15
Mar	0.65	0.75	1.00	0.50	0.19
Apr	0.40	0.34	0.50	1.00	0.29
May	0.18	0.15	0.19	0.29	1.00

As already observed from Figure 2.2, February flaws are somewhat correlated with January and March flaws (with correlation coefficients 0.78 and 0.75, respectively). January and March flaws are also marginally correlated (correlation coefficient equal to 0.65). The correlation coefficients between all the other pairs of months are below 0.50.

2.3 The Least Squares Regression Line

The scatter plot of linearly associated variables approximately follows a linear function

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$$

called *regression line*. The hats indicate that $\hat{\beta}_0$, $\hat{\beta}_0$ and $\hat{f}(x)$ are calculated from the data. In this context X and Y play different roles and are given special names. The independent variable X is called *explanatory variable* and the dependent variable Y is called *response variable*.

Least Squares

The solid line on Figure 2.4 (see Example 2.2) was obtained by the method of *least squares* (LS). According to this method, the regression coefficients (the *intercept* $\hat{\beta}_0$ and the *slope* $\hat{\beta}_1$) minimize (in b_0 and b_1) the sum of squares

$$S(b_0, b_1) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2 .$$

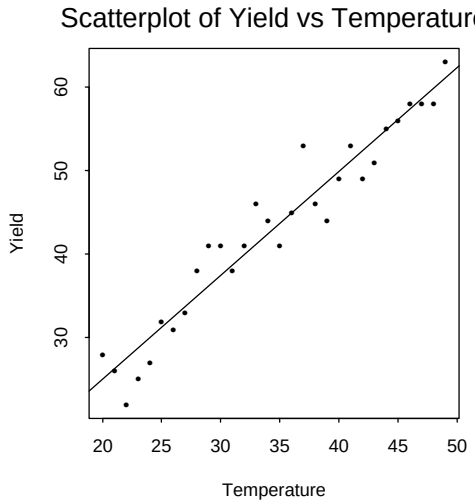


Figure 2.4: Yield vs Temperature

The LS coefficients are the solution to linear equations

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i = 0.$$

(Gauss Equations)

which are obtained by r differencing $S(b_0, b_1)$ with respect to b_0 and b_1. Carrying out the summations and dividing by n we obtain,

$$\overline{y} - \hat{\beta}_0 - \hat{\beta}_1 \overline{x} = 0 \tag{2.3}$$

$$\overline{xy} - \hat{\beta}_0 \overline{x} - \hat{\beta}_1 \overline{xx} = 0. \tag{2.4}$$

where

$$\overline{xy} = (1/n)\sum_{i=1}^{n} x_i y_i \quad \text{and} \quad \overline{xx} = (1/n)\sum_{i=1}^{n} x_i^2 \tag{2.5}$$

From (2.3), $\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$. Substituting this into (2.4) and solving for $\hat{\beta}_1$ gives

$$\hat{\beta}_1 = \frac{\overline{xy} - \overline{x}\ \overline{y}}{\overline{xx} - \overline{x}\ \overline{x}}.$$

Fitted Values and Residuals

The regression line $\hat{f}(x)$ and the *regression coefficients* $\hat{\beta}_0$ and $\hat{\beta}_1$ are good summaries for linearly associated data. In this case the *fitted value*

$$\hat{y}_i = \hat{f}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i \qquad \text{(Fitted Value)}$$

will be "close" to the observed value of y_i. How close depends on the strength of the linear association. The differences between the observed values y_i and the fitted values $\hat{y}_i$,

$$e_i = y_i - \hat{y}_i \qquad \text{(Residual)},$$

are called *regression residuals.*

Residual Plot

The regression residuals e_i are usually plotted against the fitted values $\hat{y}_i$ to determine the appropriateness of the linear regression fit. If the data are well summarized by the regression line (see Figure 2.5 (a)) the corresponding scatter plot of $(\hat{y}_i, e_i)$ has no systematic pattern (see Figure 2.5 (c)). Examples of "bad" residual plots – that is, plots that indicate that the regression line is a poor summary for the data – are given on Figure 2.5 (d) and (e). The corresponding scatter plots and linear fits are given on Figure 2.5 (b) and (c). In the case of Figure 2.5 (d), the residuals go from positive to negative and back to positive, suggesting that the relation between X and Y may not be linear. In the case of Figure 2.5 (e) larger fitted values have larger residuals (in absolute value).

2.4 Multiple Linear Regression

In practice we often use several explanatory variables to "predict" or "interpolate" the values of a single response variable. The explanatory variables may all be distinct or may include functions (powers) of the observed explanatory variables.

If for example, we have p explanatory variables (X_1, X_2, $\cdots$, X_p) and n observations or "cases", it is convenient to use double subscript notation. The first subscript (i) indicates the case and the second subscript (j) indicates the variable.

Case (i)	Response Variable (y_i)	Explanatory Variables (x_{ij})			
1	y_1	x_{11}	x_{12}	$\cdots$	x_{1p}
2	y_2	x_{21}	x_{22}	$\cdots$	x_{2p}
3	y_3	x_{31}	x_{32}	$\cdots$	x_{3p}
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
n	y_n	x_{n1}	x_{n2}	$\cdots$	x_{np}

The linear regression function is now given by

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p,$$

and the regression coefficients ($\hat{\beta}_0$, $\hat{\beta}_1$, $\cdots$,$\hat{\beta}_p$) minimize (in b_0, b_1,$\cdots$, b_p) the sum of squares

$$S(b_0, b_1, \cdots, b_p) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2} - \cdots - b_2 x_{ip})^2 .$$

The least square coefficients are the solution to the linear equations

$$\begin{aligned}
\sum_{i=1}^{n}(y_i - \hat\beta_0 - \hat\beta_1 x_{i1} - \hat\beta_2 x_{i2} - \cdots - \hat\beta_p x_{ip}) &= 0 \\
\sum_{i=1}^{n}(y_i - \hat\beta_0 - \hat\beta_1 x_{i1} - \hat\beta_2 x_{i2} - \cdots - \hat\beta_p x_{ip})x_{i1} &= 0 \\
\sum_{i=1}^{n}(y_i - \hat\beta_0 - \hat\beta_1 x_{i1} - \hat\beta_2 x_{i2} - \cdots - \hat\beta_p x_{ip})x_{i2} &= 0 \qquad \text{(Gauss Equations)} \\
&\cdots\cdots\cdots \\
&\cdots\cdots\cdots \\
&\cdots\cdots\cdots \\
\sum_{i=1}^{n}(y_i - \hat\beta_0 - \hat\beta_1 x_{i1} - \hat\beta_2 x_{i2} - \cdots - \hat\beta_p x_{ip})x_{ip} &= 0
\end{aligned}$$

which are obtained by differencing $S(b_0, b_1, \cdots, b_p)$ with respect to b_0, b_1, $\cdots$, b_p.

Carrying out the sums and dividing by n we obtain,

$$\begin{aligned}
\overline{y} - \hat\beta_0 - \hat\beta_1\overline{x_1} - \hat\beta_2\overline{x_2} - \cdots - \hat\beta_p\overline{x_p} &= 0 \\
\overline{x_1 y} - \hat\beta_0\overline{x}_1 - \hat\beta_1\overline{x_1 x_1} - \hat\beta_2\overline{x_2 x_1} - \cdots - \hat\beta_p\overline{x_p x_1} &= 0 \\
\overline{x_2 y} - \hat\beta_0\overline{x}_2 - \hat\beta_1\overline{x_1 x_2} - \hat\beta_2\overline{x_2 x_2} - \cdots - \hat\beta_p\overline{x_p x_2} &= 0 \\
&\cdots\cdots\cdots \\
&\cdots\cdots\cdots \\
&\cdots\cdots\cdots \\
\overline{x_p y} - \hat\beta_0\overline{x}_p - \hat\beta_1\overline{x_1 x_p} - \hat\beta_2\overline{x_2 x_p} - \cdots - \hat\beta_p\overline{x_p x_p} &= 0
\end{aligned}$$

where

$$\overline{yx_j} = (1/n)\sum_{i=1}^{n} x_{ij}y_i \quad \text{and} \quad \overline{x_j x_k} = (1/n)\sum_{i=1}^{n} x_{ij}x_{ik}. \tag{2.6}$$

2.5 Exercises

Problem 2.1

Problem 2.2 The following data give the logarithm (base 10) of the volume occupied by algal cells on successive days, taken over a period over which the relative growth rate was approximately constant.

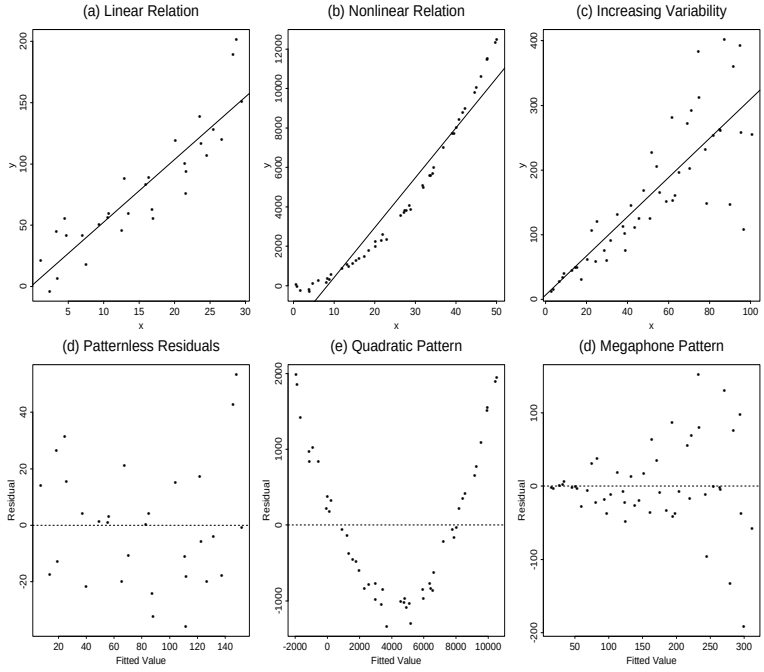


Figure 2.5: Examples of linear regression fits (above) and their residual plots (below).

Day (x)	log Volume ($\log(y)$)
1	3.592
2	3.823
3	4.174
4	4.534
5	4.956
6	5.163
7	5.495
8	5.602
9	6.087

(1) Plot $\log y$ against x. Do you think using the logarithmic scale is appropriate? Why?
(2) Calculate and interpret the sample correlation coefficient.

Problem 2.3 The maximum annual flood flows of a river, for the period 1949–1990, are given in Table 1.6.
(i) Summarize and display these data.
(ii) Compute the mean, median, standard deviation and interquartile range.
(iii) If a one–year construction project is being planned and a flow of 150000 *cfs* or greater will halt construction, what is the relative frequency (based on past relative frequencies) that the construction will be halted before the end of the project? What if it is a two-year construction project?

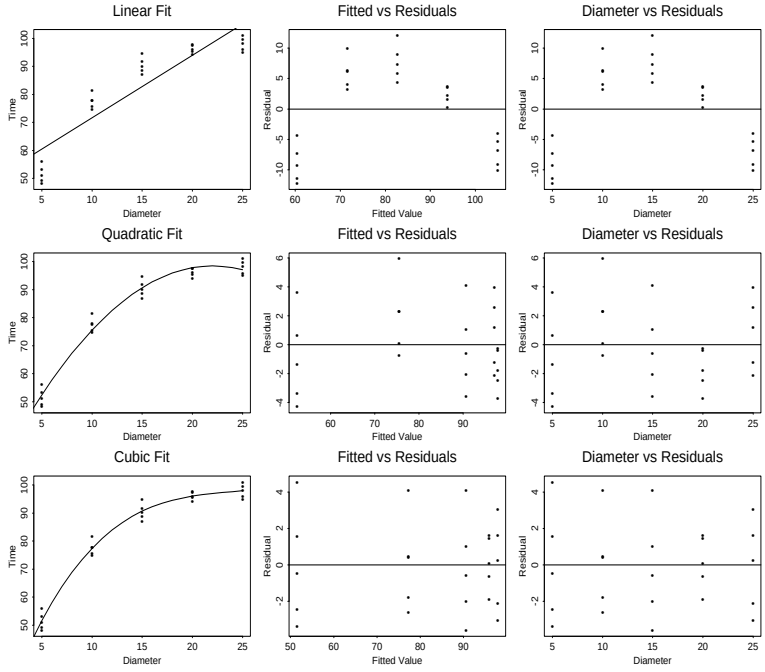


Figure 2.6: Polishing Times.

Problem 2.4 The planned and the actual times (in days) needed for the completion of 20 job orders are given in Table 1.14
(a) Calculate the average and the median planned time per order. Same for the actual time.
(b) Calculate the corresponding standard deviations and interquartile ranges.
(c) If there is a delay penalty of \$5000 per day and a before–schedule bonus of \$2500 per day, what is the average net loss (negative loss = gain) due to differences between planned and actual times? What is the standard deviation?
(d) Study the relationship between the planned and actual times.

(e) What would be your advice to the company based on the analysis of these data?

Problem 2.5 (a) Show that

$$\mathrm{Cov}(x,y) = \frac{(\sum x_i y_i) - n\overline{x}\ \overline{y}}{n-1} \quad \text{and} \quad \beta = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)},$$

(b) Show that if $u_i = a + b\ x_i$ and $v_i = c + d\ y_i$, then
(i) $\overline{u} = a + b\overline{x}$
(ii) $\mathrm{Var}(u) = b^2\mathrm{Var}(x)$
(iii) $\mathrm{r}(u,v) = \mathrm{r}(x,y)$
(iv) $\beta(u,v) = \frac{d}{b}\beta(x,y)$

Table 2.3: Fraser River Monthly Flow (cms)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1971	855	1030	841	1550	6120	7590	5590	3570	2360	1890	1550	908
1972	774	857	1500	2100	6450	10800	7330	4120	2280	1940	1500	1000
1973	984	842	850	1550	4910	6180	5000	2930	1680	2080	1620	1130
1974	987	929	927	2320	5890	8430	7470	4360	2440	1930	1290	978
1975	797	780	736	1100	3940	6830	6070	3420	2300	1950	2360	1480
1976	1140	1030	924	2300	7070	7250	7670	6440	4460	2510	1800	1480
1977	1240	1230	1130	2350	4710	5670	4830	3620	2340	1650	1260	1030
1978	881	791	952	1960	3950	5730	4540	2970	2600	2090	1590	1010
1979	801	721	957	1290	4910	6360	4860	2610	1830	1420	918	952
1980	684	649	703	1760	5120	4900	4010	2720	2600	2080	1630	1900
1981	1860	1480	1300	1880	4950	6260	4890	3620	2130	1530	1950	1140
1982	821	927	844	1010	5360	8690	7230	4850	3620	2310	1470	1110
1983	972	977	1240	1990	4090	6060	5240	3460	2210	1470	2050	878
1984	1160	1010	1160	2030	2870	6370	6580	3780	2920	2560	1370	861
1985	740	706	801	2070	5300	7390	4650	2770	1940	1980	1230	746
1986	813	809	1280	2090	3770	8390	5380	3220	1890	1470	1340	908
1987	1000	944	1300	2280	5120	5840	4070	2980	1680	1020	1210	811
1988	629	657	809	2410	5450	5940	4430	3010	1890	1540	1470	926
1989	800	685	682	1780	4860	6020	3990	3170	1840	1380	2060	1410
1990	1210	841	926	3000	5050	8760	6270	3340	1790	1520	2110	1190

Problem 2.6 The total paved area, X (in km^2), and the time, Y (in days), needed to complete the project was recorded for 25 different jobs. The data is summarized as follows:

$$\overline{x} = 12.5 \text{ km}^2 \quad , \quad \text{SD}(x) = 1.2 \text{ km}^2$$

$$\overline{y} = 30.8 \text{ days} \quad , \quad \text{SD}(y) = 3.7 \text{ days}$$

$$\text{Cov}(x, y) = 3.4 \quad , \quad r(x, y) = 0.766 \quad , \quad \beta = 2.36$$

Give the corresponding summaries when the area is measured in feet^2 and the time is measure in hours.
Hint: 1 foot = 0.305 m, and 1 km = 1000 m.

Problem 2.7 Show that $-1 \leq r(x, y) \leq 1$.
Hint: One can assume without loss of generality that

$$\overline{x} = \overline{y} = 0 \quad \text{and} \quad \text{SD}(x) = \text{SD}(y) = 1 \quad (why?)$$

Then use the fact that

$$0 \leq \sum_{i=1}^{n} (y_i - bx_i)^2$$

for all b, and in particular for $b = \text{Cov}(x, y)/\text{SD}(x)^2$.

Table 2.4: The records of maximum annual flood flows

Year	Flood, *cfs*	Year	Flood, *cfs*
1941	153000	1966	159000
1942	184000	1967	75000
1943	66000	1968	102000
1944	103000	1969	55000
1945	123000	1970	86000
1946	143000	1971	39000
1947	131000	1972	131000
1948	99000	1973	111000
1949	137000	1974	108000
1950	81000	1975	49000
1951	144000	1976	198000
1952	116000	1977	101000
1953	11000	1978	253000
1954	262000	1979	239000
1955	44000	1980	217000
1956	8000	1981	103000
1957	199000	1982	86000
1958	6000	1983	187000
1959	166000	1984	57000
1960	115000	1985	102000
1961	88000	1986	82000
1962	29000	1987	58000
1963	66000	1988	34000
1964	72000	1989	183000
1965	37000	1990	22000

Table 2.5: The planned and the actual times

Order	Planned Time	Actual Time	Order	Planned Time	Actual Time
1	22	22	11	17	18
2	11	8	12	27	34
3	11	8	13	16	14
4	16	14	14	30	35
5	21	20	15	22	18
6	12	16	16	17	16
7	25	29	17	13	12
8	20	20	18	18	14
9	13	10	19	21	19
10	34	39	20	18	17

Chapter 3

Probability

3.1 Sets and Probability

The **theory of probability**, which is briefly discussed below, is needed for the better understanding of some important statistical techniques. This theory is, roughly speaking, concerned with the assessment of the chances (or likelihood) that certain events will or will not occur. In order to give a more precise (and useful) definition of probability, we need first to introduce some technical concepts and definitions.

Random Experiment: The defining feature of a random experiment is that its outcome cannot be determined beforehand. That is, the outcome of the random experiment will only be known after the experiment has been completed. The next time the experiment is performed (seemingly under the exact same conditions) the outcome may be different. Some examples of random experiments are:

– asking a randomly selected person if she smokes,
– counting the number of defective items found in a lot,
– measuring the time elapsed between two consecutive breakdowns of a computer network,
– counting the yearly number of work-related accidents in a production plant,
– measuring the yield of a chemical reaction.

Sample Space ($\mathcal{S}$): Although we may not be able to say beforehand what the outcome of the random experiment will be, we should at least in principle to be able to make a complete "list" of all the possible outcomes. This list (set) of all the possible outcomes is called "the sample space" and denoted by $\mathcal{S}$. A generic outcome (that is, element of $\mathcal{S}$) is denoted by w. The sample spaces for the random experiment listed above are:

– $\mathcal{S} = \{\text{Yes}, \text{No}\}$,

– $\mathcal{S} = \{0, 1, 2, \ldots, n\}$ where n is the lot size,

– $\mathcal{S} = [0, \infty)$, the time (in hours) between breakdowns can be any non-negative real number.

– $\mathcal{S} = \{0, 1, 2, \ldots\}$, the number of accidents can be any non-negative integer number.

– $\mathcal{S} = [0, 100]$, the percentage yield can be any real number between zero and one hundred.

Event: The events, usually denoted by the first upper case letters of the alphabet (A, B, C, etc), are simply subsets of $\mathcal{S}$. Most events encountered in practice are meaningful and can be expressed either in words or using mathematical notation. Some examples (related to the list of random experiments given above) are:

–$A = \{$ less than four defectives$\} = \{0, 1, 2, 3\}$.

–$B = \{$ more than 200 hours$\} = (200, \infty)$.

–$C = \{2, 3, 5, 9\}$

–$D = \{$between ten and twenty percent$\} = [10, 20]$.

An important feature of the events is that they **can** or **cannot** occur, depending on the actual outcome of the random experiment. For instance, if after completing the inspection of the lot we find two defectives, the event A has occurred. On the other hand, if the actual number of defectives turned out to be five, the event A did not occur.

Two rather special events are the "impossible" event – which can never occur – denoted by the empty set $\emptyset$ and the "sure" event – which always occurs – consisting of the entire sample space, $\mathcal{S}$.

Some related mathematical notations are:

$$w \in A \quad \Longleftrightarrow \quad w \text{ belongs to } A \quad \Longleftrightarrow \quad A \text{ occurs}$$

and

$$w \notin A \quad \Longleftrightarrow \quad w \text{ doesn't belongs to } A \quad \Longleftrightarrow \quad A \text{ doesn't occur}$$

Probability Function (P): Evidently, not all the events are equally likely. For instance, the event

$$A = \{\text{more than three million accidents}\}$$

would appear to be quite unlikely, while the event

$$B = \{\text{more than three hours before the next crash}\}$$

would appear to be quite likely.

A probability function P is a function which assigns to each event a number representing the likelihood that this event will actually occur.

For self-consistency reasons, any probability function P must satisfy the following properties:

(1) $P(\phi) = 0$ and $P(\mathcal{S}) = 1$.

(2) $0 \leq P(A) \leq 1$ for all A.

(3) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Properties (4)-(6) below can be derived from (1)-(3).

(4) $P(A \cup B) = P(A) + P(B)$ if A and B are disjoint mutually exclusive) events.
In fact, if A and B are disjoint then $A \cap B = \emptyset$ and $P(A \cap B) = 0$.

(5) $P(A^c) = 1 - P(A)$, where A^c denotes the "complement of A".
In fact, since $A \cup A^c = \mathcal{S}$ and $A \cap A^c = \emptyset$, $1 = P(A \cup A^c) = P(A) + P(A^c)$ and (5) follows.

(6) If $A \subseteq B$ then $P(A) \leq P(B)$.

In fact, since $A \subseteq B$,

$$B = (B \cap A) \cup (B \cap A^c) = A \cup (B \cap A^c).$$

Since A and $(B \cap A^c)$ are disjoint, $P[A \cap (B \cap A^c)] = 0$ and so

$$P(B) = P(A) + P(B \cap A^c) \geq P(A).$$

Example 3.1 It is known from previous experience that the probability of finding zero, one, two, etc. defectives in lots of 100 items shipped by a certain supplier are as given in Table 2.1 below.
Let A, B and C be the events "less than two defectives", "more than one defective" and "one or two defectives", respectively. (a) Calculate $P(A)$, $P(B)$ and $P(C)$. (b) What is the meaning (in words) of the event A^c? Calculate $P(A^c)$ directly and using Property 4. (c) What is the meaning (in words) of the event $A \cup C$? Calculate $P(A \cup C)$ directly and using Property 3.

Table 3.1:

Defectives	Probability
0	0.50
1	0.20
2	0.15
3	0.10
4	0.03
5	0.02
6 or more	0.00

Solution
(a) From Table 2.1, $P(A) = 0.70$, $P(B) = 0.30$, and $P(C) = 0.35$

(b) $A^c = \{\text{two or more defectives}\} = \{\text{more than one defective}\} = B$, from Table 1, $P(A^c) = P(B) = 0.30$. This is consistent with the result we obtain using Property 4:

$$P(A^c) = 1 - P(A) = 1 - 0.70 = 0.30.$$

(c) $A \cup C = \{\text{less than three defectives}\}$. Therefore, directly from Table 1, $P(A \cup C) = 0.85$. To make the calculation using Property 3, we must first find $P(A \cap C)$. Since $A \cap C = \{\text{exactly one defective}\}$, it follows from Table 1 that $P(A \cap C) = 0.20$. Now,

$$P(A \cup C) = 0.70 + 0.35 - 0.20 = 0.85.$$

□

3.2 Conditional Probability and Independence

There are instances when, after obtaining some partial information regarding the outcome of a random experiment, one would like to update the probabilities of certain events, taking into account the newly acquired information.

The updated probability of the event A, when it is known that the event B has occurred, is in general denoted by $P(A|B)$ and called "the conditional probability of A given B". This conditional probability can be calculated by the formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{3.1}$$

provided that $P(B) > 0$. A simple, but nevertheless important, consequence of (2) is that

$$P(A \cap B) = P(A|B)P(B), \tag{3.2}$$

which is sometime called "the multiplication law".

Example 3.1 (continued): Suppose that we know that the lot contains two defectives or more. What is the probability that it contains three or more defectives?

Solution Let

$$B = \{ \text{ two or more defectives } \} = \{ \text{ more than one defective } \}$$

and

$$D = \{ \text{ three or more defectives } \} = \{ \text{ more than two defectives } \}.$$

Since $P(B) = 0.30$ and $P(D \cap B) = P(\{3, 4, 5\}) = 0.15$, the desired conditional probability is

$$P(D|B) = P(D \cap B)/P(B) = 0.15/0.30 = 0.50.$$

□

Posterior Probability and Bayes Formula

Suppose that we wish to investigate the occurrence of a certain event B. For example, consider the collapse of a large industrial building or the crash of a computer network.

The event B may have been caused by one of several possible **causes** or **states of nature** denoted $A_1, A_2 \dots A_m$, For example, the collapse of the industrial building may have been caused by one (and only one) of the following:

A_1 Poor design

- underestimated live load
- underestimated maximum wind speed
- etc.

A_2 Poor construction

- Low grade material
- Insufficient supervision and control
- Gross human error
- etc.

A_3 A combination of A_1 and A_2.

A_4 Other (non-assignable) causes.

Suppose that, from previous experience or some other source (for example some expert's opinion), the conditional probabilities of B given A_i are known. That is, the probabilities that the event B will occur when the the cause A_i is present are known and represented by

$$p_1, p_2, \dots, p_m.$$

We will call these conditional probabilities **risk factors**. Suppose also that the probabilities of each possible cause A_i are known. These probabilities are called **prior probabilities** and denoted

$$\pi_1, \pi_2, \dots, \pi_m.$$

In the case of our example, the prior probabilities may represent the actual fractions of industrial buildings in the country which have some design or construction problems. Or they may represent the subjective beliefs (educated guesses) of some expert consultant (perhaps the engineer hired by the insurance company to investigate the causes of the accident). In summary, we suppose that

$$p_i = P(B|A_i), \qquad \text{and} \qquad \pi_i = P(A_i),$$

are known for all $i = 1, \dots, m$. Notice that

$$\pi_1 + \pi_2 + \dots + \pi_m = 1.$$

The prior probabilities and the risks factors for the collapsed building example are given in columns 2 and 3 of Table 3.2

Table 3.2:

Cause (i)	Prior Probability (π_i)	Risk Factor (p_i)	Posterior Probability
1	0.00050	0.10	0.29
2	0.00010	0.20	0.12
3	0.00001	0.40	0.02
4	0.99939	0.0001	0.57

The engineer hired by the insurance company to investigate the accident would certainly wish to know where he can first start looking to find an assignable causes. More precisely, she would wish to know what is the most likely *assignable cause* for the collapse of the building.

The conditional probability of each possible cause, given the fact that the event has occurred, is called the **posterior probability** for this cause and can be calculated by the famous **Bayes' formula**

$$\begin{aligned} P(A_k|B) &= \frac{P(B|A_k)P(A_k)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \ldots + P(B|A_m)P(A_m)} \\ &= \frac{p_k\pi_k}{p_1\pi_1 + p_2\pi_2 + \ldots + p_m\pi_m}. \end{aligned}$$

In the case of our example the posterior probability of the cause "poor design" (A_1), for instance, is equal to

$$\begin{aligned} P(A_1|B) &= \frac{(0.00050)(0.10)}{(0.00050)(0.10) + (0.00010)(0.20) + (0.00001)(0.40) + (0.99939)(0.0001)} \\ &= 0.29. \end{aligned}$$

The other posterior probabilities are calculated analogously and the results are displayed in the fourth column of Table 3.2.

What did the engineer learn from the results of these (posterior probability) calculations? In the first place she learned that the chance of finding an assignable cause is approximately 43%. Furthermore, she learned that it is best to begin looking for flaws in the design of the building, as this cause is almost three times more likely to have caused the accident than the other assignable causes. Finally she learned that it is highly unlikely that the collapse of the building has been caused by more than one assignable cause.

Derivation of Bayes' Formula

By the definition of conditional probability,

$$P(A_k|B) = \frac{P(B \cap A_k)}{P(B)}.$$

Since in addition $\mathcal{S}$ can be expressed as the disjoint union

$$\mathcal{S} = A_1 \cup A_2 \cup \ldots \cup A_m,$$

we follow

$$B = B \cap \mathcal{S} = B \cap (A_1 \cup A_2 \cup \ldots \cup A_m) = (B \cap A_1) \cup (B \cap A_2) \cup \ldots \cup (B \cap A_m)$$

and so,

$$\begin{aligned} P(B) &= P(B \cap A_1) + P(B \cap A_2) + \ldots + P(B \cap A_m) \\ &= P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \ldots + P(B|A_m)P(A_m) \\ &= \pi_1 p_1 + \pi_2 p_2 + \cdots + \pi_m p_m. \end{aligned} \tag{3.3}$$

therefore,

$$\begin{aligned} P(A_k|B) &= \frac{P(B|A_k)P(A_k)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \cdots + P(B|A_m)P(A_m)} \\ &= \frac{\pi_k p_k}{\pi_1 p_1 + \pi_2 p_2 + \cdots + \pi_m p_m}. \end{aligned}$$

Example 3.2 A certain disease is known to affect 1% of the population. A test for the disease has the following features: if the person is contaminated the test is positive with probability 0.98. On the other hand, if the person is healthy, the test is negative with probability 0.95. (a) What is the probability of a positive test when applied to a randomly chosen subject? (b) What is the probability that an individual is affected by the disease after testing positive? (c) Explain the connections between this problem and Bayes' formula.

Solution
(a) Since B is clearly equal to the disjoint union of the events $B \cap C$ and $B \cap C^c$,

$$\begin{aligned} P(B) &= P(B \cap C) + P(B \cap C^c) \\ &= P(C)P(B|C) + P(C^c)P(B|C^c) \\ &= (0.01 \times\ 0.98) + (0.99 \times\ 0.05) \\ &= 0.0593 \end{aligned}$$

(b)

$$P(C|B) = \frac{P(B \cap C)}{P(B)} = \frac{P(B|C)P(C)}{P(B)} = \frac{0.98\ 0.01}{0.0593} = 0.1653$$

Notice that the probability of having the disease, even after testing positive, is surprisingly low (less than 0.17). Why do you think this is so?

(c) The calculation in part (a) produced the "unconditional" probability that the event "testing positive". This unconditional probability constitutes the denominator of Bayes' formula. If a person has been tested positive, given the characteristics of the test, this can be caused by two possible causes: being healthy and being contaminated. The posterior probability of the second cause is the result of part (b). □

Independence

Roughly speaking, two events A and B are independent when the probability of any one of them is not modified after knowing the results for the other (occurrence or not occurrence). In other words, knowing about the occurrence or no occurrence of any one of these events does not alter the amount of information (or uncertainty) that we initially had regarding the other event. Quite simply then, we can say that two events are independent if they do not carry any information regarding each other.

The formal definition of independence is somewhat surprising at first because it doesn't make any direct reference to the events' conditional probabilities. But see also the remarks following the definition. Probabilists prefer this formal definition, because it is easy to check and to generalize for the case of m events ($m \geq 2$).

Definition: The events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Suppose that the events A and B are such that

$$P(A|B) = P(A).$$

In this case,

$$P(A \cap B) = P(A|B)P(B) = P(A)P(B),$$

and the events A and B are independent according to the given definition.

On the other hand, if $P(B) > 0$ and A and B satisfy the given definition of independence, then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

Example 3.3 The results of the STAT 251 midterm exam can be classified as follows:

Table 3.3:

	Male	Female	
High	0.05	0.15	0.20
Medium	0.30	0.15	0.45
Low	0.30	0.05	0.35
	0.65	0.35	1.00

What is the meaning of the statement "*gender* and *performance* are independent"? Are they? Why?

Solution

Gender and performance are (intuitively) independent if for example, knowing the score of a randomly chosen test doesn't affect the probability that this test corresponds to a male (0.65, from the table) or to a female (0.35). Or vice versa, knowing the gender of the student who wrote the test doesn't modify our ability to predict its score.
Let A and B be the events "a randomly chosen student is male" and "a randomly chosen student has a high score", respectively. Is it true that $P(A|B) = P(A)$? The answer, of course, is no because

$$P(A|B) = 0.05/0.20 = 0.25 \qquad \text{and} \qquad P(A) = 0.65.$$

Before knowing that the score is high, the chances are almost two out of three that the student is a male. However, after we know that the score is high, the chances are one out of four that the student is a male. The lack of independence in this case is derived from the fact that male students are "under–represented" in the high score category and "over-represented " in the low score category. □

If Table 3.3 above is replaced by Table 3.4

Table 3.4:

	Male	Female	
High	0.13	0.07	0.20
Medium	0.29	0.16	0.45
Low	0.23	0.12	0.35
	0.65	0.35	1.00

then gender and performance are independent. (Why?).

The concept of independence also applies to three or more events and we shall now give the formal definition of independence of m events. At the same time we want to point out that, in most practical applications, the independence of certain events is often simply assumed or

derived from external information regarding the physical make up of the random experiment, as illustrated in Example 3.4 below.

Fortunately then, we will have few occasions of checking this definition throughout this course.

Definition: The events A_i $(i = 1, \ldots, m)$ are independent if

$$\begin{aligned} P(A_i \cap A_j) &= P(A_i)P(A_j) \quad \text{for all } i \neq j \text{ and} \\ P(A_i \cap A_j \cap A_k) &= P(A_i)P(A_j)P(A_k) \quad \text{for all } i \neq j \neq k \text{ and} \\ &\ldots \\ P(A_1 \cap A_2 \cap \ldots \cap A_m) &= P(A_1)P(A_2)\ldots P(A_m) \end{aligned}$$

Example 3.4 A certain system has four independent components $\{a_1, a_2, a_3, a_4\}$. The pairs of components a_1, a_2 and a_3, a_4 are in line. This means that, for instance, the subsystem $\{a_1, a_2\}$ fails if any of its two component does; similarly for the subsystem $\{a_3, a_4\}$. The subsystems $\{a_1, a_2\}$ and $\{a_3, a_4\}$ are in parallel. This means that the system works if at least one of the two subsystems does. Calculate the probability that the system fails assuming that the four components are independent and that each one of them can break down with probability 0.10. How many parallel subsystems would be needed if the probability of failure for the entire system cannot exceed 0.001?

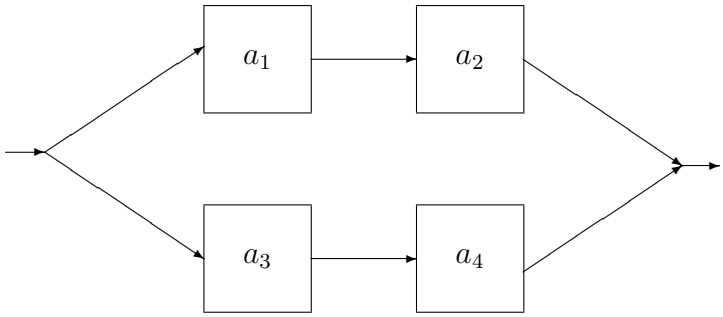


Figure 3.1: A four-component system

Solution Let A_i be the event "component a_i works" $(i = 1, \ldots, 4)$, and let C be the event "the system works".

$$\begin{aligned} P(C) &= P[(A_1 \cap A_2) \cup (A_3 \cap A_4)] = P(A_1 \cap A_2) + P(A_3 \cap A_4) - P[(A_1 \cap A_2) \cap (A_3 \cap A_4)] \\ &= P(A_1)P(A_2) + P(A_3)P(A_4) - P(A_1)P(A_2)P(A_3)P(A_4) \\ &= 0.9^2 + 0.9^2 - 0.9^4 = 0.9639 \end{aligned}$$

To answer the second question, just notice that the probability of working for each independent subsystem is $0.9^2 = 0.81$. Now, if B_i $(i = 1, \ldots, m)$ is the event "the ith subsystem works", it follows that

$$\begin{aligned} 0.001 &\geq 1 - P(B_1 \cup B_2 \cup \ldots \cup B_m) = P(B_1^c \cap B_2^c \cap \ldots \cap B_m^c) \\ &= P(B_1^c)P(B_2^c)\ldots P(B_m^c) = [1 - P(B_1)]^m = (1 - 0.81)^m. \end{aligned}$$

Therefore,

$$\log(0.001) \geq m \log(0.19) \quad \Longrightarrow \quad m \geq \frac{\log(0.001)}{\log(0.19)} \quad \Longrightarrow \quad m = 5.$$

□

3.3 Exercises

Problem 3.1 If A and B are independent events with $P(A) = 0.2$ and $P(B) = 0.5$, find the following probabilities. (a) $P(A \cup B)$; (b) $P(A \cap B)$; and (c) $P(A^c \cap B^c)$

Problem 3.2 In a certain class, 5 students obtained an A, 10 students obtained a B, 17 students obtained a C, and 6 students obtained a D. What is the probability that a randomly chosen student receive a B? If a student receives \$10 for an A, \$5 for a B, \$2 for a C, and \$0 for a D, what is the average gain that a student will make from this course?

Problem 3.3 Consider the problem of screening for cervical cancer. The probability that a women has the cancer is 0.0001. The screening test correctly identifies 90% of all the women who do have the disease, but the test is false positive with probability 0.001.
(a) Find the probability that a woman actually does have cervical cancer given the test says she does.
(b) List the four possible outcomes in the sample space.

Problem 3.4 An automobile insurance company classifies each driver as a good risk, a medium risk, or a poor risk. Of those currently insured, 30% are good risks, 50% are medium risks, and 20% are poor risks. In any given year the probability that a driver will have at least one accident is 0.1 for a good risk, 0.3 for a medium risk, and 0.5 for a poor risk.
(a) What is the probability that the next customer randomly selected will have at least one accident next year?
(b) If a randomly selected driver insured by this company had an accident this year, what is the probability that this driver was actually a good risk?

Problem 3.5 A truth serum given to a suspect is known to be 90% reliable when the person is guilty and 99% reliable when the person is innocent. In other words, 10% of the guilty are judged innocent by the serum and 1% of the innocent are judged guilty. If the suspect was selected from a group of suspects of which only 5% have ever committed a crime, and the serum indicates that he is guilty, what is the probability that he is innocent?

Problem 3.6 70% of the light aircrafts that disappear while in flight in a certain country are subsequently discovered. Of the aircrafts that are discovered, 60% have an emergency locator, whereas 80% of the aircrafts not discovered do not have an emergency locator.
(a) What percentage of the aircrafts have an emergency locator?
(b) What percentage of the aircrafts with emergency locator are discovered after they disappear?

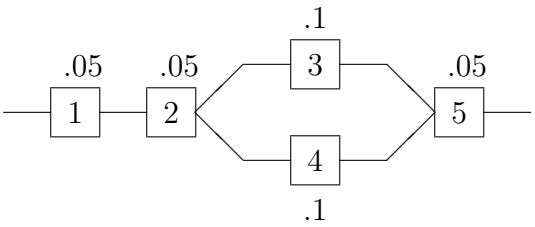
Problem 3.7 Two methods, A and B, are available for teaching a certain industrial skill. The failure rate is 20% for A and 10% for B. However, B is more expensive and hence is only used 30% of the time (A is used the other 70%). A worker is taught the skill by one of the methods, but fails to learn it correctly. What is the probability that the worker was taught by Method A?

Problem 3.8 Suppose that the numbers 1 through 10 form the sample space of a random experiment, and assume that each number is equally likely. Define the following events: A_1, the number is even; A_2, the number is between 4 and 7, inclusive.
(a) Are A_1 and A_2 mutually exclusive events? Why?
(b) Calculate $P(A_1), P(A_2), P(A_1 \cap A_2)$, and $P(A_1 \cup A_2)$.
(c) Are A_1 and A_2 independent events? Why?

Problem 3.9 A coin is biased so that a head is twice as likely to occur as a tail. If the coin is tossed three times,
(a) what is the sample space of the random experiment?
(b) what is the probability of getting exactly two tails?

Problem 3.10 Items in your inventory are produced at three different plants: 50 percent from plant A_1, 30 percent from plant A_2, and 20 percent from plant A_3. You are aware that your plants produce at different levels of quality: A_1 produces 5 percent defectives, A_2 produces 7 percent defectives, and A_3 yields 8 percent defectives. You select an item from your inventory and it turns out to be defective. Which plant is the item most likely to have come from? Why does knowing the item is defective decrease the probability that it has come from plant A_1, and increase the probability that it has come from either of the other two plants?

Problem 3.11 Calculate the reliability of the system described in the following figure. The numbers beside each component represent the probabilities of failure for this component. Note that the components work independently of one another.



Problem 3.12 A system consists of two subsystems connected in series. Subsystem 1 has two components connected in parallel. Subsystem 2 has only one component. Suppose the three components work independently and each has probability of failure equal to 0.2. What is the probability that the system works?

Problem 3.13 A proficiency examination for a certain skill was given to 100 employees of a firm. Forty of the employees were male. Sixty of the employees passed the exam, in that they scored above a preset level for satisfactory performance. The breakdown among males and females was as follows:

	Male (M)	Female (F)	
Pass (P)	24	36	
Fail	16	24	
			100

Suppose an employee is randomly selected from the 100 who took the examination.
(a) Find the probability that the employee passed, given that he was male.
(b) Find the probability that the employee was male, given that he passed.
(c) Are the events P and M independent?
(d) Are the events P and F independent?

Problem 3.14 Propose appropriate sample spaces for the following random experiments. Give also two examples of events for each case.
Counting/measuring:
1 - the number of employees attending work in a certain plant
2 - the number of days with wind speed above 50 km/hour, per year, in Vancouver
3 - the number of earthquakes in BC during any given period of two years
4 - the time between two consecutive breakdowns of a computer network
5 - the number of people leaving BC per year
6 - the percentage of STAT 241/51 students obtaining final marks above 80% in any given term
7 - the number of engineers working in BC per year
8 - the percentage of computer scientists in BC who will make more than $\$65,000$ in 1996
9 - the number of employees still working in a certain production plant after 4:30 PM on Fridays.

Problem 3.15 Let A and B be the events **"construction flaw due to some human error"** and **"construction flaw due to some mechanical problem"**.

1) What are the meaning (in words) of the following events: (a) $A \cup B$, (b) $A \cap B$, (c) $A \cap B^c$, (d) $A^c \cap B^c$, (e) $(A \cup B)^c$, (f) $A^c \cup B^c$, (g) $(A \cap B)^c$. Draw also the corresponding diagrams.

2) Show that in general $(A \cap B)^c = A^c \cup B^c$ and that $(A \cup B)^c = A^c \cap B^c$ (so the results of (f) and (g) and of (d) and (e) above were not mere coincidences).

3) Suppose that $P(A) = 0.02$, $P(B) = 0.01$ and $P(A \cup B) = 0.023$. Calculate (a) $P(A \cap B)$, (b) $P(A^c \cap B^c)$, (c) $P(A \cap B^c)$, (d) $P(A|B^c)$, (e) $P(A|B)$.

Problem 3.16 A large company hires most of its employees on the basis of two tests. The two tests have scores ranging from one to five. The following table summarizes the performance of 16,839 applicants during the last six years. From this table we learn, for example, that 3% of the applicants got a score of 2 on Test 1 and 2 on Test 2; and that 15% of the applicants got a score of 3 on Test 1 and 2 on Test 2. We also learn that, for example, 20% of the applicants got a score of 2 on Test 1 and that 25% of the applicants got a score of 2 on Test 2.

A group of 1500 new applicants have been selected to take the tests.
(a) What should the cutting scores be if between 140 and 180 applicants will be short–listed for a job interview? Assume that the company wishes to short–list people with the highest possible performances on the two tests.

Table 3.5:

Score	1	2	3	4	5	Total
1	0.07	0.03	0.00	0.00	0.00	0.10
2	0.15	0.03	0.02	0.00	0.00	0.20
3	0.08	0.15	0.09	0.02	0.01	0.35
4	0.10	0.04	0.08	0.01	0.02	0.25
5	0.00	0.00	0.06	0.02	0.02	0.10
Total	0.40	0.25	0.25	0.05	0.05	1.00

Table 3.6:

Score	Test 1	Test 2
1	0.10	0.40
2	0.20	0.25
3	0.35	0.25
4	0.25	0.05
5	0.10	0.05

(b) Same as (a) but assuming now that the company wishes to hire people with the highest possible performances on at least one of the two tests.
(c) (Continued from (a)) A manager suggests that only applicants who obtain marks above a certain bottom line in one of the tests be given the other test. Noticing that giving and marking each test costs the company $55, recommend which test should be given first. Approximately how much will be saved on the basis of your advise?
(d) Repeat (a)–(c) if the two tests performances are independent and the probabilities are given by Table 2.6.

Problem 3.17 A computer company manufactures PC compatible computers in two plants, called Plant A and B in this exercise. These plants account for 35 % and 65 % of the production, respectively. The company records show that 3 % of the computers manufactured by Plant A must be repaired under the warranty. The corresponding percentage for plant B is 2.5 %.
(a) What is the percentage of computers that are repaired under the warranty and come from Plant A?
(b) What percentage of computers repaired under the warranty come from Plant A? From Plant B?

Problem 3.18 Twenty per cent of the days in a certain area are rainy (there is some measurable precipitation during the day), one third of the days are sunny (no measurable precipitation, more than 4 hours of sunshine) and fifteen per cent of the days are cold (daily average temperature for the day below 5° C).
1 - Would you use the above information as an aid in
(i) Planning your next weekend activities (assuming that you live in this area)?
(ii) Deciding whether you want to move to this area?
(iii) Choosing the type of roofing for a large building in this area?

Justify your answers.
2 - Given that five per cent of the days are sunny and cold, and five per cent of the days are rainy and cold, calculate the probability that a given day will be either sunny, rainy or cold.
3 - Are sunny and cold days independent? What about rainy and cold days?

Problem 3.19 A company sells a (cheap) recording tape under a limited "lifetime warranty". From the company records one learns that
5% of the tapes sold by the company are defective and could be replaced under the warranty.
50% of the customers who get one of these defective tapes will claim it under the warranty and have it replaced.
90% of the tapes which are claimed to be defective are actually so. These tapes are replaced under the warranty.
(a) Which of the above are conditional probabilities?
(b) Using the above information, calculate the probability that a customer will claim the warranty.
(c) What is the maximum allowable fraction of defective tapes if the company wants to have at most 1% of the tapes returned?

Problem 3.20 Show that $P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$.

Problem 3.21 On average, 20% of the students fail the first midterm. Of those, 60% fail the second midterm. Moreover, 80% of the students that failed the two midterms fail also the final exam.
(a) What is the probability that a randomly chosen student fails the two midterms?
(b) What is the probability that a randomly chosen student fails the two midterms and the final exam?

Problem 3.22 The probability that system survives 300 hours is 0.8. The probability that a 300 hours old system survives another 300 hours is 0.6. The probability that a 600 hours old system survives another 300 hours is 0.5.
(a) What is the probability that the system survives 600 hours?
(b) What is the probability that the system survives 900 hours?

Problem 3.23 Recall the situation in Example 3.2 presented in class: the probability of infection for an individual in the general population is $\pi = .01$ and a test for the disease is such that it will be correctly positive 98% of the time and correctly negative 95% of the time. Some individuals, however, may belong to some "high risk" groups and therefore have a larger prior probability π of being infected.

1) calculate the posterior probability of infection as a function of the corresponding prior probability, π, given that the test is positive (denote this probability by $g(\pi)$) and make a plot of $g(\pi)$ versus π.

2) what is the value of π for which the posterior probability given a positive test is twice as large as the prior probability?

Problem 3.24 Suppose that we wish to determine whether an uncommon but fairly costly construction flaw is present. Suppose that in fact this flaw has only probability 0.005 of being present. A fairly simple test procedure is proposed to detect this flaw. Suppose that the probabilities of being correctly positive and negative for this test are 0.98 and 0.94, respectively.

1) Calculate the probability that the test will indicate the presence of a flaw.

2) Calculate the posterior probability that there is no flaw given that the test has indicated that there is one. Comment on the implications of this result.

Problem 3.25 One method that can be use to distinguish between granite (G) and basalt (B) rocks is to examine a portion of the infrared spectrum of the sun's energy reflected from the rock surface. Let R_1, R_2 and R_3 denote measured spectrum intensities at three different wavelengths. Normally, $R_1 < R_2 < R_3$ would be consistent with granite and $R_3 < R_1 < R_2$ would be consistent with basalt. However, when the measurements are made remotely (e.g. using aircrafts) several orderings of the $R_i's$ can arise. Flights over regions of known composition have shown that granite rocks produce

$$\begin{array}{lll} (R_1 < R_2 < R_3) & 60\% & \text{of the time,} \\ (R_1 < R_3 < R_2) & 25\% & \text{of the time, and} \\ (R_3 < R_1 < R_2) & 15\% & \text{of the time} \end{array}$$

On the other hand, basalt rocks produce these orderings of the spectrum intensities with probabilities 0.10, 0.20 and 0.70, respectively. Suppose that for a randomly selected rock from a certain region we have $P(G) = 0.25$ and $P(B) = 0.75$.
1) Calculate $P(G|R_1 < R_2 < R_3)$ and $P(B|R_1 < R_2 < R_3)$. If the measurements for a given rock produce the ordering $R_1 < R_2 < R_3$, how would you classify this rock?

2) Same as 1) for the case $R_1 < R_3 < R_2$

3) Same as 1) for the case $R_3 < R_1 < R_2$
4) If one uses the classification rule determined in 1) 2) and 3), what is the probability of a classification error (that a G rock is classified as a B rock or a B rock is classified as a G rock)?

Problem 3.26 Messages are transmitted as a sequence of zeros and ones. Transmission errors occur independently, with probability 0.001. A message of 3500 bits will be transmitted.
(a) What is the probability that there will be no errors? What is the probability that there will be more than one error?
(b) If the same message will be transmitted twice and those bits that do not agree will be revised (and therefore these "detected" transmission errors will be corrected), what is the probability that there will be no reception errors?

Problem 3.27 Suppose that the events A, B and C are independent. Show that,

(a) A^c and B^c are independent.

(b) $A \cup B$ and C are independent.

(c) $A^c \cap B^c$ and C are independent.

Problem 3.28 A test has been designed to indicate the presence of a flaw in an electronic component. The components which test positive are sent back to the production department. It is known, however, that 1% of the time the test gives either a false positive or a false negative result.
(a) What is the proportion of faulty components being produced if 2% of them are sent back to production on the basis of the test?
(b) The company produces twenty thousand components each year. The loss associated with the rejection of a sound component is \$5, that associated with the rejection of a faulty component is \$50 and that associated with the selling of a defective component is \$150. What is the total loss? How much of this loss is due to "defective" testing?

Problem 3.29 Consider the probabilities given in Table 2.7 and the events

$B_1 = \{\text{Having a low GPA}\}$, $B_2 = \{\text{Having a medium GPA}\}$, $B_3 = \{\text{Having a high GPA}\}$

$C_1 = \{\text{Having a low salary}\}$, $C_2 = \{\text{Having a medium salary}\}$, $C_3 = \{\text{Having a high salary}\}$

Table 3.7:

	Low Salary	Medium Salary	High Salary	
Low GPA	0.10	0.08	0.02	0.20
Medium GPA	0.07	0.46	0.07	0.60
High GPA	0.03	0.06	0.11	0.20
	0.20	0.60	0.20	

1) Calculate $P(B_i \cup C_j)$, $\quad i = 1, 2, 3 \quad$ and $\quad j = 1, 2, 3$

2) What is the meaning (in words), and the probability, of the event

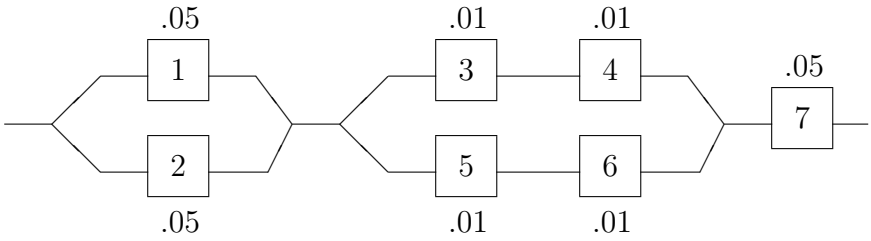
$$A = (B_1 \cap C_1) \cup (B_2 \cap C_2) \cup (B_3 \cap C_3)$$

3) Are "salary" and "GPA" independent? Why?

4) Construct a table with the same marginals (same probabilities for the six categories) but with salary and GPA being independent.

Problem 3.30 Consider the system of components connected as follows. There are two subsystems connected in parallel. Components 1 and 2 constitute the first subsystem and are connected in parallel (so that this subsystem works if either component works). Components 3 and 4 constitute the second subsystem and are connected in series (so that this subsystem works if and only if both components do). If the components work independently of one another and each component works with probability 0.85, (a) calculate the probability that the system works. (b) calculate this probability if the two subsystems are connected in series.

Problem 3.31 Calculate the reliability of the system described in the following figure. The numbers beside each component represent the probabilities of failure for this component.



Chapter 4

Random Variables and Distributions

4.1 Definition and Notation

Mathematically, a random variable X is a function defined on the sample space $\mathcal{S}$, assigning a number, $x = X(w)$, to each outcome w in the sample space. Notice that the upper case letter X represents the random variable and the lower case letter x represents one of its possible values.

Example 4.1 Let $\mathcal{S}$ the sample space associated with the inspection of four items. That is,

$$\mathcal{S} = \{w = (w_1, w_2, w_3, w_4)\}$$

where w_i, $i = 1, \ldots, 4$, is equal to D (for defective) or N (for non–defective). The random variable X is defined as "the number of D's in w" and the random variable Y is defined as "the indicator of two or more D's in w" (that is, $Y(w) = 1$ if w contains two or more defectives and $Y(w) = 0$, otherwise). For instance, $X(N, N, N, N) = 0$, $X(N, D, N, N) = 1$, $X(D, N, D, D) = 3$, and $Y(N, N, N, N) = 0$, $Y(N, D, N, N) = 0$, $Y(D, N, D, D) = 1$.

Random variables are often used to summarize the most relevant information contained in the sample space. For example, one may be interested in the total number of defectives (number of $D's$ in w) and may not care about the order in which they have been found. In this case the random variable $X(w)$ defined above would capture the most relevant information contained in w. If we will reject lots with two or more defectives (among the four inspected items) the random variable, Y would be of most interest.

Notation: The notations $\{X = x\}$ $\{X \leq x\}$ etc. will be used very often in this course. Their exact meaning is explained below. In general,

$$\{X \in A\} = \{w : X(w) \in A\}, \quad \text{where} \quad A \ \text{ is a set of numbers.}$$

This takes on different forms for different sets $A's$. For example,

$$\{X = x\} = \{w : X(w) = x\},$$

where the set $A = \{x\}$ and

$$\{X \leq x\} = \{w : X(w) \leq x\},$$

where the set $A = (-\infty, x]$. Additional examples (related to Example 5 above) are

$$\{X = 0\} = \{(N, N, N, N)\}$$

and

$$\{X \leq 1\} = \{(N, N, N, N), (D, N, N, N), (N, D, N, N), (N, N, D, N), (N, N, N, D)\}.$$

4.2 Discrete Random Variables

Discrete random variables are mainly used in relation to "counting" situations; for example,

- Counting the number of defective items in a lot
- Counting the number of yearly failures of an electrical network
- Counting the weekly number of customers arriving at a service outlet
- Counting the hourly number of cars crossing a bridge
- Counting the number of jobs interviews before finding a job

The defining feature of a discrete random variable is that its **range** (the set of all its possible values) is finite or countable. The values in the range are often integer numbers, but they don't need to be so. For instance, a random variable taking the values zero, one half and one with probabilities 0.5, 0.25 and 0.25 respectively is considered discrete.

The **probability density function** (or in short, the density), $f(x)$, of a discrete random variable X is defined as

$$f(x) = P(X = x), \quad \text{for all possible value } \ x \ \text{ of } \ X$$

That is, $f(x)$ gives the probability of each possible value x of X. It obviously has the following properties:

(1) $f(x) \geq 0$ for all x in the range R of X

(2) $\sum_{x \in R} f(x) = 1$

(3) $\sum_{x \in A} f(x) = P(X \in A)$ for all subsets A of R.

The **distribution function** of X (in short the distribution), $F(x)$, is defined as

$$F(x) = P(X \leq x) = \sum_{k \leq x} f(k), \quad \text{for all real } \ x.$$

In many engineering applications one works with $1 - F(x)$ instead of $F(x)$. Notice that $1 - F(x) = P(X > x)$ and therefore gives the probability that X will exceed the value x.
Example 4.1 (continued): Suppose that the items are independent and each one can be defective with probability p. The density and distribution of the random variable (r.v.) $X =$ "number of defectives" can then be derived as follows:

$$\begin{aligned} f(0) &= P(X=0) = P(\{N,N,N,N\}) = (1-p)(1-p)(1-p)(1-p) = (1-p)^4 \\ f(1) &= P(X=1) = P(\{D,N,N,N\},\{N,D,N,N\},\{N,N,D,N\},\{N,N,N,D\}) \\ &= p(1-p)(1-p)(1-p) + (1-p)p(1-p)(1-p) + (1-p)(1-p)p(1-p) \\ &\quad +(1-p)(1-p)(1-p)p = 4(1-p)^3 p \end{aligned}$$

In a similar way we can find that

$$f(2) = 6(1-p)^2p^2, \quad f(3) = 4(1-p)^1p^3 \quad \text{and} \quad f(4) = p^4$$

The values of the density and distribution functions of X, for the cases $p = 0.40$ and $p = 0.80$ are given in Table 2.5. A comparison of the density functions shows that smaller values of X (0, 1 and 2) are more likely when $p = 0.4$ (why?) and that higher values (3 and 4) are more likely when $p = 0.8$. Also notice that the distribution function for the case $p = 0.8$ is uniformly smaller. This is so because getting smaller values of X is always more likely when $p = 0.4$.

Table 4.1:

	$p = 0.40$		$p = 0.80$	
x	$f(x)$	$F(x)$	$f(x)$	$F(x)$
0	0.1296	0.1296	0.0016	0.0016
1	0.3456	0.4752	0.0256	0.0272
2	0.3456	0.8208	0.1536	0.1808
3	0.1536	0.9744	0.4096	0.5904
4	0.0256	1.0000	0.4096	1.0000

4.3 Continuous Random Variables

The **continuous** random variables are used in relation with "continuous" type of outcomes, as for example,

- the lifetime of a system or component

- the yield of a chemical process
- the weight of a randomly chosen item
- the difference between the specified and actual diameter of a part
- the measurement error when measuring the distance between the North and South shores of a river.

The typical events in these cases are bounded or unbounded intervals with probabilities specified in terms of the integral of a **continuous density function**, $f(x)$, over the desired interval. See property (3) below.

Since the probability of all intervals must be non-negative and the probability of the entire line should be one, it is clear that $f(x)$ must have the two following properties:

(1) Non negative:

$$f(x) \geq 0 \quad \text{for all} \quad x.$$

(2) Total mass equal to one:

$$\int_{-\infty}^{+\infty} f(x)dx = 1.$$

(3) Probability calculation:

$$P\{a < X \leq b\} = \int_a^b f(x)dx.$$

Notice that, unlike in the discrete case, the inclusion or exclusion of the end points a and b doesn't affect the probability that the continuous variable X is in the interval. In fact, the event that X will take any single value, x, can be represented by the degenerate interval $x \leq X \leq x$ and so,

$$P(X = x) = P(x \leq X \leq x) = \int_x^x f(t)dt = 0.$$

Therefore, unlike in the discrete case, $f(x)$ doesn't represent the probability of the event $X = x$. What is then the meaning of $f(x)$? It represents the *relative* probability that X will be *near* x: if $d > 0$ is small,

$$\frac{P(x - (d/2) < X < x + (d/2))}{d} = \frac{1}{d}\int_{x-(d/2)}^{x+(d/2)} f(t)dt \approx f(x).$$

Another important function related with a continuous random variable is its **cumulative distribution function** defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt, \quad \text{for all } x. \tag{4.1}$$

Notice that, in particular,

$$P(a < X < b) = F(b) - F(a).$$

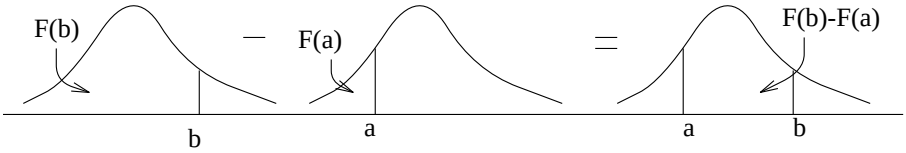


Figure 3.1: Probability on (a, b) under density function $f(x)$

By the Fundamental Theorem of Calculus,

$$f(x) = F'(x), \quad \text{for all } x. \tag{4.2}$$

Therefore, we can go back and forth from the density to the distribution function and vice versa using formulas (4.1) and (4.2).

Example 4.2 Suppose that the maximum annual flood level of a river, X (in meters), has density

$$\begin{aligned} f(x) &= 0.125(x-5), \qquad \text{if } 5 < x < 9 \\ &= 0 \qquad \text{otherwise} \end{aligned}$$

Calculate $F(x)$, $P(5 < X < 6)$, $P(6 \leq X < 7)$, and $P(8 \leq X \leq 9)$.

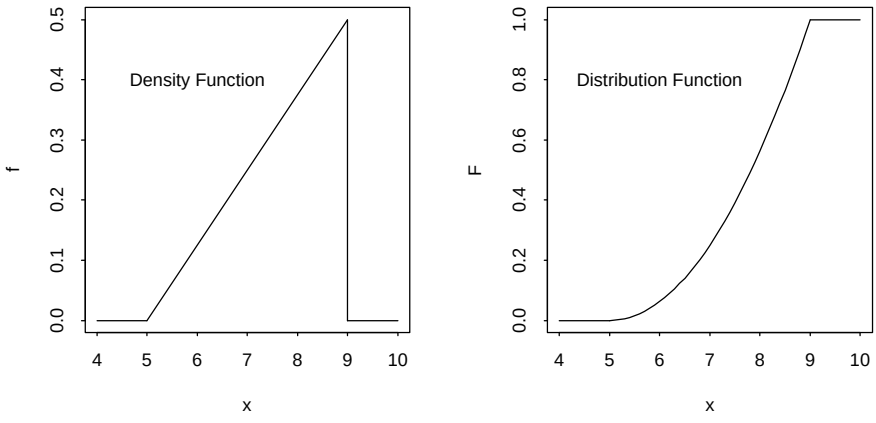


Figure 3.2: Distribution and density functions

Solution

$$\begin{aligned} F(x) &= 0, \qquad \text{if } x \leq 5 \\ &= \int_5^x 0.125(t-5)dt = (0.0625)(x-5)^2, \qquad \text{if } 5 < x < 9 \\ &= 1, \qquad \text{if } x \geq 9. \end{aligned}$$

Furthermore,

$$\begin{aligned} P(5 < X < 6) &= F(6) - F(5) \\ &= 0.0625[(6-5)^2 - (5-5)^2] \\ &= 0.0625. \end{aligned}$$

Analogously,

$$P(6 \leq X < 7) = F(7) - F(6) = 0.25 - 0.0625 = 0.1875,$$

and

$$P(8 \leq X < 9) = F(9) - F(8) = 1.0 - 0.5625 = 0.4375.$$

Notice that, since $P(X = x) = 0$, the inclusion or exclusion of the interval's boundary points doesn't affect the probability of the corresponding interval. In other words,

$$P(6 \leq X \leq 7) = P(6 < X \leq 7) = P(6 \leq X < 7) = P(6 < X < 7) = F(7) - F(6) = 0.1875.$$

Also notice that, since $f(x)$ is increasing on $(5, 9)$, $P(5 < X < 6)$, for instance, is much smaller than $P(8 < X < 9)$, despite the length of the two intervals being equal. □

Example 4.3 (Rounding-off Error and Uniform Random Variables): Due to the resolution limitations of a measuring device, the measurements are rounded-off to the second decimal place. If the third decimal place is 5 or more, the second place is increased by one unit; if the third decimal place is 4 or less, the second place is left unchanged. For example, 3.2462 would be reported as 3.25 and 3.2428 would be reported as 3.24. Let X represent the difference between the (unknown) true measurement, y, and the corresponding rounded–off reading, r. That is

$$X = y - r.$$

Clearly, X can take any value between $-0.005 < X < 0.005$. It would appear reasonable in this case to assume that all the possible values are **equally likely**. Therefore, the relative probability $f(x)$ that X will fall near any number x_0 between -0.005 and 0.005 should then be the same. That is,

$$\begin{aligned} f(x) &= c, \qquad -0.005 \leq x \leq 0.005, \\ &= 0, \qquad \text{otherwise.} \end{aligned}$$

The random variable X is said to be uniformly distributed between -0.005 and $+0.005$. By property 2

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-0.005}^{0.005} cdx = 0.01c = 1,$$

Therefore, c must be equal to $1/0.01 = 100$ and

$$\begin{aligned} f(x) &= 100, \quad -0.005 \leq x \leq 0.005, \\ &= 0, \quad \text{otherwise.} \end{aligned}$$

The corresponding distribution function is

$$\begin{aligned} F(x) &= 0, && x \leq -0.005, \\ &= 100(x + 0.005), && -0.005 \leq x \leq 0.005, \\ &= 1, && x \geq 0.005 \end{aligned}$$

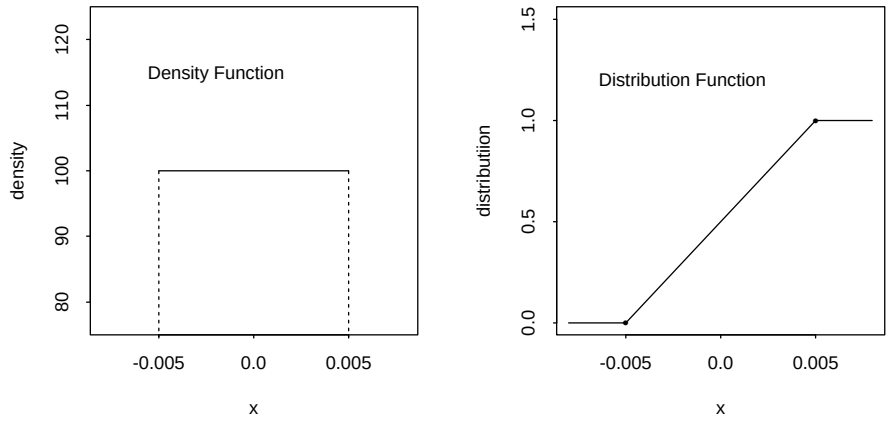


Figure 3.4: Distribution and Density of the uniform random variable

4.4 Summarizing the Main Features of $f(x)$

All the information concerning the random variable X is contained in its density function, $f(x)$, and this information can be used and displayed in the form of a picture (a graph of $f(x)$ versus x), a formula, or a table.

There are situations, however, when one would prefer to concentrate on a summary of the more complete and complex information contained in $f(x)$. This is the case, for example, if we are working with several random variables that need to be compared in order to draw some conclusions.

The summary of $f(x)$, as any other summary, should be simple and informative. The reader of such a summary should get a good idea of what are the most likely values of X and what is the degree of uncertainty regarding the prediction of future values of X.

Typical densities found in practice are approximately symmetric and unimodal. These densities can be summarized in terms of their central location and their dispersion. Therefore,

an approximately symmetric and unimodal density can be fairly well described by giving just two numbers: a measure of its central location and a measure of its dispersion.

The **median** and the **mean** are two popular measures of (central) location and **the interquartile range** and the **standard deviation** are two popular measures of dispersion. These summary measures are defined and briefly discussed below.

The Median and the Inter–Quartile Range

Given a number α between zero and one, the quantile of order α of the distribution F (or the r.v. X), denoted $Q(\alpha)$, is implicitly defined by the equation

$$P(X \leq Q(\alpha)) = \alpha.$$

Therefore $Q(\alpha)$ has the property

$$Q(\alpha) = F^{-1}(\alpha)$$

and can be found by solving (for x) the equation

$$F(x) = \alpha.$$

To find the quantile of order 0.25, for example, we must solve the equation

$$F(x) = 0.25.$$

The "special" quantiles $Q(0.25)$ and $Q(0.75)$ are often called the **first quartile** and the **third quartile**, respectively.

The **median** of X, $\text{Med}(X)$, is defined as the corresponding quantile of order 0.5, that is,

$$\text{Med}(X) = Q(0.5).$$

Evidently, $\text{Med}(X)$ divides the range of X into two sets of equal probability. Therefore, it can be used as a measure for the central location of $f(x)$.

A simple sketch showing the locations of $Q(0.25)$, $Med(X)$ and $Q(0.75)$ constitutes a good summary of $f(x)$, even if it is not symmetric. Notice that if $Q(0.75) - Med(X)$ is significantly larger (or smaller) than $Med(X) - Q(0.25)$, then $f(x)$ is fairly asymmetric.

There are situations when there is no solution or too many solutions to the defining equations above. This is typically the case for discrete random variables. In these cases the quantiles (including the median) are calculated using some "common–sense" criterion. For instance if the distribution function $F(x)$ is constant and equal to 0.5 on the interval (x_1, x_2), then the median is taken equal to $(x_1 + x_2)/2$ (see Figure 3.5 (a)). To give another example, if the distribution function $F(x)$ has a jump and doesn't take the value 0.5, the median is defined as the location of the jump (see Figure 3.5 (b))

The dispersion about the median is usually measured in terms of the **interquartile range**, denoted $\text{IQR}(X)$ and defined as:

$$\text{IQR}(X) = Q(0.75) - Q(0.25)$$

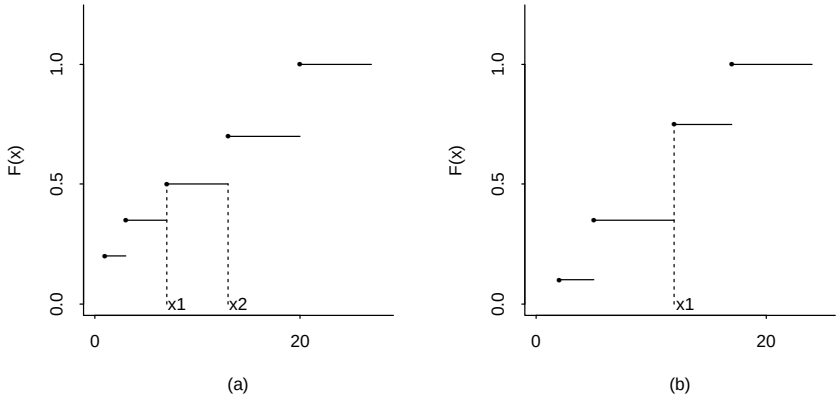


Figure 3.5: Calculation of the median

When the density $f(x)$ is fairly concentrated (around some central value) IQR(X) tends to be smaller. Roughly speaking, the size of IQR(X) is directly proportional to the degree of uncertainty that one faces in trying to predict the future values of X.

Example 4.4 (Waiting Time and Exponential Random Variables) The waiting time X (in hours) between the arrival of two consecutive customers at a service outlet is a random variable with *exponential density*

$$\begin{aligned} f(x) &= \lambda e^{-\lambda x}, \quad \text{if } x \geq 0, \\ &= 0, \quad \text{otherwise.} \end{aligned}$$

where λ is a positive parameter representing the rate at which customers arrive. For this example, take $\lambda = 2$ customers per hour. (a) Find the distribution function $F(x)$. (b) Calculate Med(X), $Q(0.25)$ and $Q(0.25)$. (c) Is $f(x)$ symmetric? (d) Calculate IQR(X).

Solution

(a)

$$\begin{aligned} F(x) &= \int_{-\infty}^{x} f(t)dt = 2\int_{0}^{x} \exp\{-2t\}dt \\ &= 2\frac{\exp\{-0\} - \exp\{-2x\}}{2} = 1 - \exp\{-2x\}. \end{aligned}$$

(b) To calculate the median,

$$1 - \exp\{-2x\} = 1/2 \quad \Rightarrow \quad \exp\{-2x\} = 1/2 \quad \Rightarrow \quad -2x = -\log(2).$$

Therefore, $\text{Med}(X) = \log(2)/2 = 0.347$.
To calculate $Q(0.25)$,

$$1 - \exp\{-2x\} = 1/4 \quad \Rightarrow \quad \exp\{-2x\} = 3/4 \quad \Rightarrow \quad -2x = \log(3) - \log(4).$$

Therefore,

$$Q(0.25) = \frac{\log(4) - \log(3)}{2} = 0.144.$$

Analogously, to calculate $Q(0.75)$,

$$1 - \exp\{-2x\} = 3/4 \quad \Rightarrow \quad \exp\{-2x\} = 1/4 \quad \Rightarrow \quad -2x = -\log(4).$$

$$Q(0.75) = \frac{\log(4)}{2} = 0.693.$$

(c) Since

$$Q(0.75) - \text{Med}(X) = 0.693 - 0.347 = 0.346$$

and

$$\text{Med}(X) - Q(0.25) = 0.347 - 0.144 = 0.203,$$

the distribution is fairly asymmetric.
(d)

$$IQR = Q(0.75) - Q(0.25) = 0.693 - 0.144 = 0.549.$$

□

The Mean, the Variance and the Standard Deviation

Let X be a random variable with density $f(x)$, and let $g(X)$ be a function of X. For example, $g(X) = \sqrt{X}$ or $g(X) = (X-t)^2$, where t is some fixed number. The notation $\text{E}[g(X)]$, read "expected value of $g(X)$", will be used very often in this course. The expected value of $g(X)$ is defined as the weighted average of the function $g(x)$, with weights proportional to the density function $f(x)$. More precisely:

$$\text{E}[g(X)] = \int_{-\infty}^{+\infty} g(x)f(x)dx \quad \text{in the continuous case, and} \tag{4.3}$$

$$\text{E}[g(X)] = \sum_{x \in R} g(x)f(x) \quad \text{in the discrete case,} \tag{4.4}$$

where R is the range of X.

Example 4.5 Refer to the random variables of Example 3.1 (number of defectives) and Example 4.3 (rounding–off error). Calculate $\mathrm{E}(X)$ and $\mathrm{E}(X^2)$.

Solution Since the random variable X of Example 3.1 is discrete, we must use formula (4.4) to obtain:

$$\mathrm{E}(X) = (0)(0.5) + (1)(0.2) + (2)(0.15) + (3)(0.10) + (4)(0.03) + (5)(0.02) = 1.02,$$

and

$$\mathrm{E}(X^2) = (0)(0.5) + (1)(0.2) + (4)(0.15) + (9)(0.10) + (16)(0.03) + (25)(0.02) = 2.68.$$

In the case of the continuous random variable X of Example 4.3 we must use formula (4.3):

$$\mathrm{E}(X) = \int_{-\infty}^{+\infty} x f(x)dx = 100 \int_{-0.005}^{0.005} x dx = 0,$$

$$\begin{aligned} \mathrm{E}(X^2) &= \int_{-\infty}^{+\infty} x^2 f(x)dx = 100 \int_{-0.005}^{0.005} x^2 dx \\ &= \frac{100[(0.005)^3 - (-0.005)^3]}{3} = \frac{(200)(0.005)^3}{3} = 0.00000833. \end{aligned}$$

□

The mean of X as a Measure of Central Location

Suppose that it is proposed that a certain number t is used as the measure of central location of X. How could we decide if this proposed value is appropriate? One way to think about this question is as follows. If t is a good measure of central location then, in principle, one would expect that the **squared residuals** $(x-t)^2$ will be fairly small for those values x of X which are highly likely (those for which $f(x)$ is large). If this is so, then one would also expect that the average of these squared residuals,

$$D(t) = \mathrm{E}\{(X-t)^2\},$$

will also be fairly small. Notice that

$$\begin{aligned} D(t) &= \int_{-\infty}^{+\infty} (x-t)^2 f(x)dx \quad \text{in the continuous case} \\ &= \sum (x-t)^2 f(x) \quad \text{in the discrete case,} \end{aligned}$$

But we could begin this reasoning from the end and say that a good measure of central location must minimize $D(t)$. This "optimal" value of t, called the **mean** of X, is denoted by the Greek letter μ.

To find μ we differentiate $D(t)$ and set the derivative equal to zero. In the continuous case,

$$D'(t) = -2\int_{-\infty}^{+\infty}(x-t)f(x)dx = -2[\mathrm{E}(X)-t] = 0 \quad \Rightarrow \quad t = \mathrm{E}(X),$$

and the discrete case can be treated similarly. Since $D''(t) = 2 > 0$ for all t, the critical point $t = \mathrm{E}(X)$ minimizes $D(t)$. Therefore,

$$\mu = \mathrm{E}(X)$$

This procedure of defining the desired summary measure by the property of minimizing the average of the squared residuals is a very important technique in applied statistics called **the method of minimum mean squared residuals**. We will come across several applications of this technique throughout this course.

The Standard Deviation of X as a Measure of Dispersion

It is clear from the above discussion that

$$D(t) \geq D(\mu) = \mathrm{E}\{(X-\mu)^2\}$$

for all values of t. The quantity $D(\mu)$ is usually denoted by the Greek symbol σ^2 (read "Sigma squared") and called **the variance of** X. An alternative notation for the variance of X, also often used in this course, is $\mathrm{Var}(X)$.

It is evident that $\mathrm{Var}(X)$ will tend to be smaller when the density of X is more concentrated around μ, since the smaller squared residuals will receive larger weights. Therefore, $\mathrm{Var}(X)$ could be taken as a measure of the dispersion of $f(x)$. A problem with $\mathrm{Var}(X)$ is that it is expressed in a unit which is the square of original unit of X. This problem is easily solved by taking the (positive) square root of $\mathrm{Var}(X)$. This is called the **standard deviation of** X and denoted by either σ or $\mathrm{SD}(X)$.

$$\sigma = \mathrm{SD}(X) = +\sqrt{\mathrm{Var}(X)} = +\sqrt{\sigma^2}.$$

Example 4.4 (continued): (a) Calculate the mean and the standard deviation for the waiting time between two consecutive customers, X. (b) How do they compare with the corresponding median waiting time and interquartile range calculated before?

Solution

(a) Using integration by–part,

$$\begin{aligned} \mathrm{E}(X) &= 2\int_0^{+\infty} x\exp\{-2x\}dx = 2[-x\exp\{-2x\}/2]_0^{+\infty} + \int_0^{+\infty}\exp\{-2x\} \\ &= [-\exp\{-2x\}/2]_0^{+\infty} = \exp\{0\}/2 = 0.5. \end{aligned}$$

More generally, if X is an exponential random variable with parameter (rate) λ, then

$$\mathrm{E}(X) = 1/\lambda. \tag{4.5}$$

Using integration by–part again, we get

$$\begin{aligned} \mathrm{E}(X^2) &= 2\int_0^{+\infty} x^2\exp\{-2x\}dx = 2[-x^2\exp\{-2x\}/2]_0^{+\infty} + 2[\int_0^{+\infty} x\exp\{-2x\}]dx \\ &= 2\int_0^{+\infty} x\exp\{-2x\}]dx = 0.5. \end{aligned}$$

Therefore,

$$\mathrm{SD}(X) = +\sqrt{\mathrm{E}(X^2) - [\mathrm{E}(X)]^2} = +\sqrt{0.5 - (0.5)^2} = 0.5.$$

More generally, if X is an exponential random variable with parameter λ, then

$$\mathrm{Var}(X) = 1/\lambda^2 \quad \text{and} \quad \mathrm{SD}(X) = 1/\lambda. \tag{4.6}$$

(b) Since the density of X is asymmetric, the median and the mean are expected to be different (as they are). Since the density is skewed to the right (longer right hand side tail) the mean expected time (0.5) is larger than the median expected time (0.347).

The two measures of dispersion ($IQR = 0.549$ and $\mathrm{SD} = 0.5$) are quite consistent. □

Properties of the Mean and the Variance

Property 1: $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$ for all constants a and b.

Proof

$$\mathrm{E}(aX + b) = \sum(ax_i + b)f(x_i) = a[\sum x_i f(x_i)] + b = a\mathrm{E}(X) + b.$$

The proof for the continuous case is identical. □

Property 2: $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$ for all pairs of random variables X and Y.

Property 3: $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ for all pairs of **independent** random variables X and Y.

Property 4: $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$ for all constants a and b.

Proof

$$\mathrm{Var}(aX + b) = \mathrm{E}\left[(aX + b) - (a\mu + b)\right]^2 = \mathrm{E}\left[a(X - \mu)\right]^2 = a^2\mathrm{E}(X - \mu)^2 = a^2\mathrm{Var}(X)$$

□

Property 5: $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ for all pairs of **independent random variables** X and Y.

All these properties will be used very often in this course. The proofs of properties 2, 3 and 5 are beyond the scope of this course, and therefore these properties must be accepted as facts and used throughout the course.

The formula

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 = \mathrm{E}(X^2) - \mu^2,$$

is often used for calculations. The derivation of this formula is very simple, using the properties of the mean listed above. In fact,

$$\begin{aligned} \mathrm{Var}(X) &= \mathrm{E}\{(X-\mu)^2\} = \mathrm{E}(X^2 + \mu^2 - 2\mu X) = \mathrm{E}(X^2) + \mu^2 - 2\mu\mathrm{E}(X) \\ &= \mathrm{E}(X^2) + \mu^2 - 2\mu^2 = \mathrm{E}(X^2) - \mu^2. \end{aligned}$$

4.5 Sum and Average of Independent Random Variables

Random experiments are often independently repeated many times generating a sequence $X_1, X_2, \ldots, X_n$ of n independent random variables. We will consider linear combinations of these variables,

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n,$$

where the coefficients $a_1, a_2, \ldots, a_n$ are some given constants. For example, $a_i = 1$, for all i, produces the total

$$T = X_1 + X_2 + \cdots + X_n,$$

and $a_i = 1/n$, for all i, produces the average

$$\overline{X} = (X_1 + X_2 + \cdots + X_n)/n.$$

Using the properties of the expected value and variance we have

$$\mathrm{E}(Y) = a_1\mathrm{E}(X_1) + a_2\mathrm{E}(X_2) + \cdots + a_n\mathrm{E}(X_n)$$

and

$$\mathrm{Var}(Y) = a_1^2\mathrm{Var}(X_1) + a_2^2\mathrm{Var}(X_2) + \cdots + a_n^2\mathrm{Var}(X_n).$$

Typically, the n random variables X_i will have a common mean μ and an common variance σ^2. In this case the sequence $\{X_1, X_2, \ldots, X_n\}$ is said to be a *random sample*. In this case,

$$\mathrm{E}(Y) = (a_1 + a_2 + \cdots + a_n)\mu$$

and

$$\mathrm{Var}(Y) = (a_1^2 + a_2^2 + \cdots + a_n^2)\sigma^2.$$

Example 4.6 Twenty randomly selected students will be asked the question "do you regularly smoke?". (a) Calculate the expected number of smokers in the sample if 10% of the students smoke; (b) what is your "estimate" of the proportion, p, of smokers if six students answered "Yes"?; (c) What are the expected value and the variance of your estimate?

Solution

(a) Let X_i be equal to one if the ith student answers "Yes" and equal to zero otherwise. Let p be equal to the proportion of smokers in the student population. Then the X_i are independent discrete random variables with density $f(0) = 1 - p$ and $f(1) = p$. Therefore,

$$\mathrm{E}(X_i) = \mathrm{E}(X_i^2) = 0f(0) + 1f(1) = f(1) = p = 0.1$$

and

$$\mathrm{Var}(X_i) = \mathrm{E}(X_i^2) - [\mathrm{E}(X_i)]^2 = p - p^2 = p(1-p) = 0.09.$$

Hence, the expected number of smokers in a sample of 20 students is

$$\mathrm{E}(X_1 + X_2 + \cdots + X_{20}) = 20p = 2.$$

The corresponding variance is

$$\mathrm{Var}(X_1 + X_2 + \cdots + X_{20}) = 20p(1-p) = 1.8$$

(b) A reasonable estimate for the fraction, p, of smokers in the population is given by the corresponding fraction of smokers in the sample, $\overline{X}$. In the case of our sample, the observed value, $\overline{x}$, of $\overline{X}$ is $\overline{x} = 6/20 = 0.3$.

(c) The expected value of the estimate in (b) is p and its variance is $p(1-p)/20$. Why? □

Example 4.7 The independent random variables X, Y and Z represent the monthly sales of a large company in the provinces of BC, Ontario and Quebec, respectively. The mean and standard deviations of these variables are as follows (in hundred of dollars):

$$\begin{array}{lll} \mathrm{E}(X) = 1,435 & \mathrm{E}(Y) = 2,300 & \mathrm{E}(Z) = 1,500 \\ \mathrm{SD}(X) = 120 & \mathrm{SD}(Y) = 150, & \mathrm{SD}(Z) = 150. \end{array}$$

(a) What are the expected value and the standard deviation of the total monthly sales?

(b) Sales manager J. Smith is responsible for the sales in BC and 2/3 of the sales in Ontario. Sales manager R. Campbell is responsible for the sales in Quebec and the remaining 1/3 of the sales in Ontario. What are the expected values and standard deviations of Mr. Smith's and Mrs. Campbell's monthly sales?

(c) What are the expected values and standard deviations of the annual sales for each province? Assume for simplicity that the monthly sales are independent.

Solution

(a) The total monthly sales are

$$S = X + Y + Z.$$

By Property 2

$$\mathrm{E}(S) = \mathrm{E}(X) + \mathrm{E}(Y) + \mathrm{E}(Z) = 1,435 + 2,300 + 1,500 = 5,235.$$

By Property 5

$$\mathrm{Var}(S) = \mathrm{Var}(X) + \mathrm{Var}(Y) + \mathrm{Var}(Z) = 120^2 + 150^2 + 150^2 = 59,400.$$

Therefore,

$$\mathrm{SD}(S) = \sqrt{59,400} = 243.72$$

(b) First, notice that

$$S_1 = X + (2/3)Y \qquad \text{and} \qquad S_2 = Z + (1/3)Y,$$

are Mr. Smith's and Mrs. Campbell's monthly sales. By Property 2

$$\mathrm{E}(S_1) = \mathrm{E}(X) + (2/3)\mathrm{E}(Y) = 1,435 + (2/3)2,300 = 2968.33.$$

Analogously

$$\mathrm{E}(S_2) = 2266.67$$

By Property 5

$$\mathrm{Var}(S_1) = \mathrm{Var}(X) + (2/3)^2\mathrm{Var}(Y) = 120^2 + (2/3)^2 150^2 = 24,400,$$

and so

$$\mathrm{SD}(S_1) = \sqrt{24,400} = 156.20.$$

Analogously

$$\mathrm{SD}(S_2) = 158.11.$$

(c) If X_i $(i = 1, \ldots, 12)$ represent BC's monthly sales, the annual sales for BC are

$$T = \sum_{i=1}^{12} X_i$$

Therefore,

$$\mathrm{E}(T) = \mathrm{E}\left[\sum_{i=1}^{12} X_i\right] = \sum_{i=1}^{12} \mathrm{E}(X_i) = (12)(1,435) = 17,220.$$

The variance and the standard deviation of the annual sales in BC (assuming independence) are:

$$\mathrm{Var}(T) = \mathrm{Var}\left[\sum_{i=1}^{12} X_i\right] = \sum_{i=1}^{12} \mathrm{Var}(X_i) = (12)(120^2) = 172,800.$$

$$\mathrm{SD}(T) = \sqrt{172,800} = 415.69.$$

The student can now calculate the expected values and the standard deviations for the annual sales in Toronto and Quebec. □

Question: The total monthly sales can be obtained as the sum of Mr. Smith's ($S_1 = \overline{X+(2/3)Y}$) and Mrs. Campbell's ($S_1 = Z+(1/3)Y$) monthly sales, with variances (calculated in part (b)) equal to $24,400$ and $25,000$, respectively. Why is it then true that the total sales variance ($\mathrm{Var}(X+Y+Z)$), calculated in part (b), is not equal to the sum of $24,400+25,000 = 49,400$?

4.6 Max and Min of Independent Random Variables

The maximum, V, and the minimum, Λ, of a sequence of n independent random variables are of practical interest. They can be used to represent (or model) a number of random quantities which naturally appear in practice. For example, the maximum,

$$V = \max\{X_1, X_2, \ldots, X_n\}.$$

can be used to model

1. The lifetime of a system of n components connected in parallel. In this case

$$X_i \;=\; \text{Lifetime of the } \; i^{th} \; \text{ component.}$$

2. The completion time of a project made up of n subprojects which can be pursued simultaneously. In this case

$$X_i \;=\; \text{Completion time for the } \; i^{th} \; \text{ subproject .}$$

3. The maximum flood level of a river in the next n years. In this case

$$X_i \;=\; \text{Maximum flood level in the } \; i^{th} \; \text{ year .}$$

On the other hand, the minimum,

$$\Lambda = \min\{X_1, X_2, \ldots, X_n\}.$$

can be used to model

1. The lifetime of a system of n components connected in series. In this case

$$X_i \;=\; \text{Lifetime of the } \; i^{th} \; \text{ component.}$$

2. The completion time of a project independently pursued by n competing teams. In this case

$$X_i \;=\; \text{Completion time by the } \; i^{th} \; \text{ team .}$$

3. The minimum flood level of a river in the next n years. In this case

$$X_i \;=\; \text{Minimum flood level in the } \; i^{th} \; \text{ year .}$$

4.6.1 The Maximum

Suppose that $F_i(x)$ and $f_i(x)$ are the distribution and density functions of the random variable X_i, and let $F_V(v)$ and $f_V(v)$ be the distribution and density functions of the maximum V. Since the maximum, V, is less than a given value, v, if and only if each random variable X_i is less than v we have

$$\begin{aligned} F_V(v) &= P\{V \le v\} = P\{X_1 \le v, X_2 \le v, \ldots, X_n \le v\} \\ &= P\{X_1 \le v\}P\{X_2 \le v\}\cdots P\{X_n \le v\} \qquad [\text{ since the variables } X_i \text{ are independent}] \\ &= F_1(v)F_2(v)\cdots F_n(v) \qquad\qquad [\text{since } \; P\{X_i \le v\} = F_i(v), \;\; i = 1, \ldots, n] \end{aligned}$$

This formula is greatly simplified when the X_i's are identically distributed, that is, when

$$F_1(x) = F_2(x) = \cdots = F_n(x) = F(x)$$

for all values of x. In this case,

$$F_V(v) = [F(v)]^n \tag{4.7}$$

and

$$f_V(v) = F_V'(v) = n[F(v)]^{n-1} f(v). \tag{4.8}$$

Example 4.8 A system consists of five components connected in parallel. The lifetime (in thousands of hours) of each component is an exponential random variable with mean $\mu = 3$. See Example 4.4 and Example 4.4 (continued) for the definition of exponential random variables and formulas for their mean and variance.
(a) Calculate the median life (often called "half–life") and standard deviation for each component.
(b) Calculate the probability that a component fails before 3500 hours.
(c) Calculate the probability that the system will fail before 3500 hours. Compare this with the probability that a component fails before 3500 hours.
(d) Calculate the half–life (median life), mean life and standard deviation for the system.

Solution

(a) Using equation (4.5) and the fact that the lifetime X of each component is exponentially distributed with mean $\mu = 3$ we obtain that $\lambda = 1/3$ and that the density and distribution functions of X are

$$f(x) = (1/3)\exp\{-x/3\} \qquad \text{and} \qquad F(x) = 1 - \exp\{-x/3\}, \quad x \geq 0,$$

respectively. The half-life of each component can be obtained as follows

$$1 - \exp\{-x/3\} = 0.5 \Rightarrow \exp\{-x/3\} = 0.5 \Rightarrow x_0 = -3\log(0.5) = 2.08.$$

Therefore, the half-life of each component is equal to 2,080 hours. To obtain the standard deviation, recall that from equation (4.6) the standard deviation of an exponential random variable is equal to its mean, that is,

$$\text{SD}(X) = \text{E}(X) = \mu.$$

Therefore, the standard deviation of the lifetime of each component is equal to 3.
(b) The probability that a component will fail before 3500 is

$$P\{X \leq 3.5\} = F(3.5) = [1 - \exp\{-3.5/3\}] = 0.6886.$$

(c) Using formula (4.7)

$$F_V(v) = [1 - \exp\{-v/3\}]^5$$

and so the probability that the system will fail before 3,500 hours is

$$P\{V \leq 3.5\} = F_V(3.5) = [1 - \exp\{-3.5/3\}]^5 = (0.6886)^5 = 0.1548.$$

The probability that a single component fails (calculated in part (b)) is more than four times larger.
(c) To calculate the median life of the system we must use formula (1) once again:

$$F_V(v) = 0.5 \;\; \Rightarrow \;\; [1 - \exp\{-v/3\}]^5 = 0.5 \Rightarrow \exp\{-v/3\}] = 1 - (0.5)^{1/5} = 0.12945$$

$$\Rightarrow \;\; v_0 = -3\log(0.12945) = 6.133.$$

Therefore, the median life of the system is equal to 6,133 hours.
To calculate the mean life we must first obtain the density function of V. Using formula (2) above we obtain

$$\begin{aligned} f_V(v) &= (5)[1 - \exp\{-v/3\}]^4(1/3)\exp\{-v/3\} \\ &= (5/3)[\exp\{-v/3\} - 4\exp\{-2v/3\} + 6\exp\{-v\} - 4\exp\{-4v/3\} + \exp\{-5v/3\}]. \end{aligned}$$

Since, for any $\beta > 0$,

$$\int_0^\infty v\exp\{-\beta v\}dv = (1/\beta)\left[\int_0^\infty v\beta\exp\{-\beta v\}dv\right] = \text{E}(V)(1/\beta) = (1/\beta)^2,$$

the mean life, E(V), is equal to

$$\begin{aligned}
\mathrm{E}(V) &= \int_0^\infty v f_V(v)dv = (5/3)\left[\int_0^\infty v\exp\{-v/3\}dv - 4\int_0^\infty v\exp\{-2v/3\}dv\right. \\
&+ \left. 6\int_0^\infty v\exp\{-v\}dv - 4\int_0^\infty v\exp\{-4v/3\}dv + \int_0^\infty v\exp\{-5v/3\}dv\right] \\
&= (5/3)[\ (9) - (4)(9/4) + (6)(1) - (4)(9/16) + (9/25)\ ] = 6.85.
\end{aligned}$$

To calculate SD(V) we must first find

$$\mathrm{Var}(V) = \mathrm{E}(V^2) - [\mathrm{E}(V)]^2 = \mathrm{E}(V^2) - (6.85)^2.$$

Since, for any $\beta > 0$,

$$\int_0^\infty v^2\exp\{-\beta v\}dv = 2/(\beta^3), \qquad [\textbf{why?}]$$

we have that

$$\begin{aligned}
\mathrm{E}(V^2) &= \int_0^\infty v^2 f_V(v)dv = (5/3)[\int_0^\infty v^2\exp\{-v/3\}dv - 4\int_0^\infty v^2\exp\{-2v/3\}dv \\
&+ 6\int_0^\infty v^2\exp\{-v\}dv - 4\int_0^\infty v^2\exp\{-4v/3\}dv + \int_0^\infty v^2\exp\{-5v/3\}dv \\
&= (2)(5/3)[\ (27) - (4)(27/8) + (6)(1) - (4)(27/64) + (27/125)\ ] = 60.095.
\end{aligned}$$

Therefore,

$$\mathrm{SD}(V) = \sqrt{60.095 - (6.85^2)} = \sqrt{13.1725} = 3.63.$$

□

4.6.2 The Minimum

Now we turn our attention to the distribution of the minimum, Λ. Let $F_\Lambda(u)$ and $f_\Lambda(u)$ denote the distribution and density functions of Λ. Since the minimum, Λ, is greater than a given value, u, if and only if each random variable X_i is greater than u we have

$$\begin{aligned}
F_\Lambda(u) &= P\{\Lambda \le u\} = 1 - P\{\Lambda > u\} = 1 - P\{X_1 > u, X_2 > u, \ldots, X_n > u\} \\
&= 1 - P\{X_1 > u\}P\{X_2 > u\}\cdots P\{X_n > u\} \quad [\text{ since the variables } X_i \text{ are independent}] \\
&= 1 - [1 - F_1(u)][1 - F_2(u)]\cdots[1 - F_n(u)] \quad [\text{since } \ P\{X_i > u\} = 1 - F_i(u), \ \ i = 1, \ldots, n]
\end{aligned}$$

As before, this formula can be greatly simplified when the X_i's are equally distributed, that is, when

$$F_1(x) = F_2(x) = \cdots = F_n(x) = F(x)$$

for all values of x. In this case,

$$F_\Lambda(u) = 1 - [1 - F(u)]^n \tag{4.9}$$

and

$$f_\Lambda(u) = F'_\Lambda(u) = n[1 - F(u)]^{n-1} f(u). \tag{4.10}$$

Example 4.9 A system consists of five components connected in series. The lifetime (in thousands of hours) of each component is an exponential random variable with mean $\mu = 3$.
(a) Calculate the probability that the system will fail before 3500 hours. Compare this with the probability that a component fails before 3500 hours.
(b) Calculate the median life, the mean life and the standard deviation for the system.

Solution

(a) Using formula (4.9) above we obtain

$$F_\Lambda(u) = 1 - [\exp\{-u/3\}]^5 = 1 - \exp\{-(5/3)u\}$$

and so Λ is also exponentially distributed with parameter $5 \times (1/3) = 5/3$. In general, the minimum of n exponential random variables with parameter λ is also exponential with parameter $n\lambda$. Finally,

$$P\{\Lambda \leq 3.5\} = F_\Lambda(3.5) = 1 - \exp\{-(5/3)3.5\} = 0.9971$$

The probability that a component will fail before 3500 has been found (in Example 4.8) to be 0.6886. Therefore, the probability that the system will fail before $3,500$ hours is almost 45% larger.
(b) Since Λ is exponentially distributed, its mean and standard deviation can be obtained directly from the distribution function found in (a), using equations (4.5) and (4.6). That is,

$$\mathrm{E}(\Lambda) = \mathrm{SD}(\Lambda) = (3/5) = 0.6.$$

Therefore, the mean life of the system, 600 hours, is 5 times smaller than that of the individual components. Finally, the median life of the system can be found as follows:

$$1 - \exp\{-(5u)/3\} = 0.5 \Rightarrow \exp\{-(5u)/3\} = 0.5 \Rightarrow u_0 = -3\log(0.5)/5 = 0.416.$$

Therefore, the median life of the system is equal to 416 hours. □

4.7 Exercises

4.7.1 Exercise Set A

Problem 4.1 A system consists of five identical components all connected in series. Suppose each component has a lifetime (in hours) that is *exponentially* distributed with the rate $\lambda = 0.01$, and all the five components work independently of one another.

Define T to be the time at which the system fails. Consider the following questions:
(a) Obtain the distribution of T. Can you tell what type of distribution it is?
(b) Compute the IQR (interquartile range) for the distribution obtained in part (a).
(c) What is the probability that the system will last at least 15 hours?

Problem 4.2 Are the following functions density functions? Why?
(a) $f_1(x) = 1, 1 \leq x \leq 3$; 0, otherwise.
(b) $f_2(x) = x, -1 \leq x \leq 1$; 0, otherwise.
(c) $f_3(x) = \exp(-x), x \geq 0$; 0, otherwise.

Problem 4.3 Suppose that the response time X at a certain on-line computer terminal (*the elapsed time between the end of a user's inquiry and the beginning of the system's response to that inquiry*) has an **exponential** distribution with expected response time equal to 5 seconds (i.e. the exponential rate is $\lambda = 0.2$).
(a) Calculate the median response time.
(b) What is the probability that the next three response times exceed 5 seconds? (Assume that all the response times are independent).

Problem 4.4 The hourly volume of traffic, X, for a proposed highway has density proportional to $g(x)$, where

$$g(x) = \begin{cases} x(100 - x) & \text{if } 0 < x < 100 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Derive the density and the distribution functions of X.
(b) The traffic engineer may design the highway capacity equal to the mean of X. Determine the **design capacity** of the highway and the corresponding **probability of exceedence** (i.e. traffic volume is greater than the capacity).

Problem 4.5 A discrete random variable X has the density function given below.

x	-1	0	1	2
$f(x)$	0.2	c	0.2	0.1

(a) Determine c;
(b) Find the distribution function $F(x)$;
(c) Show that the random variable $Y = X^2$ has the density function $g(y)$ given by

y	0	1	4
$g(y)$	0.5	0.4	0.1

(d) Calculate expectation $\mathrm{E}(X)$, variance $\mathrm{Var}(X)$ and the mode of X (the value x with the highest density).

Problem 4.6 A continuous random variable X has the density function $f(x)$ which is proportional to cx on the interval $0 \le x \le 1$, and 0 otherwise.
(a) Determine the constant c;
(b) Find the distribution function $F(x)$ of X;
(c) Calculate $\mathrm{E}(X)$, $\mathrm{Var}(X)$ and the median, $Q(0.5)$;
(d) Find $P(|X| \ge 0.5)$.

Problem 4.7 Show that
(a) Any distribution function $F(x)$ is non-decreasing, i.e. for any real values $x_1 < x_2$, $F(x_1) \le F(x_2)$.
(b) Suppose X is a random variable with finite variance. Then, $\mathrm{Var}(X) \le \mathrm{E}(X^2)$.
(c) If a density function $f(x)$ is symmetric around 0, i.e. $f(-x) = f(x)$ for all $x \in R$, then $F(0) = P(X \le 0) = 0.5$.

Problem 4.8 If the probability density of a random variable is given by

$$f(x) = \begin{cases} kx & \text{for } 0 < x < 2 \\ 2k(3-x) & \text{for } 2 \le x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Find the value of k such that $f(x)$ is a probability density function.
(b) Find the corresponding distribution function.
(c) Find the mean and median.

Problem 4.9 Suppose a random variable X has a probability density function given by

$$f(x) = \begin{cases} kx(1-x) & \text{for } 0 \le x \le 1 \\ 0 & \text{elsewhere.} \end{cases}$$

(a) Find the value of k such that $f(x)$ is a probability density function.
(b) Find $P(0.4 \le X \le 1)$.
(c) Find $P(X \le 0.4 | X \le 0.8)$.
(d) Find $F(b) = P(X \le b)$, and sketch the graph of this function.

Problem 4.10 Suppose that random variables X and Y are independent and have the same mean 3 and standard deviation 2. Calculate the mean and variance of $X - Y$.

Problem 4.11 Suppose X has exponential distribution with a unknown parameter λ, i.e. its density is that

$$f(x) = \begin{cases} \lambda \exp(-\lambda x) & \text{if } x \ge 0 \\ 0 & \text{otherwise.} \end{cases}$$

If $P(X \ge 1) = 0.25$, determine λ.

Problem 4.12 Suppose an enemy aircraft flies directly over the Alaska pipeline and fires a single air-to-surface missile. If the missile hits anywhere within 10 feet of the pipeline, a major structural damage will occur and the oil flow will be disrupted. Let X be the distance from the pipeline to the point of impact. Note that X is a continuous random variable. The probability function describing the missile's point of impact is given by

$$f(x) = \begin{cases} \frac{60+x}{3600} & \text{for } -60 < x < 0 \\ \frac{60-x}{3600} & \text{for } 0 \leq x < 60 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the distribution function, $F(x)$.
(b) Let A be the event "flow is disrupted." Find $P(A)$.
(c) Find the mean and the standard deviation of X.
(d) Find the median and the interquartile range of X.

Problem 4.13 Consider a random variable X which follows the uniform distribution on the interval $(0, 1)$. (a) Give the density function $f(x)$ and obtain the cumulative distribution function$F(x)$ of X;
(b) Calculate the mean (expectation) $\mathrm{E}(X)$ and variance $\mathrm{Var}(X)$;
(c) Let $Y = \sqrt{X}$. Find the $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$;
(d) Obtain the distribution function $G(y)$ and furthermore the density function $g(y)$ of random variable Y.

Problem 4.14 The reaction time (in seconds) to a certain stimulus is a continuous random variable with density given below

$$f(x) = \begin{cases} \frac{3}{2x^2} & \text{for } 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Obtain the distribution function.
(b) Take next two observations X_1 and X_2 (we can assume they are *i.i.d*). Then consider $V = \max\{X_1, X_2\}$. What is the density and distribution functions of V?
(c) Compute the expectation $\mathrm{E}(V)$ and the standard deviation $\mathrm{SD}(V)$.
(d) Compute the difference between the expectation and the median for the distribution of V.

4.7.2 Exercise Set B

Problem 4.15 The continuous random variable X takes values between -2 and 2 and its density function is proportional to
(a) $4 - x^2$
(b) x^2
(c) $2 + x$
(d) $\exp\{-|x|\}$

Find, in each case, the density function, the distribution function, the mean, the standard deviation, the median and the interquartile range of X.

Problem 4.16 Find the density functions corresponding to the pictures in Figure 3.7. For each case also calculate the distribution function, the mean, the median, the interquartile range and the standard deviation.

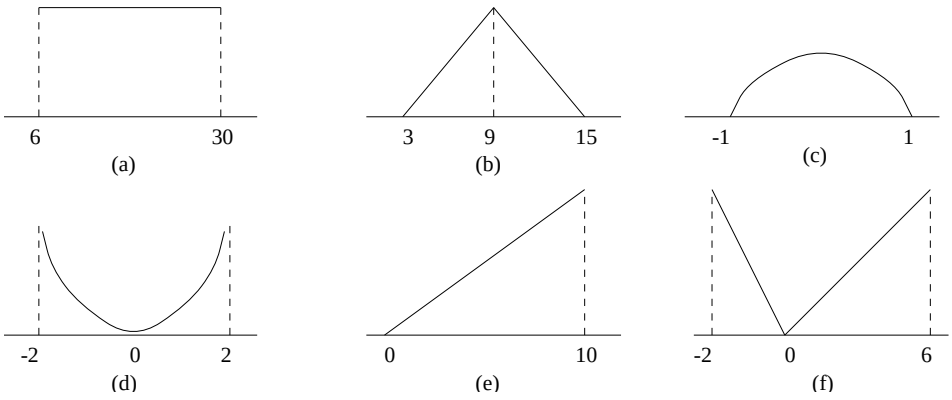


Figure 3.7: Pictures of densities

Problem 4.17 The density function for the lifetime of a part, X, decays exponentially fast. If the half–life of X is equal to fifty weeks, find the mean and standard deviation of X.

Problem 4.18 The density function for the measurement error, X, is uniform on the interval $(-0.5, 0.8)$. What is the distribution function of X^2? What is the density of X^2?

Problem 4.19 The hourly volume of traffic, X, for a proposed highway has density proportional to $d(x)$, where

$$\begin{aligned} d(x) &= x && \text{if} \quad 0 < x < 300 \\ &= (3/2)(500 - x) && \text{if} \quad 300 \leq x < 500 \\ &= 0 && \text{otherwise.} \end{aligned}$$

(a) Derive the density and the distribution functions of X.

(b) The traffic engineer may design the highway capacity equal to one of the following:

(i) the mode of X (defined as the value x with highest density)

(ii) the mean of X

(iii) the median of X

(iv) the quantile of order 0.90 of X ($Q(0.90)$).

Determine the design capacity of the highway and the corresponding probability of exceedance (that is, capacity is less than traffic volume) for each of the four cases.

Problem 4.20 The company has 20 welders with the following "performances":

Welder	0	1	2	3	4
1	0.10	0.20	0.40	0.20	0.10
2	0.20	0.20	0.20	0.20	0.20
3	0.50	0.30	0.10	0.05	0.05
4	0.05	0.05	0.10	0.30	0.50
5	0.50	0.00	0.00	0.00	0.50
6	0.85	0.00	0.00	0.00	0.15
7	0.30	0.25	0.20	0.10	0.15
8	0.20	0.30	0.20	0.10	0.20
9	0.10	0.10	0.50	0.20	0.10
10	0.20	0.50	0.10	0.20	0.00
11	0.30	0.30	0.40	0.00	0.00
12	0.10	0.10	0.50	0.15	0.15
13	0.35	0.25	0.20	0.15	0.05
14	0.40	0.30	0.10	0.10	0.10
15	0.20	0.30	0.50	0.00	0.00
16	0.60	0.30	0.10	0.00	0.00
17	0.70	0.10	0.10	0.10	0.00
18	0.10	0.80	0.10	0.00	0.00
19	0.40	0.40	0.10	0.10	0.00
20	0.15	0.60	0.15	0.10	0.00

1) How would you rank these twenty welders (e.g. for promotion) on the basis of this information alone?

2) Would you change the "ranking" if you know that items with one, two, three and four cracks must be sold for \$6, \$15, \$40, and \$60 less, respectively? What if the associated losses are \$6, \$15, \$40, and \$80. **Suggestion:** Use the computer.

Problem 4.21 Suppose that the maximum annual wind velocity near a construction site, X, has exponential density

$$f(x) = \lambda \exp\{-\lambda x\}, \qquad x > 0.$$

(a) If the records of maximum wind speed show that the probability of maximum annual wind velocities less than 72 mph is approximately 0.90, suggest an appropriate estimate for λ.

(b) If the annual maximum wind speeds for different years are statistically independent, calculate the probability that the maximum wind speed in the next three years will exceed 75 mph. What about the next 15 years?

(c) Plot the distribution function of the maximum wind speed for the next year, for the next 3 years and for the next 15 years. Briefly report your conclusions.

(d) Let $Q_m(p)$ $(m = 1, 2, \ldots)$ the quantile of order p for the maximum wind speed on the next m years. Show that

$$Q_m(p) = Q_1\left[p^{1/m}\right], \qquad \text{for all} \quad m = 1, 2, \ldots$$

Use this formula to plot $Q_m(0.90)$ versus m. Same for $Q_m(0.95)$. Briefly report your conclusions. **Suggestion:** Use the computer.

Problem 4.22 A system has two independent components A and B connected in parallel. If the operational life (in thousand of hours) of each component is a random variable with density

$$\begin{aligned} f(x) &= \frac{1}{36}(x-4)(10-x) && 4 < x < 10 \\ &= 0 && \text{otherwise} \end{aligned}$$

(a) Find the median and the mean life of each component. Find also the standard deviation and IQR.
(b) Calculate the distribution and density functions for the lifetime of the system. What is the expected lifetime of the system?

(c) Same as (b) but assuming that the components are connected "in series" instead of "in parallel".

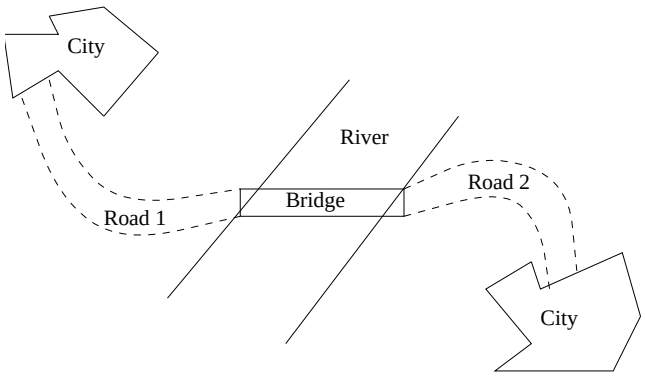
Problem 4.23 A large construction project consists of building a bridge and two roads linking it to two cities (see the picture below). The contractual time for the entire project is 18 months.

The construction of each road will require between 15 and 20 months and that of the bridge will require between 12 and 19 months. The three parts of the projects can be done simultaneously and independently. Let X_1, X_2 and Y represent the construction times for the two roads and the bridge, respectively and suppose that these random variables are uniformly distributed on their respective ranges.
(a) What is the expected time for completion of each part of the project? What are the corresponding standard deviations?

(b) What is the expected time for the completion of the entire project? What is the corresponding standard deviation?

(c) What is the probability that the project will be completed within the contractual time?

Problem 4.24 Same as Problem 2.51, but assuming that the variables X_1, X_2 and Y have triangular distributions over their ranges.



Chapter 5

Normal Distribution

5.1 Definition and Properties

Normal Distribution $N(\mu, \sigma^2)$

The Normal distribution is, for reasons that will be evident as we progress in this course, the most popular distribution among engineers and other scientists. It is a continuous distribution with density,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

where μ and σ are "parameters" which control the central location and the dispersion of the density, respectively. The normal density is perfectly symmetric about the center, μ, and this bell-shaped function is "shorter" and "fatter" as σ increases.

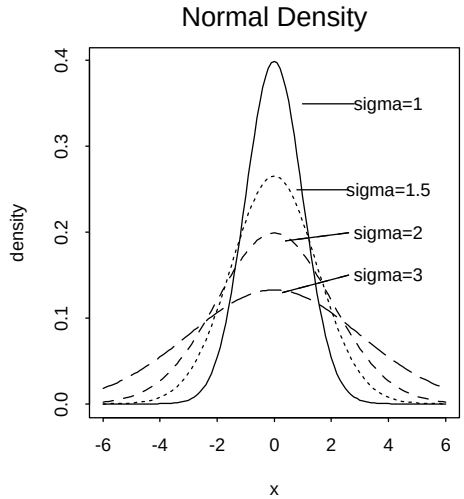


Figure 4.1: Normal density functions

The density steadily decreases as we move away from its highest value

$$f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}.$$

Therefore, the relative (and also the absolute) probability that X will take a value near μ is the highest. Since $f(x) \to 0$ as $x \to \infty$, exponentially fast,

$$g(k) = P\{|X - \mu| \leq k\sigma\} \to 1, \quad \text{as } k \to \infty,$$

very fast. In fact, it can be shown that $g(1) = 0.6827$, $g(2) = 0.9544$, $g(3) = 0.9973$ and $g(4) = 0.9999$. For practical purposes $g(k) = 1$ for $k \geq 4$.

Some Important Facts about the Normal Distribution

Fact 1: If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, where a and b are two constants with $a \neq 0$, then

$$Y \sim N(a\mu + b, a^2\sigma^2).$$

For example, if $X \sim N(2, 9)$ and $Y = 5X + 1$, then $\mathrm{E}(Y) = (5)(2) + 1 = 11$, $\mathrm{Var}(Y) = (5^2)(9) = 225$ and $Y \sim N(11, 225)$.

Proof We will consider the case $a > 0$. The proof for the $a < 0$ case is left as an exercise. The distribution function of Y, denoted here by G is given by

$$G(y) = P(Y \leq y) = P(aX + b \leq y) = P\left(X \leq \frac{y-b}{a}\right) = F\left(\frac{y-b}{a}\right),$$

where F is the distribution function of X. The density function $g(y)$ of Y can now be found by differentiating $G(y)$. That is,

$$\begin{aligned} g(y) &= G'(y) = \frac{d}{dy}F\left(\frac{y-b}{a}\right) = \frac{1}{a}f\left(\frac{y-b}{a}\right) \\ &= \frac{1}{a\sigma\sqrt{2\pi}} \exp\left\{-\frac{[y-(a\mu+b)]^2}{2a^2\sigma^2}\right\}, \end{aligned}$$

□

Standardized Normal

An important particular case emerges when $a = (1/\sigma)$ and $b = -(\mu/\sigma)$. In this case the transformed variable is denoted by Z and called "standard normal". Since

$$Z = (1/\sigma)X - (\mu/\sigma) = \frac{X - \mu}{\sigma},$$

by Property 1, the parameters of the new normal variable, Z, can be obtained from those of the given normal variable, X, (μ and σ^2) as follows:

$$\mu \longrightarrow a\mu + b = (1/\sigma)\mu - (\mu/\sigma) = 0$$

and

$$\sigma^2 \longrightarrow a^2\sigma^2 = (-1/\sigma)^2\sigma^2 = 1.$$

That is, any given normal random variable $X \sim N(\mu, \sigma^2)$ can be transformed into a standard normal $Z \sim N(0, 1)$ by the equation

$$Z = \frac{X - \mu}{\sigma}. \tag{5.1}$$

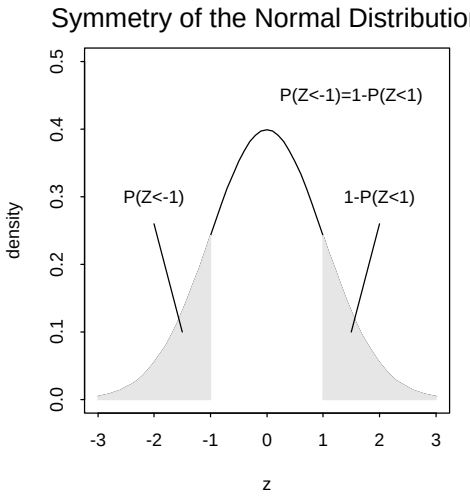


Figure 4.2: Symmetry of normal distribution

Fact 2: The standard normal density is denoted by the Greek letter ϕ (pronounced Phi) and the standard normal distribution function is denoted by the corresponding upper case Greek letter Φ. In symbols,

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}$$

and

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{t^2}{2}\right\} dt.$$

Since $\phi(z)$ is symmetric about zero $[\phi(-z) = \phi(z)$ for all $z]$ we have the important identity

$$\Phi(-z) = 1 - \Phi(z)$$

for all z. See Figure 4.2.

For example,

$$\Phi(-1) = 1 - \Phi(1) = 1 - 0.8413447 = 0.1586553,$$

and

$$\begin{aligned} P(-1.5 < Z < 1.2) &= \Phi(1.2) - \Phi(-1.5) = \Phi(1.2) - [1 - \Phi(1.5)] \\ &= \Phi(1.2) + \Phi(1.5) - 1 = 0.8181231. \end{aligned}$$

Fact 3: The normal density cannot be integrated in closed form. That is, there are no simple formulas for calculating expressions like

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

or

$$P(a < X < b) = \int_{a}^{b} f(t)dt = F(b) - F(a).$$

These expressions can only be calculated by numerical methods (numerical integration or quadrature). Fortunately, however, we can use Fact 1 to reduce calculations involving any normal random variable to the standard normal case (see Table 1 in the Appendix). The basic formula for these calculations is

$$\begin{aligned} F(x) &= P(X \leq x) = P[(X-\mu)/\sigma < (x-\mu)/\sigma] \\ &= P[Z < (x-\mu)/\sigma] = \Phi[(x-\mu)/\sigma]. \end{aligned}$$

The application of this "reduction method" is illustrated in the following example.

Example 5.1 Let $X \sim N(2, 9)$. Calculate (a) $P(X < 5)$, (b) $P(-3 < X < 5)$ (c) $P(X > 5)$ (d) $P(|X-2| < 3)$ (e) The value of c such that $P(X < c) = 0.95$ (f) The value of c such that $P(|X-2| > c) = 0.10$

Solution

(a) $P(X < 5) = F(5) = \Phi[(5-2)/3)] = \Phi(1) = 0.8413447$, from Table 1 in the Appendix.

(b) $P(-3 < X < 5) = F(5) - F(-3) = \Phi[(5-2)/3] - \Phi[(-3-2)/3] = \Phi(1) - \Phi(-5/3) = 0.8413447 - 0.04779035 = 0.7935544$, from Table 1 in the Appendix.

(c) $P(X > 5) = 1 - P(X \leq 5) = 1 - F(5) = 1 - \Phi(1) = 0.1586553$.

(d) To solve this question we must first remember that a number has absolute value smaller than 3 if and only if this number is between -3 and 3. In other words, to say that $|X-2| < 3$ is equivalent to saying that $-3 < X - 2 < 3$. Therefore,

$$\begin{aligned} P[|X-2| < 3] &= P[-3 < X-2 < 3] = P[-1 < (X-2)/3 < 1] = P[-1 < Z < 1] \\ &= \Phi(1) - \Phi(-1) = \Phi(1) - [1 - \Phi(1)] = 2\Phi(1) - 1 \\ &= 0.6826895. \end{aligned}$$

One useful result to point out here is

$$P[|Z| \leq z)] = 2\Phi(z) - 1$$

(e) To solve this question we first notice that

$$P(X < c) = P[Z < (c-2)/3] = \Phi[(c-2)/3].$$

Second, we see from the Normal Table that $\Phi(d) = 0.95$ if $d \approx 1.64$. Therefore

$$\frac{c-2}{3} = 1.64 \quad \Rightarrow \quad c = (3)(1.64) + 2 = 6.92.$$

(f) The value of c such that $P(|X-2| > c) = 0.10$ is calculated as follows,

$$P(|X-2| > c) = P[|Z| > c/3] = 1 - P[|Z| \leq c/3] = 1 - \{2\Phi(c/3) - 1\} = 2[1 - \Phi(c/3)] = 0.10$$

Therefore,

$$\Phi(c/3) = 0.95 \quad \Rightarrow \quad c/3 = 1.64 \quad \Rightarrow \quad c = (3)(1.64) = 4.92$$

□

Fact 4: If $X \sim N(\mu, \sigma^2)$, then

$$\mathrm{E}(X) = \mu \qquad \text{and} \qquad \mathrm{Var}(X) = \sigma^2.$$

Proof It suffices to prove that $\mathrm{E}(Z) = 0$ and $\mathrm{Var}(Z) = 1$, because from (9)

$$X = \sigma Z + \mu,$$

and then we would have $\mathrm{E}(X) = \mathrm{E}(\sigma Z + \mu) = \sigma \mathrm{E}(Z) + \mu = \mu$ and $\mathrm{Var}(\sigma Z + \mu) = \sigma^2 \mathrm{Var}(Z) = \sigma^2$. By symmetry, we must have $\mathrm{E}(Z) = 0$. In fact, since $\phi'(z) = (-z/\sqrt{2\pi}) \exp\{-z^2/2\} = -z\phi(z)$, it follows that

$$\int_{-\infty}^{\infty} z\phi(z)dz = -\int_{-\infty}^{\infty} \phi'(z)dz = \phi(z)|_{-\infty}^{\infty} = 0.$$

Finally, using by–part integration [$u = z$ and $dv = \phi'(z)$] we obtain

$$\int_{-\infty}^{\infty} z^2\phi(z)dz = -\int_{-\infty}^{\infty} z\phi'(z)dz = -\left\{z\phi(z)|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \phi(z)dz\right\} = 1.$$

□

Fact 5: Suppose that $X_1, X_2, \ldots, X_n$ are independent normal random variables with mean $\mathrm{E}(X_i) = \mu_i$ and variance $\mathrm{Var}(X_i) = \sigma_i^2$. Let Y be a linear combination of the X_i, that is,

$$Y = a_1X_1 + a_2X_2 + \ldots + a_nX_n,$$

where a_i ($i = 1, \cdots, n$) are some given constant coefficients. Then,

$$Y \quad \sim \quad N(a_1\mu_1 + a_2\mu_2 + \ldots + a_n\mu_n, \quad a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \ldots + a_n^2\sigma_n^2)$$

Proof The proof that Y is normal is beyond the scope of this course. On the other hand, to show that

$$\mathrm{E}(Y) = a_1\mu_1 + a_2\mu_2 + \ldots + a_n\mu_n,$$

and

$$\mathrm{Var}(Y) = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \ldots + a_n^2\sigma_n^2,$$

is very easy, using Properties 2 and 5 for the mean and the variance of sums of random variables. □

Example 5.2 Suppose that X_1 and X_2 are independent, $X_1 \sim N(2, 4)$, $X_2 \sim N(5, 3)$ and

$$Y = 0.5X_1 + 2.5X_2.$$

Find the probability that Y is larger than 15.

Solution By Fact 5, Y is a normal random variable, with mean

$$\mu = (0.5 \times 2) + (2.5 \times 5) = 13.5,$$

and variance

$$\sigma^2 = (0.5^2 \times 4) + (2.5^2 \times 3) = 19.75.$$

Therefore,

$$P(Y > 15) = 1 - \Phi\left(\frac{15 - 13.5}{\sqrt{19.75}}\right) = 1 - \Phi(0.34) = 1 - 0.6331 = 0.3669.$$

□

An important particular case arises when $X_1, \ldots, X_n$ is a normal sample, that is, when the variables $X_1, \ldots, X_n$ are independent, identically distributed, normal random variables, with mean μ and variance σ^2. One can think of the $X_i's$ as a sequence of n independent measurements of the normal random variable, $X \sim N(\mu, \sigma^2)$. μ is usually called **the population mean** and σ^2 is usually called the **population variance**.

If the coefficients, a_i, are all equal to $1/n$. then Y is equal the sample average:

$$Y = \sum_{i=1}^{n}(1/n)X_i = \frac{1}{n}\sum_{i=1}^{n}X_i = \overline{X}.$$

By Fact 5, then, the normal sample average is also a normal random variable, with mean

$$\sum_{i=1}^{n} a_i\mu = \sum_{i=1}^{n}\frac{1}{n}\mu = \frac{n\mu}{n} = \mu,$$

and variance

$$\sum_{i=1}^{n} a_i^2\sigma^2 = \sum_{i=1}^{n}\frac{1}{n^2}\sigma^2 = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Example 5.3 Suppose that $X_1, X_2, \ldots, X_{16}$ are independent $N(\mu, 4)$ and $\overline{X}$ is their average. (a) Calculate $P(|X_1 - \mu| < 1)$ and $P(|\overline{X} - \mu| < 1)$. (b) Calculate $P(|\overline{X} - \mu| < 1)$ when the sample size is 25 instead of 16. (c) Comment on the result of your calculations.

Solution

(a) Since $X_1 \sim N(\mu, 4)$, $X_1 - \mu \sim N(0, 4)$ and so,

$$P(|X_1 - \mu| < 1) = 2\Phi(1/2) - 1 = 2\Phi(0.5) - 1 = 0.383.$$

Moreover, since $\overline{X} \sim N(\mu, 4/16)$, $\overline{X} - \mu \sim N(0, 1/4)$ and

$$P(|\overline{X} - \mu| < 1) = 2\Phi\left[1/\sqrt{1/4}\right] - 1 = 2\Phi(2) - 1 = 0.954.$$

(b) Since $\overline{X} \sim N(\mu, 4/25)$, $\overline{X} - \mu \sim N(0, 4/25)$ and

$$P(|\overline{X} - \mu| < 1) = 2\Phi(5/2) - 1 = 2\Phi(2.5) - 1 = 0.9876.$$

(c) The probability that the sample mean, $\overline{X}$, is close to the population mean, μ, (0.954 when $n = 16$, and 0.9876 when $n = 25$) is much larger than the probability that any single observation, X_i, is close to μ (0.383). The probability that the sample mean is close to the population mean depends on the sample size, n, and gets larger when the n gets larger. □

5.2 Checking Normality

A data set, $x_1, x_2, \ldots, x_n$, is a **sample** if the $x_i's$ are a sequence of independent observations of a random variable, X. The sample is called **normal** if $X \sim N(\mu, \sigma^2)$. The statistical analysis of many data sets is based on the assumption the data set is a normal sample. The validity of this assumption must be carefully examined because the conclusions of the analysis may be seriously distorted in the absence of the assumed normality. The most common types departures from normality are **asymmetry, heavy tailness and the presence of outliers**.

One simple method for checking the normality of the sample $x_1, x_2, \ldots, x_n$ is the so called normal Q–Q plot. A normal Q–Q plot is a plot of the theoretical standard normal quantiles of order $(i - 0.50/n$, d_i, versus the corresponding empirical sample quantiles, $\hat{q}_i = x_{(i)}$. If the sample is normal, then the points $(d_i, \hat{q}_i)$ must be close to a straight line. Therefore, departures from a straight–line pattern in the Q–Q plot indicate lack of normality.

Several normal Q–Q plots are displayed on Figure 4.3. The sample for case (a) is normal. The samples for the other five cases depart from normality in different ways.

The Q–Q plot technique is based on the following rational. The theoretical quantile of order $(i - 0.5)/n$ for the random variable X, denoted q_i, is defined by the equation

$$P(X \leq q_i) = F(q_i) = (i - 0.5)/n.$$

That is,

$$q_i = F^{-1}[(i - 0.5)/n],$$

where F^{-1} denotes the inverse of F. In the special case of the standard normal the theoretical quantiles will be denoted by d_i. They are given by the formula

$$d_i = \Phi^{-1}[(i - 0.5)/n],$$

where, as usual, Φ denotes the standard normal distribution function. In the case of a normal random variable, X, with mean μ and variance σ^2, we have

$$P(X \leq q_i) = \Phi[(q_i - \mu)/\sigma] = (i - 0.5)/n.$$

and therefore,

$$(q_i - \mu)/\sigma = \Phi^{-1}[(i - 0.5)/n] = d_i \quad \Longrightarrow \quad q_i = \mu + \sigma d_i.$$

Given the sample

$$x_1, x_2, \ldots, x_n,$$

the corresponding empirical quantiles, $\hat{q}_i$ are simply given by the sorted sample, $x_{(1)}, x_{(2)}, \ldots, x_{(n)}$, that is,

$$\hat{q}_1 = x_{(1)}, \ \hat{q}_2 = x_{(2)}, \ \ldots, \ \hat{q}_n = x_{(n)}.$$

If this sample comes from a $N(\mu, \sigma^2)$ distribution then one would expect that

$$\hat{q}_i \approx q_i = \mu + \sigma d_i,$$

and therefore the plot of $x_{(i)}$ versus d_i will be close to a straight line, with slope σ and intercept μ.

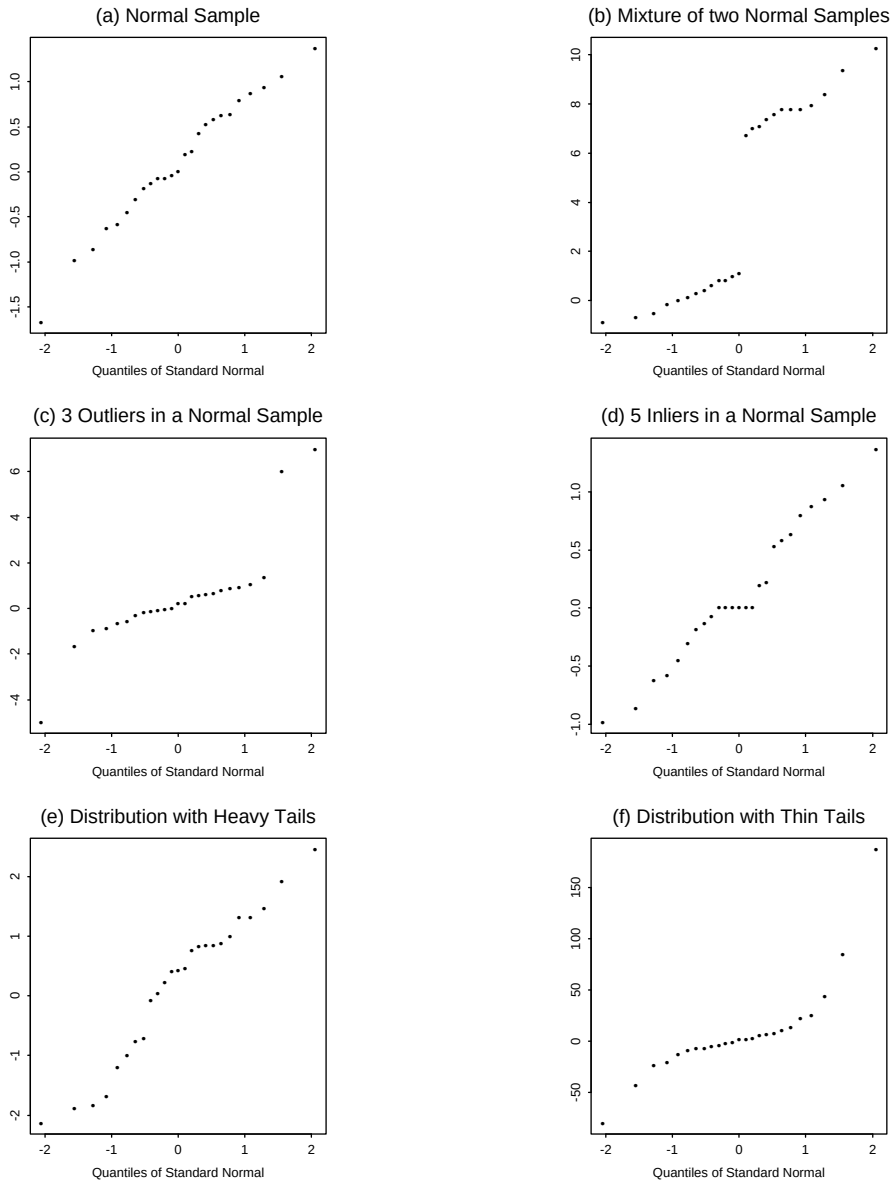


Figure 4.3: Q-Q plots for checking normality

5.3 Exercises

5.3.1 Exercise Set A

Problem 5.1 A machine operation produces steel shafts having diameters that are normally distributed with a mean of 1.005 inches and a standard deviation of 0.01 inch. Specifications call for diameters to fall within the interval 1.00 ± 0.02 inches. What percentage of the output of this operation will fail to meet specifications? What should be the mean diameter of the shafts produced in order to minimize the fraction not meeting specifications?

Problem 5.2 Extruded plastic rods are automatically cut into nominal lengths of 6 inches. Actual lengths are normally distributed about a mean of 6 inches and their standard deviation is 0.06 inch.
(a) What proportion of the rods exceeds the tolerance limits of 5.9 inches to 6.1 inches?
(b) To what value does the standard deviation need to be reduced if 99% of the rods must be within tolerance?

Problem 5.3 Suppose X_1 and X_2 are independent and identically distributed $N(0, 4)$, and define $Y = \max(X_1, X_2)$. Find the density and the distribution functions of Y.

Problem 5.4 Assume that the height of UBC students is a normal random variable with mean 5.65 feet and standard deviation 0.3 feet.
(a) Calculate the probability that a randomly selected student has height between 5.45 and 5.85 feet.
(b) What is the proportion of students above 6 feet?

Problem 5.5 The raw scores in a national aptitude test are normally distributed with mean 506 and standard deviation 81.
(a) What proportion of the candidates scored below 574?
(b) Find the 30th percentile of the scores.

Problem 5.6 Scores on a certain nationwide college entrance examination follow a normal distribution with a mean of 500 and a standard deviation of 100.
(a) If a school admits only students who scores over 670, what proportion of the student pool will be eligible for admission?
(b) What admission requirements would you see if only the top 15% are to be eligible?

Problem 5.7 A machine is designed to cut boards at a desired length of 8 feet. However, the actual length of the boards is a normal random variable with standard deviation 0.2 feet. The mean can be set by the machine operator. At what mean length should the machine be set so that only 5 per cent of the boards are under cut (that is, under 8 feet)?

Problem 5.8 The temperature reading X from a thermocouple placed in a constant-temperature medium is normally distributed with mean μ, the actual temperature of the medium, and standard deviation σ.
(a) What would the value of σ have to be to ensure that 95% of all readings are within 0.1°

of μ?
(b) Consider the difference between two observations X_1 and X_2 (here we could assume that X_1 and X_2 are *i.i.d.*), what is the probability that the absolute value of this difference is at most 0.075°?

Problem 5.9 Suppose the random variable X follows a normal distribution with mean $\mu = 50$ and standard deviation $\sigma = 5$.
(a) Calculate the probability $P(|X| > 60)$.
(b) Calculate $\mathrm{E}X^2$ and the interquartile range of X.

5.3.2 Exercise Set B

Problem 5.10 Let Z be a standard normal random variable. Find:

(a) $P(Z < 1.3)$

(b) $P(0.8 < Z < 1.3)$

(c) $P(-0.8 < Z < 1.3)$

(d) $P(-1.3 < Z < -0.8)$

(e) c such that $P(Z < c) = 0.9032$

(f) c such that $P(Z < c) = 0.0968$

(g) c such that $P(-c < Z < c) = 0.90$

(h) c such that $P(|Z| < c) = 0.95$

(i) c such that $P(|Z| > c) = 0.80$

Problem 5.11 Let X be a normal random variable with mean 10 and variance 25 Find:

(a) $P(X < 13)$

(b) $P(11 < X < 13)$

(c) $P(8 < X < 13)$

(d) $P(6 < X < 8)$

(e) c such that $P(X < c) = 0.9032$

(f) c such that $P(X < c) = 0.0968$

(g) c such that $P(-c < X - 10 < c) = 0.90$

(h) c such that $P(-c < X < c) = 0.95$

(j) c such that $P(|X - 10| > c) = 0.80$

(k) c such that $P(|X| > c) = 0.80$

Problem 5.12 A scholarship is offered to students who graduate in the top 5% of their class. Rank in the class is based on GPA (4.00 being perfect). A professor tells you the marks are distributed normally with mean 2.64 and variance 0.5831. What GPA must you get to qualify for the scholarship?

Problem 5.13 If the test scores of 40 students are normally distributed with a mean of 65 and a standard deviation of 10.
(a) Calculate the probability that a randomly selected student scored between 50 and 80;
(b) If two students are randomly selected, calculate the probability that the difference between their scores is less than 10.

Problem 5.14 The length of trout in a lake is normally distributed with mean $\mu = 0.93$ feet and standard deviation $\sigma = 0.5$ feet.
(a) What is the probability that a randomly chosen trout in the lake has a length of at least 0.5 feet;
(b) Suppose now that the σ is unknown. What is the value of σ if we know that 85% of the trout in the lake are less than 1.5 feet long. Use the same mean 0.93.

Problem 5.15 The life of a certain type of electron tube is normally distributed with mean 95 hours and standard deviation 6 hours. Four tubes are used in a electronic system. Assume that these tubes alone determine the operating life of the system and that, if any one fails, the system is inoperative.
(a) What is the probability of a tube living at least 100 hours?
(b) What is the probability that the system will operate for more than 90 hours?

Problem 5.16 A product consists of an assembly of three components. The overall weight of the product, Z, is equal to the sum of the weights X_1, X_2 and X_3 of its components. Because of variability in production, they are independent random variables, each normally distributed as $N(2, 0.02)$, $N(1, 0.010)$ and $N(3, 0.03)$, respectively. What is the probability that Z will meet the overall specification 6.00 ± 0.30 inches?

Problem 5.17 Due to variability in raw materials and production conditions, the weight (in hundred of pounds) of a concrete beam is a normal random variable with mean 31 and standard deviation 0.50.

(a) Calculate the probability that a randomly selected beam weights between 3000 and 3200 pounds.

(b) Calculate the probability that 25 randomly selected beams will
weight more than 79,500 pounds.

Problem 5.18 A machine fills 250-pound bags of dry concrete mix. The actual weight of the mix that is put in the bag is a normal random variable with standard deviation $\sigma = 0.40$ pound. The mean can be set by the machine operator. At what mean weight should the machine be set so that only 10 per cent of the bags are underweight? What about the larger 500-pound bags?

Problem 5.19 Check if the following samples are normal. Describe the type of departure from normality when appropriate.

(a) 2.52 3.06 2.41 3.98 2.63 4.11 4.66 5.83 4.80 6.17 4.44 5.38 5.02 1.09 3.31 2.72 1.75 3.81 4.45 2.93

(b) 2.15 -3.46 1.12 0.25 -1.42 0.06 -1.16 -2.24 -1.50 0.37 0.66 -0.76 6.24 0.36 -0.40 0.52 -0.97 0.36 1.74 -0.65

(c) 1.79 -0.65 1.16 1.23 2.80 0.92 -2.62 -5.48 0.75 -2.64 -6.41 0.92 1.14 0.18 0.06 -1.49 -3.99 -10.36 7.12 -1.86

(d) -0.53 0.71 1.40 0.28 -0.65 1.02 -0.71 0.70 1.55 -0.52 -0.73 -1.04 -2.39 0.39 5.71 6.39 4.28 6.70 6.05 5.62

(e) -1.61 -1.29 0.59 -0.33 0.14 1.16 2.02 -0.52 0.69 -0.30 -0.56 0.43 -1.01 0.83 -0.95 0.24 0.01 0.10 0.12 0.07

Chapter 6

Some Probability Models

6.1 Bernoulli Experiments

Some random experiments can be viewed as a sequence of identical and independent trials, on each of which one of two possible outcomes occurs. Some examples of random experiments of this kind are

- Recording the number of times the maximum annual wind speed exceeds a certain level v_0 (during a fixed number of years).
- Counting the number of years until v_0 is exceeded for the first time.
- Testing (pass–no–pass) a number of randomly chosen items.
- Polling some randomly (and independently) chosen individuals regarding some yes–no question, for instance, "did you vote in the last provincial election?"

Each trial is called *Bernoulli trial* and a set of independent Bernoulli trials is called *Bernoulli process* or *Bernoulli experiment*. The defining features of a Bernoulli experiment are:

- **The experiment consists of a sequence of trials**
- **The trials are independent**
- **Each trial has only two possible outcomes**.

These outcomes refer to the occurrence or not of a certain event, A. They are arbitrarily called *success* (when A occurs) and *failure* (when A^c occurs) and denoted by

$$\text{S (for success)} \qquad \text{and} \qquad \text{F (for failure)}$$

- **The probability of** S **is the same for all trials**.

This constant probability is denoted by p, that is

$$P(S) = p$$

and so

$$P(F) = 1 - P(S) = 1 - p = q.$$

The number of trials in a Bernoulli experiment can either be fixed or random. For example, if we are considering the number of maximum annual wind speed exceedances of v_0 in the next fifteen years, the number of trials is fixed and equal to 15. On the other hand, if we are considering the number of years until v_0 is first exceeded, the number of trials is random.

6.2 Bernoulli and Binomial Random Variables

Given a Bernoulli experiment of size n (n independent Bernoulli trials), there are n Bernoulli random variables $Y_1, Y_2, \ldots, Y_n$ associated with it. The random variable Y_i ($i = 1, \ldots, n$) depends only on the outcome of the ith trial and is defined as follows

$$\begin{aligned} Y_i &= 1, \quad \text{if the } i^{th} \text{ trial ends in } S \\ &= 0, \quad \text{if the } i^{th} \text{ trial ends in } F. \end{aligned}$$

That is, Y_i is a "counter" for the number of S's in the outcome of the random experiment.

The variables Y_i are very simple. By definition, they are independent and their common density function is

$$f(y) = p^y(1-p)^{1-y}, \quad y = 0, 1.$$

The mean and the variance of Y_i (they are, of course, the same for all $i = 1, \ldots, n,$) are given by,

$$\mathrm{E}(Y_i) = (0)f(0) + (1)f(1) = p,$$

and

$$\mathrm{Var}(Y_i) = (0-p)^2 f(0) + (1-p)^2 f(1) = (-p)^2 q + q^2 p = pq, \quad \text{where } \; q = 1 - p.$$

The student can check that the variance is maximized when $p = q = 0.5$. This result is hardly surprising as the uncertainty is clearly maximized when S and F are equally likely. On the other hand, the uncertainty is clearly smaller for smaller or larger values of p. For example, if $p = 0.01$ we can feel very confident that most of the trials will result in failures. Similarly, if $p = 0.99$ we can confidently predict that most of the trials will result in successes.

Binomial Random Variable ($B(n, p)$).

Given a Bernoulli experiment of fixed size n, the corresponding Binomial random variable X is defined as the total number of S's in the sequence of F's and S's that constitutes the outcome of the experiment. That is,

$$X = \sum_{i=1}^{n} Y_i.$$

Using properties (2) and (4) of the mean and variance of random variables,

$$\mathrm{E}(X) = \mathrm{E}\left(\sum_{i=1}^{n} Y_i\right) = \sum_{i=1}^{n} \mathrm{E}(Y_i) = \sum_{i=1}^{n} p = np.$$

and

$$\mathrm{Var}(X) = \mathrm{Var}(\sum_{i=1}^{n} Y_i) = \sum_{i=1}^{n} \mathrm{Var}(Y_i) = \sum_{i=1}^{n} pq = npq, \quad \text{where} \;\; q = 1 - p.$$

The probability density function of X is

$$f(x) = \binom{n}{x} p^x q^{n-x}, \qquad \text{for all} \quad x = 0, 1, \ldots, n, \tag{6.1}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!} = \frac{[n(n-1)\ldots(2)(1)]}{[x(x-1)\ldots(2)(1)]\;[(n-x)(n-x-1)\ldots(2)(1)]}.$$

For example, if $n = 5$ and $x = 3$ we have

$$\binom{5}{3} = \frac{5!}{3!2!} = \frac{[(5)(4)(3)(2)(1)]}{[(3)(2)(1)][(2)(1)]} = 10$$

To derive the density (6.1) first notice that X takes the value x only if x of the Y_i are equal to one and the remainder are equal to zero. The probability of this event is $p^x q^{n-x}$. In addition, the n variables Y_i can be divided into two groups of x and $n - x$ variables in $\binom{n}{x}$ many different ways.

The distribution function of X doesn't have a simple closed form and can be obtained from Table A5 for a limited set of values of n and p.

Example 6.1 Suppose that the logarithm of the operational life of a machine, T (in hours), has a normal distribution with mean 15 and standard deviation 7. If a plant has 20 of these machines working independently, (a) what is the probability that more than one machine will breakdown before 1500 hours of operation? (b) how many more machines are needed if the expected number of machines that will not break down before 1500 hours of operation must be larger than 18?

Solution The number of machines breaking down before 1500 hours of operation, X, is a binomial random variable with $n = 20$ and

$$\begin{aligned} p &= P(T < 1500) = P(\log(T) < \log(1500)) = \Phi\left[\frac{\log(1500) - 15}{7}\right] \\ &= 1 - \Phi(1.1) = 1 - 0.8643 = 0.14. \end{aligned}$$

(a) First we notice that

$$P(X > 1) = 1 - P[X \leq 1].$$

Since

$$\begin{aligned} P[X \leq 1] &= P(X=0) + P(X=1) = \binom{20}{0}(0.86)^{20} + \binom{20}{1}(0.14)(0.86)^{19} \\ &= 0.04897 + 0.15945 = 0.21, \\ P(X > 1) &= 1 - 0.21 = 0.79 \end{aligned}$$

(b) The expected number of machines in operation (out of n machines) after 1500 hours of operation is $n(1-0.14)$. For this expected value to be larger than 18, n must be larger than $18/0.86 = 20.93$. Therefore, the company needs to acquire one additional machine.

□

6.3 Geometric Distribution and Return Period

The expected value, τ, of the number of trials before the first occurrence of a certain event, A, is called the **return period** of that event. For example, the return period of the event "maximum annual wind speed exceeding v_0" is equal to the expected number of years before v_0 is exceeded for the first time.

The number of trials itself is a discrete random variable, X, with **Geometric** density,

$$f(x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots \tag{6.2}$$

where p is the probability of A and $q = (1-p)$ is the probability of the complementary event, A^c. The distribution function of X has a simple closed form (see Problem 6.6)

$$F(x) = 1 - (1-p)^x, \quad x = 1, 2, \ldots$$

The derivation of (6.2) is fairly straightforward: first of all, it is clear that the range of X is equal to $\{1, 2, \ldots\}$. Furthermore, we can have $X = x$ only if the event A^c occurs during the first $x-1$ trials and A occurs in the xth trial. In other words, we must have a sequence of $x-1$ failures followed by a success. Because of the independence of the trials in a Bernoulli experiment, it is clear that the probability of such a sequence is equal to $p(1-p)^{x-1}$.

To check that $f(x) = pq^{x-1}$ is actually a probability density function we must verify that

$$\sum_0^\infty f(x) = 1.$$

In fact, using the well known formula for the sum of a geometric series with rate $0 < q < 1$,

$$[1 + q + q^2 + \cdots] = 1/(1-q),$$

we obtain

$$\sum_0^\infty f(x) = \sum_1^\infty pq^{x-1} = p[1 + q + q^2 + \cdots] = \frac{p}{1-q} = 1.$$

Finally, the return period, τ, of the event A is given by

$$\begin{aligned}\tau &= \mathrm{E}(X) = p\sum_1^\infty x(1-p)^{x-1} = p\sum_1^\infty\{-\frac{d}{dp}[(1-p)^x]\} \\ &= -p\{\frac{d}{dp}[\sum_1^\infty(1-p)^x]\} = -p\{\frac{d}{dp}(1-p)[1+(1-p)+(1-p)^2+\cdots]\} \\ &= -p\{\frac{d}{dp}\frac{1-p}{p}\} = p\frac{1}{p^2} = \frac{1}{p}.\end{aligned}$$

The return period of A is then inversely proportional to $p = P(A)$. If $p = P(A)$ is small then we must wait, on average, a large number of periods τ until the first occurrence of A. On the other hand, if p is large then we must wait, on average, a small number τ of periods for the first occurrence of A.

The student will be asked to show (see Problem 6.6) that the variance of X is given by

$$\mathrm{Var}(X) = (1-p)/p^2 = \tau(\tau-1).$$

One may well ask the question: why is τ called "return period"? The reason for this becomes clear after we notice that, because of the assumed independence, the expected number of trials before the first occurrence of A is the same as the expected number of trials between any two consecutive occurrences of A.

Example 6.2 Suppose that a structure has been designed for a "25–year rain" (that is, a rain that occurs on average every 25 years).
(a) What is the probability that the design annual rainfall will be exceeded for the first time on the sixth year after completion of the structure?
(b) If the annual rainfall Y (in inches) is normal with mean 55 and variance 16, what is the corresponding design rainfall?

Solution
(a) To say that a certain structure has been designed for a "25–year rain" means that it has been designed for an annual rainfall with return period of 25 years.

The return period, τ, is equal to 25, and therefore the probability of exceeding the design annual rainfall is

$$p = 1/\tau = 1/25 = 0.04.$$

If X represents the number of years until the first time the design annual rainfall is exceeded, then

$$P(X=6) = (0.04)(0.96)^{6-1} = (0.04)(0.96)^5 = 0.033$$

is the required probability.
(b) The design rainfall, v_0, must satisfy the equation

$$P(Y > v_0) = 0.04$$

or equivalently,

$$\Phi\left[\frac{v_0 - 55}{4}\right] = 0.96.$$

From the Standard Normal Table we find that $\Phi(1.75) = 0.96$. Therefore,

$$\frac{v_0 - 55}{4} = 1.75 \quad \Rightarrow \quad v_0 = (4)(1.75) + 55 = 62.$$

□

6.4 Poisson process and associated random variables

Many physical problems of interest to engineers and other applied scientists involve the possible occurrences of an event A at some points in time and/or space. For example:

– earthquakes can occur at any time over a seismically active region

– traffic accidents can occur at any time along a busy highway

– fatigue cracks can occur at any point along a continuous weld

– flaws can occur at any point over a wood panel

– phone calls can arrive at any time at a telephone switchboard

– crashes can occur at any time on a computer network

An important feature of these processes is the expected number of occurrences of the event A per unit of time (or space). This average number of occurrences is represented by λ and called **the rate of the process**. We will see that this parameter determines the main features of the entire process.

To fix ideas, suppose that we are studying the sequence of crashes of a computer network and that we are using a week as the unit of time. In this case λ is the average number of crashes per week.

There are at least two main features of the sequence of occurrences (e.g. crashes) which are of interest: the number of occurrences in an interval of length Δ and the time between consecutive occurrences of A. These features are represented by the random variables X and T below.

- X is the number of occurrences in an interval of length Δ,

and

- T is the time between two consecutive occurrences

The process is called **a Poisson Process** if A is a "rare" event, that is, has the following properties:

1) The number of occurrences of A on non–overlapping intervals are independent.

2) The probability of **exactly one occurrence** of A on any interval of length Δ is approximately equal to $\lambda\Delta$ when Δ is small.

3) The probability of **more than one occurrence** of A on any interval of length Δ is approximately equal to $\lambda\Delta^2$ when Δ is small (that is, A is a "rare" event).

The discrete random variable X described above (number of occurrences on an interval of fixed length Δ) has the so called Poisson density function

$$f(x) = \frac{\exp\{-\lambda\Delta\}(\lambda\Delta)^x}{x!}, \quad x = 0, 1, 2, \ldots$$

and the continuous random variable T (time between consecutive occurrences of A, or inter-arrival time) has the so called exponential density

$$f(t) = \lambda \exp\{-\lambda t\}, \quad t > 0.$$

The derivation of these densities from assumptions 1) 2) and 3) is not very difficult. The interested student can read the heuristic derivation given at the end of this chapter.

Example 6.3 In Southern California there is on average one earthquake per year with Richter magnitude 6.1 or greater (big earthquakes).

(a) What is the probability of having three or more big earthquakes in the next five years?

(b) What is the most likely number of big earthquakes in the next 15 months?

(c) What is the probability of having a period of 15 months without a big earthquake?

(d) What is the probability of having to wait more than three and a half years until the occurrence of the next four big earthquakes?

Solution We assume that the sequence of big earthquakes follows a Poisson process with (average) rate $\lambda = 1$ per year.

(a) The number X of big earthquakes in the next five years is a Poisson random variable with average rate 5 and so, using the Poisson Table we get

$$P(X \geq 3) = 1 - P(X < 3) = 1 - F(2) = 1 - 0.125 = 0.875.$$

(b) In general, a Poisson density $f(x)$ with parameter δ is increasing at x ($x \geq 1$) if and only if the ratio $f(x)/f(x-1) > 1$. Since

$$\frac{f(x)}{f(x-1)} = \frac{\exp\{-\delta\}\delta^x}{x!} \div \frac{\exp\{-\delta\}\delta^{(x-1)}}{(x-1)!} = \frac{\delta}{x}.$$

it follows that

$f(x) > f(x-1)$ when $x < \delta$

$f(x) = f(x-1)$ when $x = \delta$

$f(x) < f(x-1)$ when $x > \delta$

Therefore, the largest value of $f(x)$ is achieved when $x = [\delta]$, where

$$[\delta] = \text{integer part of } \delta$$

So, the most likely number of big earthquakes is $[1.25] = 1$ (notice that 15 months = 1.25 years).

(c) The waiting time T to the next big earthquake is an exponential random variable with rate $\lambda = 1$ year, with distribution function

$$F(t) = 1 - \exp\{-t\}.$$

Therefore,

$$P\{T > 1.25\} = 1 - F(1.25) = 1 - [1 - \exp\{-1.25\}] = 0.287.$$

(d) Let Y represent the number of big earthquakes in the next three and a half years and let W represent the waiting time (in years) until the occurrence of the next four big earthquakes.

We notice that Y is a Poisson random variable with rate 3.5 and that W is larger than 3.5 years if and only if Y is less than 4. So,

$$P[W > 3.5] = P[Y < 4] = F(3) = 0.5366.$$

□

Means and Variances The means of X and T are of practical interest, as they represent the expected number of occurrences on a period of length Δ and the expected waiting time between consecutive occurrences, respectively. We will see that, not surprisingly,

$$\mathrm{E}(X) = \lambda\Delta \qquad \text{and} \qquad \mathrm{E}(T) = \frac{1}{\lambda}.$$

We will also see that

$$\mathrm{Var}(X) = \lambda\Delta \qquad \text{and} \qquad \mathrm{Var}(T) = \frac{1}{\lambda^2}.$$

First, let's calculate $\mathrm{E}(X)$.

$$\begin{aligned}\mathrm{E}(X) &= \sum_{x=0}^{\infty} x f(x) = \sum_{x=0}^{\infty} x \frac{\exp\{-\lambda\Delta\}(\lambda\Delta)^x}{x!} = \sum_{x=1}^{\infty} x \frac{\exp\{-\lambda\Delta\}(\lambda\Delta)^x}{x!} \\ &= \exp\{-\lambda\Delta\}(\lambda\Delta) \sum_{x=1}^{\infty} \frac{(\lambda\Delta)^{x-1}}{(x-1)!} = \exp\{-\lambda\Delta\}(\lambda\Delta)\exp\{\lambda\Delta\} = \lambda\Delta.\end{aligned}$$

Analogously, it follows that

$$\begin{aligned}\mathrm{E}[X(X-1)] &= \sum_{x=0}^{\infty} x(x-1) f(x) = \sum_{x=0}^{\infty} x(x-1) \frac{\exp\{-\lambda\Delta\}(\lambda\Delta)^x}{x!} \\ &= \sum_{x=2}^{\infty} x(x-1) \frac{\exp\{-\lambda\Delta\}(\lambda\Delta)^x}{x!} = \exp\{-\lambda\Delta\}(\lambda\Delta)^2 \sum_{x=2}^{\infty} \frac{(\lambda\Delta)^{x-2}}{(x-2)!} \\ &= \exp\{-\lambda\Delta\}(\lambda\Delta)^2 \exp\{\lambda\Delta\} = (\lambda\Delta)^2.\end{aligned}$$

Therefore,

$$\mathrm{E}(X^2) = \mathrm{E}[X(X-1)] + \mathrm{E}(X) = (\lambda\Delta)^2 + \lambda\Delta,$$

and

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 = (\lambda\Delta)^2 + \lambda\Delta - (\lambda\Delta)^2 = \lambda\Delta.$$

To calculate $\mathrm{E}(T)$, we use integration–by–part with

$$u = t \quad \text{and} \quad dv = \exp\{-\lambda t\}dt,$$

to get

$$\mathrm{E}(T) = \int_0^{\infty} t f(t) dt = \int_0^{\infty} t\lambda \exp\{-\lambda t\} dt = \int_0^{\infty} \exp\{-\lambda t\} dt = 1/\lambda.$$

To calculate $\mathrm{E}(T^2)$, we use integration–by–part with

$$u = t^2 \quad \text{and} \quad dv = \exp\{-\lambda t\}dt,$$

to get

$$\begin{aligned} \mathrm{E}(T^2) &= \int_0^\infty t^2 f(t)dt = \int_0^\infty t^2 \lambda \exp\{-\lambda t\}dt = 2\int_0^\infty t \exp\{-\lambda t\}dt \\ &= (2/\lambda)[\int_0^\infty t\lambda \exp\{-\lambda t\}dt] = (2/\lambda)(1/\lambda) = (2/\lambda^2). \end{aligned}$$

Finally,

$$\mathrm{Var}(T) = \mathrm{E}(T^2) - [\mathrm{E}(T)]^2 = (2/\lambda^2) - (1/\lambda^2) = (1/\lambda^2).$$

Example 6.3 (continued):

(e) What is the expected number of big earthquakes in the next five years? Fifteen months? What are the corresponding standard deviations?

(f) What is the expected waiting time (in years) between two consecutive big earthquakes?

(g) What is the expected waiting time (in years) until the 25th big earthquake? The standard deviation?

(h) What is the approximate probability that the waiting time until the 25th big earthquake will exceed 27 years? This question will be answered in the next chapter.

Solution

(e) Since $X =$ "number of big earthquakes in the next five years" is Poisson(5), we have that $\mathrm{E}(X) = 5$ and $\mathrm{SD}(X) = \sqrt{5} = 2.24$. In the case of fifteen months (1.25 years) the mean is 1.25 and the standard deviation is $\sqrt{1.25} = 1.12$.

(f) Since $T =$ "waiting time (in years) between two consecutive big earthquakes" is an exponential random variable with rate $\lambda = 1$, its expected value ($\mathrm{E}(T) = 1/\lambda$) and standard deviation ($\mathrm{Var}(T) = 1/\lambda^2$) are both equal to one.

(g) Let

$$W = \text{"waiting time (in years) until the } 25^{th} \text{ big earthquake"}$$

and let

$T_i =$ "waiting time (in years) between the $(i-1)^{th}$ and the ith big earthquakes", $\quad i = 1, 2, \ldots, 25.$

Notice that

$$W = \sum_{i=1}^{25} T_i$$

where, because of the Poisson process assumptions,

$$T_1, \quad T_2, \quad \ldots, \quad T_{25} \qquad \text{are iid Exp(1)},$$

where Exp(λ) means "exponential distribution with parameter (rate) λ". Therefore,

$$\mathrm{E}(W) = \mathrm{E}\sum_{i=1}^{25} T_i = \sum_{i=1}^{25} \mathrm{E}T_i = \sum_{i=1}^{25} 1 = 25.$$

$$\mathrm{Var}(W) = \mathrm{Var}(\sum_{i=1}^{25} T_i) = \sum_{i=1}^{25} \mathrm{Var}(T_i) = \sum_{i=1}^{25} 1 = 25.$$

and,

$$\mathrm{SD}(W) = 5.$$

□

6.5 Poisson Approximation to the Binomial

Let $X \sim B(n, p)$ be a binomial random variable with parameters n and p. If n is large ($n \geq 20$) and p is small ($np < 5$) then we can use a Poisson random variable with rate $\lambda = np$, $Y \sim \mathcal{P}(np)$, to approximate the probabilistic behavior of X. In other words, we can use the approximation

$$P\left[B(n, p] = x\right) \approx P\left[\mathcal{P}(np) = x\right] = \exp\{-np\}(np)^x/x!, \qquad \text{for all} \quad x = 0, \ldots, n.$$

Example 17: On average, one per cent of the 50-kg dry concrete bags are underfilled below 49.5 kg. What is the probability of finding 4 or more of these underfilled bags in a lot of 200?

Solution: Since $n = 200$ and $p = 0.01$,

$$\min\{np, n(1-p)\} = \min\{2, 198\} = 2 < 5.$$

Since n is large and $np = 2$ is small, we can use the Poisson approximation

$$\begin{aligned} P[B(200, 0.01) \geq 4] &\approx P[\mathcal{P}(2) \geq 4] = 1 - P[\mathcal{P}(2) < 4] \\ &= 1 - F(3) = 1 - 0.857 = 0.143 \quad \text{from the Poisson table.} \end{aligned}$$

6.6 Heuristic Derivation of the Poisson and Exponential Distributions

Let m be some fixed integer number. If Y_i is the number of occurrences of the event A in the interval $\left(\frac{i-1}{m}, im\right)$, then the total number of occurrences, X, in the interval $(0, 1]$ (we are taking $\Delta = 1$ for simplicity) can be written as

$$X = Y_1 + Y_2 + \ldots + Y_m.$$

Because of the independence of the number of occurrences on non-overlapping intervals, the variables $Y_1, Y_2, \ldots, Y_m$ are independent. Moreover, because of the assumption that A is a rare event, the probability that the variables Y_i will take values other than zero and one is nearly zero,

$$P(Y_i > 1) \approx 0, \quad \text{when } m \text{ is large,}$$

and so the variables Y_i are approximately Bernoulli random variables when m is large.

By the above remarks, the random variable X is approximately Binomial, $B(m, \lambda/m)$, when m is large. Of course, the larger m, the better the approximation, and in the limit (when $m \to \infty$) the approximation becomes exact. Therefore, the probability that X will take any fixed value x can be obtained from the limit, as $m \to \infty$, of the binomial expression

$$P(X = x) = \binom{n}{x}[\lambda/m]^x[1 - \lambda/m]^{m-x}$$

Since, as $m \to \infty$, we have

$$\binom{n}{x}/m^x = \frac{m}{m}\frac{m-1}{m}\frac{m-2}{m}\ldots\frac{m-x+1}{m} \to 1,$$

$$[1 - \lambda/m]^{-x} \to 1,$$

and

$$[1 - \lambda/m]^m \to \exp\{-\lambda\},$$

we obtain that, as $m \to \infty$,

$$\begin{aligned}\binom{n}{x}[\lambda/m]^x[1 - \lambda/m]^{m-x} &= \frac{m}{m}\frac{m-1}{m}\frac{m-2}{m}\ldots\frac{m-x+1}{m}[1 - \lambda/m]^{-x}[1 - \lambda/m]^m\lambda^x/x! \\ &\to \exp\{-\lambda\}\lambda^x/x!, \quad \text{the Poisson density function.}\end{aligned}$$

In particular, this justifies the $\mathcal{P}(np)$ approximation to the $B(n,p)$, when n is large and p is small. The requirement that n is large corresponds to m being large and the requirement that p is small corresponds to λ/m being small.

To derive the Exponential density of T, we reason as follows: The waiting time T until the first occurrence of A will be larger than t if and only if the number of occurrences X in the period $(0,t)$ is equal to zero. Since $X \sim \mathcal{P}(\lambda t)$,

$$\begin{aligned} P(T \leq t) &= 1 - P(T > t) = 1 - P(X = 0) = 1 - [\exp\{-\lambda t\}(\lambda t)^0/0!] \\ &= 1 - \exp\{-\lambda t\}, \quad \text{the exponential distribution with parameter } \lambda. \end{aligned}$$

6.7 Exercises

6.7.1 Exercise Set A

Problem 6.1 A weighted coin is flipped 200 times. Assume that the probability of a head is 0.3 and the probability of a tail is 0.7. Each flip is independent from the other flips. Let X be the total number of heads in the 200 flips.
(a) What is the distribution of X?
(b) What is the expected value of X and variance of X?
(c) What is the probability that X equals 35?
(d) What is the approximate probability that X is less than 45?
Note: Come back to this question after you learned about normal approximations in the next chapter.

Problem 6.2 Suppose it is known that a treatment is successful in curing a muscular pain in 50% of the cases. If it is tried on 15 patients, find the probabilities that:
(a) At most 6 will be cured.
(b) The number cured will be no fewer than 6 and no more than 10.
(c) Twelve or more will be cured.
(d) Calculate the mean and the standard deviation.

Problem 6.3 The office of a particular U.S. Senator has on average five incoming calls per minute. Use the Poisson distribution to find the probabilities that there will be:
(a) exactly two incoming calls during any given minute;
(b) three or more incoming calls during any given minute;
(c) no incoming calls during any given minute.
(d) What is the expected number of calls during any given period of five minutes?

Problem 6.4 A die is colored blue on 5 of its sides, and green on the other 1 side. This die is rolled 8 times. Assume each roll of the die is independent from the other rolls. Let X be the number of times blue comes up n the 8 rolls of the die.
(a) What is the expected value of X and the variance of X?
(b) What is the probability that X equals 6?
(c) What is the probability that X is greater than 6?

Problem 6.5 A factory produced $10,000$ light bulbs in February, in which there are 500 defectives. Suppose 20 bulbs are randomly inspected. Let X denote the number of defectives in the sample.
(a) Calculate $P(X = 2)$.
(b) If the sample size, i.e., the number of the inspected bulbs, is large, how would you calculate $P(X \geq 2)$ approximately? For $n = 200$, calculate this probability approximately.

6.7.2 Exercise Set B

Problem 6.6 Let X be a random variable with geometric density (6.2). Show that

(a) $F(x) = 1 - P(X > x) = 1 - (1-p)^x$.
(b) $\mathrm{E}[X(X-1)] = 2(1-p)/p^2$, and therefore $\mathrm{E}(X^2) = (2-p)/p^2$.
(c) $\mathrm{Var}(X) = (1-p)/p^2 = \tau(\tau-1)$.

Problem 6.7 The Statistical Tutorial Center has been designed to handle a maximum of 25 students per day. Suppose that the number X of students visiting this center each day is a normal random variable with mean 15 and variance 16.
(a) What is the return period τ for this center?
(b) What is the probability that the "design" number of visits will not be exceeded before the 10th day?

Problem 6.8 A transmission tower has been designed for a 30–year wind.
(a) What is the probability that the design maximum annual wind velocity will be exceeded for the first time on the 7th year after completion of the project?
(b) What is the probability that the design maximum annual wind velocity will be exceeded during the first 7 years after completion of the project?
(c) If the maximum annual wind velocity (in miles per hour) is an exponential random variable with mean 35, what is the design maximum annual wind velocity?
(d) What is the return period if the design maximum annual wind velocity is decreased by 15%?

Problem 6.9 (a) Let X_1 and X_2 be two Binomial random variables with $n = 14$ and $p = 0.30$. Calculate
(i) $P(X_1 = 4)$, $P(X_1 < 6)$ and $P(2 < X_1 < 6)$ (use the Binomial table)
(ii) $\mathrm{E}(X_1)$, $\mathrm{SD}(X_1)$, $\mathrm{E}(X_1 + X_2)$, $\mathrm{SD}(X_1 + X_2)$, $\mathrm{E}(X_1X_2)$ and $\mathrm{SD}(X_1X_2)$
(iii) $P(X_1 + X_2 = 8)$, $P(X_1 + X_2 < 12)$ and $P(4 < X_1 + X_2 < 12)$.

Problem 6.10 The arrival of customers to a service station is well approximated by a Poisson Process with rate $\lambda = 5$ per hour.
(a) What is the expected number of customers per day? (the service station is open eight hours per day)
(b) What is the most likely number of customers in any given hour?
(c) What is the probability that more than seven customers will arrive in the next hour?
(d) What is the probability that the waiting time between two consecutive arrivals will be 25 minutes or more?
(e) What is the expected time until the arrival of the next 25 customers? The standard deviation?

Problem 6.11 A bag contains 4 red balls and 6 white balls. One ball was drawn with equal probability and replaced in the bag before next draw was made. Let X be the number of red balls out of 100 draws from the bag.
(a) Give a general expression for $P(X = k)$, $k = 0, 1, ..., 100$;
(b) Calculate the mean and variance of X;
(c) Calculate the probability $P(X \leq 38)$.

Problem 6.12 The number of killer whales arriving at the Pacific Rim Observatory Station follows a Poisson Process with rate $\lambda = 4$ per hour.
(a) What is the expected number and variance during the next hour?
(b) What is the probability that the waiting time T between two consecutive arrivals will be 30 minutes or more?
(c) What is the expected time and variance until the next 20 killer whales arriving at the Observatory Station.

Problem 6.13 Car accidents are random and can be said to follow a Poisson distribution. At a certain intersection in East Vancouver there are, on average, 4 accidents a week. Answer the following questions:
(a) What is the probability of there being no accidents at this intersection next week?
(b) The record for accidents in one month at a single intersection is 20. Find the probability that this record will be broken, at this intersection, next month. (Assume 30 days in one month)
(c) What is the expected waiting time for 20 accidents to occur?

Problem 6.14 A test consists of ten multiple-choice questions with five possible answers. For each question, there is only one correct answer out of five possible answers. If a student randomly chooses one answer at each question, calculate the following probabilities that
(a) at most three questions are answered correctly?
(b) five questions are answered correctly?
(c) all questions are answered correctly?
And (d) calculate the mean and the standard deviation of number of correct answers.

Problem 6.15 The number of meteorites hitting Mars follows a Poisson process with parameter $\lambda = 6$ per month.
(a) What is the probability that at least 2 meteorites hit Mars in any given month?
(b) Find the probability that exactly 10 meteorites hit Mars in the next 6 months.
(c) What is the expected number of meteorites hitting the Mars in the next year?

Problem 6.16 A biased coin is flipped 10 times independently. The probability of tails is 0.4. Let X be the total number of heads in the 10 flips.
(a) Use a computer to find $P(X = 4)$;
(b) Use the Binomial table to find $P(1 < X < 5)$;
(c) What is the probability that one has to flip at least 5 times to get the first head?

Problem 6.17 Three identical fair coins are tossed simultaneously until all three show the same face.
(a) What is the probability that they are tossed more than three times?
(b) Find the mean for the number of tosses.

Chapter 7

Normal Probability Approximations

7.1 Central Limit Theorem

By Fact 5 in Chapter 4, if $X_1, X_2, \ldots, X_n$ is a sample from a normal population with mean μ and standard deviation σ then

$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} \quad \sim \quad N(\mu, \frac{\sigma^2}{n}).$$

Often, however, one has to deal with non–normal samples. For example, the population variable, X, may represent the lifetime of a part and X_i may represent the lifetime of the ith randomly chosen part. Since the lifetime of a part cannot be negative, X cannot be normal. A more reasonable assumption may be that X is exponentially distributed ($X \sim \mathcal{E}(\lambda)$) with unknown parameter λ. Or more generally, one may simply assume that X is positive with mean $\mu = 1/\lambda$, and variance $1/\lambda^2$. The sample, $X_1, X_2, \ldots, X_n$, would then be typically obtained in order to estimate the expected life, μ, for the parts.

Let $X_1, X_2, \ldots, X_n$ be a sample from an arbitrary population with mean μ and variance σ^2. A very important result, called the **Central Limit Theorem** (CLT), states that, when n is large, $\overline{X}$ is **approximately normal**, with mean μ and variance σ^2/n, regardless of the actual shape of the population distribution. This remarkable result will be extensively used throughout this course.

The CLT is a limit (asymptotic) result, and the distribution of the average is not exactly normal for any finite value of n. An obvious question at his point is **when should n be considered large enough** for practical applications? Unfortunately, the size of n for which the normal approximation is good depends on the distribution of the variables X_i being averaged. If this distribution is symmetric and has light tails then the CLT approximation may be quite good for small values of n (n equal to five or six). If the distribution of the $X_i's$ is very asymmetric, then it will take longer for the CLT approximation to provide a reasonable approximation. In many practical situations, the CLT normal approximation can be used when $n \geq 20$.

Example 7.1 A system consists of 25 independent parts connected in such a way that the ith part automatically turns–on when the $(i-1)^{th}$ part burns out. The expected lifetime of each part is 10 weeks and the standard deviation is equal to 4 weeks. (a) Calculate the expected lifetime and standard deviation for the the system. (b) Calculate the probability that the system will last more than its expected life. (c) Calculate the probability that the system will last more than 1.1 times its expected life. (d) What are the (approximate) median life and interquartile range for the system?

Solution

(a) Let X_i denote the lifetime of the ith component and let

$$T = \sum_{i=1}^{25} X_i,$$

denote the lifetime of the system. Then,

$$\mathrm{E}(T) = \sum_{i=1}^{25} \mathrm{E}(X_i) = 25 \times 10 = 250 \text{ weeks},$$

and, using the assumption of independence,

$$var(T) = \sum_{i=1}^{25} \mathrm{Var}(X_i) = 25 \times 16 = 400.$$

Therefore,

$$\mathrm{SD}(T) = \sqrt{400} = 20 \text{ weeks},$$

Notice that the mean of T is 25 times larger than that of each X_i while the standard deviation of T is only $\sqrt{25} = 5$ times larger.

(b) First observe that

$$\frac{T}{25} = \overline{X} \quad \simeq \quad N(10, \frac{16}{25}),$$

where the symbol $\simeq$ means "approximately distributed as", and so

$$T \quad \simeq \quad 25\ N(10, \frac{16}{25}) \ = \ N(250, 400) = N(\mathrm{E}(T), \mathrm{Var}(T)).$$

Therefore,

$$P(T > \mathrm{E}(T)) = P(T > 250) \approx 1 - \Phi\left[\frac{250 - 250}{20}\right] = 0.5.$$

(c) First of all notice that $1.1 \times \mathrm{E}(T) = 1.1 \times 250 = 275$. Now, by the discussion in (b),

$$P(T > 275) \approx 1 - \Phi\left[\frac{275 - 250}{20}\right] = 1 - \Phi(1.25) = 0.1056.$$

(d) Let Z denote the standard normal random variable. Using that $T \simeq N(250, 400)$, it follows that

$$Q_1(T) \approx Q_1(N(250, 400)) = 250 + (20 \times Q_1(Z)) = 250 - (20 \times 0.675) = 236.5.$$

Analogously,

$$Q_2(T) = \text{Median}(T) = 250 + (20 \times Q_2(Z)) = 250,$$

and

$$Q_3(T) \approx Q_3(N(250, 400)) = 250 + (20 \times Q_3(Z)) = 250 + (20 \times 0.675) = 263.5.$$

Therefore,

$$\text{IQR}(T) \approx Q_3(T) - Q_1(T) = 263.5 - 236.5 = 27.0$$

□

Table 7.1

Rainfall Intensity (in.)	Midpoint	Frequency
38–42	40	15
42–46	44	34
46–50	48	26
50–54	52	23
54–58	56	17
58–62	60	16
62–66	64	4
66–70	68	10
Total		145

Example 7.2 Consider Table 7.1 with data on the annual (cumulative) rainfall intensity (X) on a certain watershed area. The average annual rainfall intensity can be calculated from Table 7.1 as:

$$\begin{aligned} \overline{X} &= \frac{(40)(15) + (44)(34) + \ldots + (68)(10)}{15 + 34 + \ldots + 10} \\ &= \frac{7388}{145} = 50.952. \end{aligned}$$

Since the average has been calculated from a frequency table, using the midpoint of each class to represent all the points in each class, there is an approximation error to be considered. How likely is that this approximation error is (a) larger than 0.05? (b) larger than 0.10? (c) larger than 0.5?

Solution

To make the required probability calculations we will assume that the rainfall intensities are uniformly distributed on each interval. This is a reasonable assumption given that we do not have any additional information on the distribution of values on each class.

Let r_i represent the actual annual rainfall intensity ($i = 1, 2, \ldots, 145$) and let m_i be the midpoint of the corresponding class. For instance, if $r_5 = 50.35$ (a value in the class 50–54), then $m_i = 52.0$. Let

$$U_i = r_i - m_i, \quad i = 1, 2, \ldots, 145.$$

Given our "uniformity" assumption, the $U_i's$ are uniform random variables on the interval $(-2, 2)$.

To proceed with our calculation, we will assume that the variables $U_i's$ (which represent the approximation errors) are independent.

Let

$$\overline{r} = \frac{r_1 + r_2 + \ldots + r_{145}}{145}.$$

The approximation error, D, in the calculation of $\overline{X}$ can now be written as

$$D = \emptyset\overline{r} - verlineX = \frac{r_1 + r_2 + \cdots + r_{145}}{145} - \frac{m_1 + m_2 + \ldots + m_{145}}{145} = \frac{U_1 + U_2 + \ldots + U_{145}}{145}$$

Since D is the average of 145 independent, identically distributed random variables with zero mean and variance equal to

$$\sigma^2 = \frac{1}{4}\int_{-2}^{2} t^2 dt = \frac{4}{3},$$

we can use the (CLT) normal approximation. That is, we can use a normal distribution with zero mean and variance equal to $(4/3)/145$, to approximate the distribution of D. The corresponding standard deviation is $\sqrt{4/435} = 0.095893$.

(a)

$$P(|D| > 0.05) = P(|D|/0.095893 > 0.05/0.095893) \approx 2[1 - \Phi(0.52)] = 0.6084.$$

(b)

$$P(|D| > 0.1) = P(|D|/0.095893 > 0.1/0.095893) \approx 2[1 - \Phi(1.04)] = 0.2984.$$

(c)

$$P(|D| > 0.5) = P(|D|/0.095893 > 0.5/0.095893) \approx 2[1 - \Phi(5.21)] = 0.$$

□

Example 6.3 (continued from Chapter 6):

Recall part (h) of Example 6.3 from the previous chapter which was left unanswered

(h) What is the approximate probability that the waiting time until the 25th big earthquake will exceed 27 years?

Solution
(h) Since W is a sum of iid random variables, we can use the Central Limit Theorem to approximate $P(W > 27)$. Since $\mathrm{E}(W) = 25$ and $\mathrm{SD}(W) = 5$ we have

$$P(W > 27) = 1 - P(W \leq 27) = 1 - \Phi\left[\frac{27 - 25}{5}\right] = 1 - \Phi[0.40] = 1 - 0.6554 = 0.3446.$$

□

7.2 Normal Approximation to the Binomial Distribution

Let X be a binomial random variable with parameters n and p. When n is large so that

$$\min\{np, n(1-p)\} \geq 5,$$

we can use the following approximation:

$$P(X = k) = P[k - 0.5 < X < k + 0.5] = \Phi\left[\frac{k - np + 0.5}{\sqrt{npq}}\right] - \Phi\left[\frac{k - np - 0.5}{\sqrt{npq}}\right]. \quad (7.1)$$

The justification for the approximation above is given by the Central Limit Theorem. In fact, we have seen before that

$$X = Y_1 + Y_2 + \cdots + Y_n$$

where $Y_1, Y_2, \ldots, Y_n$ are independent Bernoulli random variables with parameter p. Therefore,

$$\frac{X}{n} = \frac{Y_1 + Y_2 + \cdots + Y_n}{n} = \overline{Y}$$

which is approximately $N(p, pq/n)$ when n is large. Therefore,

$$X = n\overline{Y}$$

is approximately distributed as $N(p, pq/n)$ multiplied by n, that is, $N(np, npq)$. The *continuity correction* 0.5 which is added and subtracted to k is needed because we are approximating a discrete random variable with a continuous random variable.

For example, if $n = 15$ and $p = 0.4$, then

$$\min\{np, n(1-p)\} = \min\{6, 9\} = 6 \geq 5,$$

$$np = 6 \qquad \sqrt{npq} = 1.9$$

and

$$\begin{aligned} P(X = 8) &= \Phi\left[\frac{8 - 6 + 0.5}{1.9}\right] - \Phi\left[\frac{8 - 6 - 0.5}{1.9}\right] \\ &= \Phi(1.32) - \Phi(0.79) = 0.9065825 - 0.7852361 = 0.1213. \end{aligned}$$

Using the Binomial Table on the Appendix we have that the exact probability is equal to

$$P(X = 8) = F(8) - F(7) = 0.9050 - 0.7869 = 0.1181.$$

Therefore, the approximation error is equal to 0.0032.

The student can verify, as an exercise, the entries in Table 6.2, where $P(X = k)$ is being approximated using formula (7.1).

Table 7.2

k	Approximated	Exact	Error
0	0.0016	0.0005	0.0011
1	0.0070	0.0047	0.0023
2	0.0240	0.0219	0.0021
3	0.0605	0.0634	-0.0029
4	0.1213	0.1268	-0.0055
5	0.1827	0.1859	-0.0032
6	0.2051	0.2066	-0.0015
7	0.1827	0.1771	0.0056
8	0.1213	0.1181	0.0032
9	0.0605	0.0612	-0.0007
10	0.0240	0.0245	-0.0005
11	0.0070	0.0074	-0.0004
12	0.0016	0.0016	0.0000
13	0.0003	0.0003	0.0000
14	0.0000	0.0000	0.0000
15	0.0000	0.0000	0.0000

7.3 Normal Approximation to the Poisson Distribution

The Central Limit Theorem can also be used to approximate Poisson probabilities when the expected number of counts, α, is large. As a rule of thumb, we will use this approximation when $\alpha \geq 20$.

The Poisson random variable, $X \simeq \mathcal{P}(\alpha)$, is approximated by the normal random variable, $N(\alpha, \alpha)$, with the same mean and variance. In other words,

$$P(X = x) \approx \Phi\left[\frac{x + .5 - \alpha}{\sqrt{\alpha}}\right] - \Phi\left[\frac{x - .5 - \alpha}{\sqrt{\alpha}}\right].$$

provided that $\alpha \geq 20$. The *continuity correction* 0.5 added and subtracted to x is needed because we are approximating a discrete random variable with a continuous random variable.

This approximation is justified by the following argument: consider a Poisson process with rate $\lambda = 1$, and suppose that X represents the number of occurrences in a period of length α. We can divide α into n subintervals of length α/n and denote by Y_i the number of occurrences in the ith subinterval. It is clear that $Y_1, \ldots, Y_n$ are independent Poisson random variables with mean α/n and that

$$X = Y_1 + Y_2 + \cdots + Y_n = n\overline{Y}.$$

Therefore, by the CLT,

$$X = n\overline{Y} \simeq nN(\alpha/n, \alpha/n^2) = N(\alpha, \alpha).$$

Intuitively, the requirement that α is large is necessary because one needs to represent X as the sum of a a large number, n, of independent random variables, Y_i, and the common distribution of these random variables becomes very asymmetric when α/n is very small.

As an example, let $X \simeq \mathcal{P}(25)$ and calculate (a) $P(X = 27)$, (b) $P(X > 27)$ and (c) $P(24 \leq X < 27)$. In the case of (a),

$$P(X = 27) \approx \Phi\left[\frac{27 + .5 - 25}{\sqrt{25}}\right] - \Phi\left[\frac{27 - .5 - 25}{\sqrt{25}}\right] = \Phi(0.5) - \Phi(0.3) = 0.6915 - 0.6179 = 0.0736.$$

The exact probability is $\exp -25 \times 25^{27}/(27!) = 0.07080$. In the case of (b),

$$P(X > 27) = 1 - P(X \leq 27) \approx 1 - \Phi((27.5 - 25)/5) = 1 - \Phi(0.5) = 1 - 0.6915 = 0.3085.$$

The exact probability in this case is 0.2998. Finally, in the case of (c),

$$P(24 \leq X < 27) \approx \Phi((26.5 - 25)/5) - \Phi((23.5 - 25)/5) = \Phi(0.3) - \Phi(-0.3) = 2\Phi(0.3) - 1 = 0.2358.$$

The corresponding exact probability is 0.2355.

7.4 Exercises

7.4.1 Exercise Set A

Problem 7.1 Two types of wood (Elm and Pine) are tested for breaking strength. Elm wood has an expected breaking strength of 56 and a standard deviation of 4. Pine wood has an expected breaking strength of 72 and a standard deviation of 8. Let $\bar{X}$ be the sample average breaking strength of an Elm sample of size 30, and $\bar{Y}$ be the sample average breaking strength of a Pine sample of size 40.
(a) What is the approximate distribution of $\bar{X}$?
(b) What is the approximate distribution of $\bar{Y}$?
(c) Calculate (approximately) $P(\bar{X} + \bar{Y} < 110)$.

Problem 7.2 Consider a population with mean 82 and standard deviation 12.
(a) If a random sample of size 64 is selected, what is the probability that the sample mean will lie between 80.8 and 83.2?
(b) With a random sample of size 100, what is the probability that the sample mean will lie between 80.8 and 83.2?
(c) What assumption(s) have you used in (a) and (b)?

Problem 7.3 Suppose that the population distribution of the gripping strengths of industrial workers is known to have a mean of 110 and a standard deviation of 10. For a random sample of 75 workers, what is the probability that the sample mean gripping strength will

be:
(a) between 109 and 112?
(b) greater than 111?
(c) What assumption(s) have you made?

Problem 7.4 The expected amount of sulfur in the daily emission from a power plant is 134 pounds with a standard deviation of 22 pounds. For a random sample of 40 days, find the approximate probability that the total amount of sulfur emissions will exceed $5,600$ pounds.

Problem 7.5 Suppose we draw two samples of equal size n from a population with unknown mean μ but a known standard deviation 3.5, respectively. Let $\bar{X}$ and $\bar{Y}$ be the corresponding sample averages. How large would the sample size n be required to be to ensure that $P(-1 \leq \bar{X} - \bar{Y} \leq 1) = 0.90$?

Problem 7.6 Suppose $X_1, \ldots, X_{30}$ are independent and identically distributed random variables with mean $\mathrm{E}X_1 = 10$ and variance $\mathrm{Var}(X_1) = 5$.
(a) Calculate the mean of $\bar{X} = \frac{1}{30}\sum_{i=1}^{30} X_i$ and the standard deviation of $X_1 - X_2$.
(b) Calculate the interquartile range of $\bar{X}$ approximately.

7.4.2 Exercise Set B

Problem 7.7 Show that if U has uniform distribution on the interval $(0, 1)$ and F is any given continuous distribution function, then $X = F^{-1}(U)$ has distribution F. This result can be used to generate random variables with any given distribution.

Problem 7.8 (a) Generate $m = 100$ samples of size $n = 10$, of independent random variables with uniform distribution on the interval $(0, 1)$. Let X_{ij} denote the jth element of the ith sample ($i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$).

Construct the histogram and $Q - Q$ plot for the sample means

$$\overline{X}_i = \frac{1}{n}\sum_{i=1}^{n} X_{ij}.$$

(b) Same as (a) but with $n = 20$ and $n = 40$. What are your conclusions?

(c) Repeat (a) and (b) but with the X_{ij} having density

$$\begin{aligned} f(x) &= \frac{1}{18}(x-4) \qquad 4 < x < 10 \\ &= 0 \qquad\qquad\qquad\quad \text{otherwise} \end{aligned}$$

What are your conclusions?

Hint: See Problem 7.7.

Problem 7.9 Solve part (a) of Problem 7.8 but with $p = 0.7$, instead of 0.3.

Problem 7.10 Referring to Problem 6.10, find the probability that more than 800 customers will come during the next 20 business days?

Problem 7.11 The expected tensile strength of two types of steel (types A and B, say) are 106 ksi and 104 ksi. The respective standard deviations are 8 ksi and 6 ksi. Let $\overline{X}$ and $\overline{Y}$ be the sample average tensile strengths of two samples of 40 specimens of type A and 35 specimens of type B, respectively.

(a) What is the approximate distribution of $\overline{X}$? Of $\overline{Y}$?

(b) What is the approximate distribution of $\overline{X} - \overline{Y}$? Why?

(c) Calculate (approximately) $P[|\overline{X} - \overline{Y}| < 1]$.

(d) Suppose that after completing all the sample measurements you find $\overline{x} - \overline{y} = 6$. What do you think now of the "population" assumptions made at the beginning of this problem? Why?

Problem 7.12 (a) There are 75 defectives in a lot of 1500. Twenty five items are randomly inspected (the inspection is non-destructive and the items are returned to the lot immediately after inspection). If two or more items are defective the lot is returned to the supplier (at the supplier's expense). Otherwise, the lot is accepted. What is the probability that the lot will be rejected?
(b) Suppose that the actual number of defectives is unknown and that five out of twenty five independently inspected items turned out to be defectives. Estimate the total number of defectives in the lot (of 1500 items). What is the expected value and standard deviation of your estimate? What is the (approximated) probability that your estimate is within a distance of 10 from the actual total number of defectives?

Problem 7.13 A sequence of n independent pH determinations of a chemical compound will be made. Each determination can be viewed as a random variable, X_i, with mean μ (the unknown "true" pH of the compound) and standard deviation $\sigma = 0.15$. How many independent determinations are required if we wish that the sample average $\overline{X}$ is within 0.01 of the true pH with probability 0.95? What is the necessary n if $\sigma = 0.30$?

Problem 7.14 Bits are independently received in a digital communication channel. The probability that a received bit is in error is 0.00001.
(a) If 16 million bits are transmitted, calculate the (approximate) probability that more than 150 errors occur.
(b) If 160,000 bits are transmitted, calculate the (approximate) probability that more than 1 error occurs.

Chapter 8

Statistical Modeling and Inference

8.1 Introduction

One is often interested on random quantities (variables Y, T, N, etc.) such as the strength Y of a concrete block, the time T of a chemical reaction, the number N of visits to a website, etc. Engineers and applied scientists use **statistical models** to represent these random quantities. Statistical models are a set of mathematical equations involving random variables and other unknown quantities called **parameters**.

For example, the compressive strength of a concrete block can be modeled as

$$Y = \mu + \sigma\varepsilon \tag{8.1}$$

where μ is a parameter that represents the "true" average compressive strength of the concrete block, ε is a random variable with zero mean and unit variance that accounts for the "block-to-block" variability and σ is a parameter that determines the average size of the "block-to-block" variability. Notice that according to this model the compressive strength of a concrete block is a random variable that results from the sum of two components: a systematic component or *signal* (μ) and a random component or *noise* ($\sigma\varepsilon$).

Independent measurements are often taken to "adjust the model", that is, to estimate the unknown parameters that appear in the model equations. For example, the compressive strength of several concrete blocks can be measured to get information about μ and σ.

Before the measurements are actually performed they can be thought of as independent replicates of the random quantity of interest. For example, the future measurements of the compressive strengths can be represented as

$$Y_i = \mu + \sigma\varepsilon_i, \qquad i = 1, \ldots, n, \tag{8.2}$$

where n is the number of measurements.

Population and Sample: The complete set of items or individuals on which we are interested and on which we could, in principle, measure the variable (s) of interest is called *population*. Some examples of populations are a lot of concrete blocks, the websites on a certain topic, the most recent 300 days of operation of a retail store, etc. It is often impossible (or impractical) to measure the quantity of interest on all the units that comprise the

population under study. In practice some units are randomly chosen and the measurements are performed only on them. The set of selected units is called *sample*. The corresponding set of measurements is also called a *sample*.

Given a statistical model and a set of measurements (sample) one can carry on some some statistical procedures called **statistical inference** which are aimed at extrapolating from the sample to the population. The most typical statistical procedures are:

- Point estimation of the model parameters.
- Confidence intervals for the model parameters.
- Testing of hypotheses about the model parameters.

These procedures will be described and further discussed in the context of the simple situations considered below.

8.2 One Sample Problems

Sometimes it can be assumed that the quantity of interest is *homogeneous* for all the units in the population and that the measurements are the sum of a systematic and a random part (signal plus noise). In these cases we normally assume that the sample is a set of homogeneous measurements

$$Y_i = \mu + \sigma\varepsilon_i, \qquad i = 1, \ldots, n. \tag{8.3}$$

where μ and σ are as described in the Introduction above and n is the number of measurements or *sample size*. It is often assumed that the measurements are independent and therefore that the random variables $\varepsilon_i, \quad i = 1, \ldots, n$ are independent. Finally, we assume that the random variables ε_i are normal with mean zero and variance one.

Note: *Multiplicative models* where the measurements are the product of a systematic factor and a random factor

$$X_i = \gamma U_i$$

can be transformed on additive models like (8.3) by taking the log of the measurements

$$Y_i = \ln(X_i) = \ln(\gamma) + \ln(U_i).$$

8.2.1 Point Estimates for μ and σ

A **point estimate** is a certain combination of the sample measurements (a function of the sample) which is expected to take values "reasonably close" to the parameter it is supposed to estimate. The point estimate is usually denoted by the same letter as the parameter but with an added hat to indicate that it is an estimate (e.g. $\hat{\theta}$ is a point estimate for the parameter θ).

Of course, there are in principle many ways of combining the data to obtain a point estimate. The particular combination is chosen in order to minimize some function the estimation error

$$\hat{\theta} - \theta,$$

for example the expected squared estimation error or the expected absolute estimation error.

Estimation of μ: A good point estimate for μ, the main parameter of model (8.3), can be obtained by the method of **least squares** which consists of minimizing (in m) the sum of squares

$$S(m) = \sum_{j=1}^{n} (Y_j - m)^2$$

Differentiating with respect to m and setting the derivative equal to zero gives the equation

$$S'(m) = -2\sum_{j=1}^{n} (Y_j - m) = 0, \quad \text{or} \quad m = \frac{1}{n}\sum_{j=1}^{n} Y_j = \overline{Y} = \hat{\mu}.$$

Estimation Error: Being functions of the random variables, the point estimate $\overline{Y}$ and the estimation error $\overline{Y} - \mu$ are also random variables. Obviously, we would like that the estimation error is small. To have some idea of the behavior of the estimation error we can calculate its expected value (mean) and its variance:

$$\mathrm{E}[\overline{Y} - \mu] = \mathrm{E}(\overline{Y}) - \mu = \frac{1}{n}\sum_{j=1}^{n} \mathrm{E}(Y_j) - \mu = \frac{1}{n}\sum_{j=1}^{n} \mu - \mu = 0 \quad (\ \overline{Y} \text{ is unbiased}),$$

and

$$\mathrm{Var}(\overline{Y} - \mu) = \mathrm{Var}(\overline{Y}) = \frac{1}{n^2}\sum_{j=1}^{n} \mathrm{Var}(Y_j) = \frac{1}{n^2}\sum_{j=1}^{n} n\sigma^2 = \frac{\sigma^2}{n}.$$

In this case, the estimation error has then a distribution centered at zero and a variance inversely proportional to n. In other words, if n is sufficiently large, likely values of $\overline{Y}$ will all be close to μ.

Estimation of σ^2: The point estimate for σ^2 is based on the minimized sum of squares, $S(\overline{Y})$, divided by a quantity d so that the $\mathrm{E}[S(\overline{Y})/d] = \sigma^2$. The simple derivation outlined in Problem 8.9 shows that $d = n - 1$, and so

$$\hat{\sigma}^2 = S^2 = \frac{\sum_{j=1}^{n-1}(Y_j - \overline{Y})^2}{n-1}.$$

The Standard Error of $\overline{y}$: The precision of $\overline{y}$ as an estimate of μ can be measured in terms of its estimated standard deviation,

$$\mathrm{SE}(\overline{y}) = s/\sqrt{n},$$

called *standard error of* $\overline{y}$.

Example 8.1 A scientists wishes to detect small amounts of contamination in the environment. To test her measurement procedure, she spiked 12 specimens with a known concentration (2.5 $\mu g/l$ of lead). The readings for the 12 specimens are

1.9 2.4 2.2 2.1 2.4 1.5 2.3 1.7 1.9 1.9 1.5 2.0

The sample mean and variance are $\overline{y} = 1.9833$ and $s^2 = 0.09787879$, respectively. The standard error of $\overline{y}$ is then $\text{SE}(\overline{y}) = \sqrt{0.09787879/12} = 0.09031371$. It would appear that the scientist's measurement procedure is biased giving values below the true concentration. The bias can be estimated as $1.9833 - 2.5 = -0.5166667$, give or take 0.181 ($0.181 = 2 \times \text{SE}(\overline{y})$).

8.2.2 Confidence Interval for μ

Consider the *absolute estimation error* $|\overline{Y} - \mu|$. We wish to find a value d such that there is a large probability (0.95 or 0.99) that the absolute estimation error is below d. That is, we wish to find d such that, for some small value of α (typically $\alpha = 0.05$ or 0.01) we have

$$P[|\overline{Y} - \mu| < d] = 1 - \alpha$$

The resulting d can be, then, added to and subtracted from the observed average $\overline{y}$ to obtain the upper and lower limits of an interval called $(1 - \alpha)100\%$ *confidence interval*:

$$(\overline{y} - d, \overline{y} + d)$$

Typical values of α are $\alpha = 0.05$ and $\alpha = 0.01$ yielding 95% and 99% confidence intervals, respectively. To fix ideas we will take $\alpha = 0.05$ in what follows.

Assuming that the model (8.3) is correct, the probability that μ and $\overline{Y}$ differ by more than d is only 0.05. In other words, if we repeatedly obtain samples of size n and construct the corresponding 95% confidence intervals for μ, on average, 95% of these intervals will include the (unknown) value of μ.

Using that $\overline{Y} \sim N(\mu, \sigma^2/n)$ we have

$$0.95 \;=\; P[\ |\overline{Y} - \mu| < d\ ] = P\left[\ |\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}| < \frac{d\sqrt{n}}{\sigma}\ \right] = 2\Phi\left[\frac{\sqrt{n}d}{\sigma}\right] - 1.$$

That is,

$$\Phi\left[\frac{d\sqrt{n}}{\sigma}\right] = 0.975.$$

Using the standard normal table we get,

$$\frac{d\sqrt{n}}{\sigma} = 1.96,$$

from which we have

$$d \;=\; 1.96\ \frac{\sigma}{\sqrt{n}}.$$

Unfortunately, in most practical applications, the value of σ is unknown and must be estimated from the data. To estimate σ we can use, for instance, the sample standard deviation s. The corresponding estimate for d is now

$$\hat{d} \;=\; 1.96 \;\; \frac{s}{\sqrt{n}} = 1.96 \times \mathrm{SE}(\overline{y}).$$

The precision of s as an estimate of σ increases with the sample size. Therefore, replacing σ by s has little effect when the sample size is large ($n \geq 20$, say). However, when n is small the added level of uncertainty is somewhat increased and an adjustment is needed. To adjust for the increased level of uncertainty the value from the normal table (1.96 when $\alpha = 0.05$) must be replaced by a slightly larger value, $t_{df}(\alpha)$, obtained from the Student's t table. The precise Student's t value, $t_{df}(\alpha)$, depends on two parameters: the **significance level**, α, and the **degrees of freedom**, df.

The significance level, α, is equal to one minus the desired confidence level . In our case, the confidence level (desired precision) is 0.95 and so $\alpha = 0.05$. In this simple case the degrees of freedom parameter, df is simply equal to the sample size minus one, that is $df = n - 1$. More generally, (for future applications) the degrees of freedom are given by the formula

$$df = n - k,$$

where

$$n = \text{ number of squared terms appearing in the variance estimate}$$

and

$$k = \text{ number of additional estimated parameters appearing in the variance estimate}$$

Table A.2 in Appendix gives the values of $t_{(df)}(\alpha)$ for several values of α and df.

In summary, the estimated value of d is

$$\hat{d} \;=\; t_{df}(\alpha) \;\; \frac{s}{\sqrt{n}} = \; t_{df}(\alpha) \;\; \mathrm{SE}(\overline{Y}).$$

Notice that for most values of n that appear in practice, $t_{n-1}(0.05) \approx 2$, justifying the common practice of adding and subtracting $2 \times \mathrm{SE}(\overline{y})$ from the observed average $\overline{y}$.

Example 8.2 Refer to the data in example 8.1. A 95% confidence interval for the actual mean of the scientist's measurements is

$$1.9833 \pm t_{(11)}(0.05) \times \mathrm{SE}(\overline{y})$$

or

$$1.9833 \pm 2.20 \times 0.09031371.$$

That is, the systematic part of the scientist's measurement is likely to lay between 1.8 and 2.2.

8.2.3 Testing of Hypotheses about μ

There are situations when one wishes to determine if a certain statement or *hypothesis* about a model parameter is consistent with the given data. That is, one wishes to confront the statement against the empirical evidence (data). For example, the scientist of Examples 8.1 and 8.2 may wish to *test* the hypothesis that the given measurement method is unbiased, using her collected data.

The procedure for rejecting a hypothesis about a certain unknown population parameter, on the basis of statistical evidence, is called **testing of hypothesis**. The hypothesis to be tested is denoted by H_0.

Typical hypotheses, H_0, about μ are

$$\text{(i)}\ \ H_0 : \mu = \mu_0 \quad \text{or} \quad \text{(ii)}\ \ H_0 : \mu \leq \mu_0 \quad \text{or} \quad \text{(iii)}\ \ H_0 : \mu \geq \mu_0,$$

where μ_0 is some specified value. In the case of the scientist of examples 8.1 and 8.2, the statement "the measurement method is unbiased" corresponds to (i) with $\mu = 2.5$. On the other hand, the statement "the measurement method does not consistently under-estimates the true concentration" corresponds to (iii) with $\mu = 2.5$. What statement would correspond to (ii) with $\mu = 2.5$?

Significance Level of a Test: When testing a hypothesis one can incur in two possible errors: Rejecting a hypothesis that is true (*Error of type I*) or non-rejecting a hypothesis that is false (*Error of type II*). Errors of type I are considered more important and kept under tight control. Therefore, usual testing procedures insure that the probability of rejecting a true hypothesis is rather small (0.01 or 0.05). The probability of error of type I is usually denoted by α and called *significance level* of the test.

Taking that into consideration, the hypothesis H_0 is constructed in a such a way that its incorrect rejection has a small probability. H_0 states, then, the most conservative statement. A statement that one would like to reject only in the presence of strong empirical evidence. Because of that H_0 is called as the "null hypothesis".

The Testing Procedure: The testing procedures learned in this course are simply derived from confidence intervals. Suppose we wish to test H_0 at level α. Then we distinguish two cases:

Two sided tests: Hypotheses of the form $H_0 : \mu = \mu_0$ give rise to *two sided tests* because in this case we reject H_0 if we have evidence indicating that μ is smaller or larger than μ_0.

The two–sided level α testing procedure consists of the following two steps:

Step 1. Construct a $(1 - \alpha)100\%$ confidence interval for μ.

Step 2. Reject $H_0 : \mu = \mu_0$ if μ_0 lies outside that interval.

One sided tests: Hypotheses of the form $H_0 : \mu \geq \mu_0$ ($H_0 : \mu \leq \mu_0$) are called *directional hypotheses* and give rise to *one–sided tests*. Notice that in this case we reject H_0 only if we suspect that $\mu < \mu_0$ ($\mu > \mu_0$).

The one–sided level α testing procedure consists of the following two steps:

Step 1. Construct a $[1 - (2 \times \alpha)]100\%$ confidence interval for μ.

Step 2. Reject $H_0 : \mu \geq \mu_0$ ($H_0 : \mu \leq \mu_0$) if μ_0 is larger (smaller) than the upper (lower) end of that interval.

That is, we reject H_0 if the confidence interval is completely contained in the complement of the interval assumed under H_0.

Example 8.3 Refer to the data in example 8.1. Test at level $\alpha = 0.05$ the following hypotheses: (a) $H_0 : \mu = 2.5$; and (b) $H_0 : \mu \geq 2.3$.
(a) Since the 95% confidence interval $(1.785, 2.182)$ (see example 8.2) does not include 2.5, we reject H_0. There is statistical evidence indicating that the measurement procedure is not unbiased.
(b) We must first construct a 90% confidence interval for μ. From example 8.1 we have that $\overline{y} = 1.9833$ and $\text{SE}(\overline{y}) = 0.0931371$. Moreover, from the Student-t Table we have $t_{(11)}(0.10) = 1.80$. Therefore, the 90% confidence interval for μ is

$$(1.9833 - 1.80 \times 0.0931371, 1.9833 - 1.80 \times 0.0931371) = (1.82, 2.15).$$

Since $2.15 < 2.3$ we reject the H_0. There is statistical evidence indicating that the measurement procedure systematically underestimates the true lead concentration by at least 0.2 $\mu g/l$.

Example 8.4 A shipyard must order a large shipment of lacquer from a supplier. Besides other design requirements, the lacquer must be durable and dry quickly. The average drying time must not exceed 25 minutes. Supplier A claims that, on average, its product dries in 20.5 minutes. A sample of 30 20-liter cans from supplier A yields an average drying time of 22.3 minutes and standard deviation of 2.9 minutes.

(a) Is there statistical evidence to distrust supplier A's claim that its product has an average drying time of 20.5 minutes?

(b) Can we say that, on average, supplier A's lacquer dries before 24 minutes?

Solution to Example 8.4:

(a) To answer this question we must assess the precision of $\overline{y}$ as an estimate of μ. Evidently, $\overline{y} = 22.3$ is different from the claimed value 20.5 for μ. However, we need still to determine if the observed difference of 1.8 is within the normal range of variability of $\overline{Y}$.

To answer the question we can test the hypothesis

$$H_0 : \mu = 20.5.$$

at level $\alpha = 0.05$, say. Since it is a non-directional hypothesis (two–sided test) we must construct a 95% confidence interval for μ and check if it contain the value 20.5. In the

present case $\alpha = 0.05$ and $df = 30 - 1$ Hence, from Table A.2, $t_{29}(0.05) = 2.05$. Moreover, $\text{SE}(y) = 2.9/\sqrt{30} = 0.529465$. Therefore,

$$\hat{d} \;=\; 2.05 \times 0.529465 \;=\; 1.085,$$

and the 95% confidence interval for μ is

$$(\overline{y} \pm \hat{d}) \;=\; (22.3 \pm 1.084) \;=\; (21.21 \;,\; 23.39).$$

Since this interval doesn't include the value $\mu = 20.5$, we reject supplier's A claim that $\mu = 20.5$. That is, we reject the hypothesis $\mu = 20.5$ on the basis of the given data and statistical model.

(b) One way to answer this question is to test the hypothesis

$$H_0 : \mu \geq 24.0$$

at some (small) level α. To take advantage of the calculations already made we may choose $\alpha = 0.025$. Since the upper limit 95% confidence interval for μ is smaller than 24.0, we reject H_0 and answer question (b) in a positive way. □

8.3 Two Sample Problems

There are practical situations where we are interested in comparing several populations. In this section we will consider the simplest case of two populations. In Chapter 10 we will consider the general case of two or more populations.

Example 8.5 Refer to the situation described in Problem 8.4. Another supplier, called Supplier B, could also supply the lacquer. A sample of 10 20-liter cans from supplier B yields an average drying time of 20.7 minutes and standard deviation of 2.5 minutes. Does the data support supplier B's claim that, on average, its product dries faster than A's? What if the sample size from supplier B were 100 instead of 10?

This example illustrates a fairly common situation: one must take or recommend an important decision involving a large number of items (or individuals) on the basis of a relatively small number of measurements performed on some of these items. Recall that the set of all the items under study is called the population and the subset of items used to obtain the measurements (and often the measurements themselves) is called the sample.

Example 8.5 includes two populations, namely the 3,000 20-liter cans of lacquer that can be acquired from either supplier A or B. In the following these two populations will be called population A and population B, respectively.

Although we are concerned with the entire populations, we will only be able to test the items in the samples. Therefore, we must try to investigate and exploit the mathematical connections between the samples and the population from which they came. This can be

done with the help of an statistical model, that is, a set of probability assumptions regarding the sample measurements. The two sample measurements can be modeled as

$$Y_{ij} = \mu_i + \sigma_i \varepsilon_{ij}, \qquad i = 1, 2 \text{ and } j = 1, \ldots, n_i. \tag{8.4}$$

where the first subscript (i) indicates the population and the second subscript (j) indicates the observation. Thus, μ_i and σ_i are the population means and variances, respectively, and n_1 and n_2 are sample sizes. In the case of Example 8.5, $n_1 = 30$ and $n_2 = 10$. It is often assumed that the measurements are independent and therefore that the random variables ε_{ij}, $i = 1, 2$ and $j = 1, \ldots, n_i$ are independent. Finally, as in the case of one sample, we assume that the random variables ε_{ij} are normal with mean zero and variance one.

Similarly to the one-sample case, the population means μ_1 and μ_2 can be estimated by the corresponding sample means:

$$\overline{Y}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} Y_{1j} \quad \text{and} \quad \overline{Y}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} Y_{2j},$$

Notice that $\overline{Y}_1$ and $\overline{Y}_2$ are normal random variables with means μ_1 and μ_2 and variances σ_1^2/n_1 and σ_2^2/n_2, respectively. Furthermore, the population variances σ_1^2 and σ_2^2 can be estimated by the sample variances

$$S_1^2 = \frac{1}{n_1} \sum_{j=1}^{n_1} [Y_{1j} - \overline{Y}_1]^2 \quad \text{and} \quad S_2^2 = \frac{1}{n_2} \sum_{j=1}^{n_2} [Y_{2j} - \overline{Y}_2]^2$$

Notice that $\mathrm{E}(S_i^2) = \sigma_i^2$ (see Problem 8.9).

The Pooled Variance Estimate If the variances of the two populations are approximately equal it then makes sense to compare their means. On the other hand, if the variances are very different, comparing the population means may be a gross oversimplification. A practical solution in these cases is to apply a transformation (e.g. use $\log(Y_{ij})$ instead of Y_{ij}) that *stabilizes* (equalizes) the variances.

In this course we will only consider the simple situation where

$$\sigma_1^2 = \sigma_2^2 = \sigma^2.$$

An unbiased estimate for the common variance σ^2, based on the individual unbiased estimates S_1^2 and S_2^2, is given by the pooled variance estimate

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} = \frac{\sum_{i=1}^{2} \sum_{j=1}^{n_i} [Y_{ij} - \overline{Y}_i]^2}{n_1 + n_2 - 2}.$$

Linear Combinations of the Population Means: In practice one often wishes to estimate linear combinations of the population means and to test hypotheses about them. In such cases we say that the *parameter of interest* is a linear combination of μ_1 and μ_2.

The most common linear combination of μ_1 and μ_2 is the simple difference:

$$\theta = \mu_1 - \mu_2.$$

Other examples are

$$\theta = \mu_1 - 2\mu_2, \quad \theta = 3\mu_1 - \mu_2, \quad \theta = 1.2\mu_1 + 0.5\mu_2,$$

etc. In general, θ can be written as

$$\theta = a\mu_1 + b\mu_2$$

where a and b are given constants.
The parameter of interest, θ can be unbiasedly estimated by

$$\hat{\theta} = a\overline{Y}_1 + b\overline{Y}_2$$

In fact,

$$\mathrm{E}(\hat{\theta}) = \mathrm{E}(a\overline{Y}_1 + b\overline{Y}_2) = a\mathrm{E}(\overline{Y}_1) + b\mathrm{E}(\overline{Y}_2) = a\mu_1 + b\mu_2 = \theta.$$

The variance of $\hat{\theta}$ is equal to

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}(a\overline{Y}_1 + b\overline{Y}_2) = a^2\mathrm{Var}(\overline{Y}_1) + b^2\mathrm{Var}(\overline{Y}_2) = a^2\frac{\sigma^2}{n_1} + b^2\frac{\sigma^2}{n_2} = \sigma^2\left[\frac{a^2}{n_1} + \frac{b^2}{n_2}\right].$$

Therefore, the standard error of $\hat{\theta}$ is

$$\mathrm{SE}(\hat{\theta}) = \hat{\sigma}\sqrt{\frac{a^2}{n_1} + \frac{b^2}{n_2}}.$$

In the case of Example 8.5 the parameter of interest is $\theta = \mu_1 - \mu_2$, estimated as

$$\hat{\theta} = \overline{y}_1 - \overline{y}_2 = 22.3 - 20.7 = 1.6,$$

The pooled variance estimate is

$$s^2 = \frac{(29)(2.9^2) + (9)(2.5^2)}{30 + 10 - 2} = 7.90$$

and so

$$\mathrm{SE}(\hat{\theta}) = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 2.8106\sqrt{\frac{1}{30} + \frac{1}{10}} = \sqrt{1.053} = 1.026.$$

Degrees of Freedom: Notice that we are using $n_1 + n_2$ observations to calculate s^2, and that we estimated two unknown parameters (μ_1 and μ_2), therefore

$$df = n_1 + n_2 - 2.$$

Confidence Interval for θ: A $(1-\alpha)100\%$ confidence interval for θ is given by

$$\hat{\theta} \pm t_{(n_1+n_2-2)}(\alpha) \times \text{SE}(\hat{\theta})$$

In the case of Example 8.5, a 95% confidence interval for $\mu_1 - \mu_2$ is given by

$$(\overline{y}_1 - \overline{y}_2) \ \pm \ t_{(38)}(0.05) \times \ s\sqrt{\frac{n_1+n_2}{n_1 n_2}} = 1.6 \ \pm \ 2.02 \times 1.026 = (-0.47 \ , \ 3.67),$$

We have used the approximation $t_{(38)}(0.05) \approx t_{(38)}(0.05) = 2.02$, because $t_{(38)}(0.05)$ is not included in the table.

Solution to Example 8.5: The statement of Supplier B is consistent with the hypothesis

$$H_0 : \mu_1 \geq \mu_2$$

or equivalently

$$H_0 : \mu_1 - \mu_2 \geq 0.$$

We may answer the question by testing this (directional) hypothesis at some (small) level α. For example, we may take $\alpha = 0.05$. The 90% confidence interval for $\theta = \mu_1 - \mu_2$ is

$$(\overline{y}_1 - \overline{y}_2) \ \pm \ t_{(40)}(0.10) \times \ s\sqrt{\frac{n_1+n_2}{n_1 n_2}} = 1.6 \ \pm \ 1.68 \times 1.026 = 1.6 \ \pm \ 1.724 = (-0.124 \ , \ 3.324),$$

Since the value $\mu_1 - \mu_2 = 0$ falls in the interval, we conclude that there is no statistically significant difference between the two means. There is, then, statistical evidence against Supplier B's claim of having a superior product. □

Example 8.6 Either 20 large machines or 30 small ones can be acquired for approximately the same cost. One large and one small machines have been experimentally run for 20 days with the following results:

$$\overline{y}_{large} = \overline{y}_1 = 31.0, \quad s_{large} = s_1 = 2.1$$
$$\overline{y}_{large} = \overline{y}_1 = 31.0, \quad s_{large} = s_1 = \sout{2.1}\ 1.9$$

Is there statistical evidence in favor of either type of machine? Use $\alpha = 0.05$.

Solution: Since the total cost of 20 large machines equals the cost of 30 small machines, it is reasonable to compare the total outputs:

$$\text{Total output of 20 large machines} = 20\mu_1$$
$$\text{Total output of 30 small machines} = 30\mu_2$$

where μ_1 and μ_2 are the average daily outputs for each type of machine.

Therefore, the parameter of interest is the linear combination

$$\theta = 20\mu_1 - 30\mu_2.$$

From the information given we have $n_1 = n_2 = 20$ and θ can be estimated by

$$\hat{\theta} = 20\ \overline{y}_1 - 30\ \overline{y}_2 = 20 \times 31 - 30 \times 22.7 = -61.0.$$

The pooled estimate of σ^2 is

$$\frac{19 \times 2.1^2 + 19 \times 1.9^2}{20 + 20 - 2} = 4.01$$

and so $s = 2.0$. Since $df = 20 + 20 - 2 = 38$, from the Student's t table we have

$$t_{(38)}(0.05) \approx t_{(40)}(0.05) = 2.02.$$

Therefore, the 95% confidence interval for θ is

$$-61.0 \pm 2.02 \times 2.0 \times \sqrt{\frac{20^2}{20} + \frac{30^2}{20}} = -61.0 \pm 35.57 = (-93.57, -28.43).$$

Therefore we reject (at level $\alpha = 0.05$) the hypothesis that both alternatives are equally convenient. It appears that it would be more convenient to acquire 30 small machines.

8.4 Exercises

8.4.1 Exercise Set A

Problem 8.1 Given that $n_1 = 15, \bar{x} = 20, \sum(x_i - \bar{x})^2 = 28$, and $n_2 = 12, \bar{y} = 17, , \sum(y_i - \bar{y})^2 = 22$.
(a) Calculate the pooled variance s^2.
(b) Determine a 95% confidence interval for $\mu_1 - \mu_2$.
(c) Test $H_0 : \mu_1 = \mu_2$ with $\alpha = .05$.

Problem 8.2 The time for a worker to repair an electrical instrument is a normally distributed $N(\mu, \sigma^2)$ random variable measured in hours, where both μ and σ^2 are unknown. The repair times for 10 such instruments chosen at random are as follows:

212,234,222,140,280,260,180,168,330,250

(1) Calculate the sample mean and the sample variance of the 10 observations.
(2) Construct a 95% confidence interval for μ.
(3) Suppose the worker claims that his average repair time for the instrument is no more than 200 hours. ~~Test if his claim conforms with the data.~~ Test if there is sufficient evidence to dispute the worker's claim.

Problem 8.3 (Hypothetical) The effectiveness of two STAT251/241 labs which were conducted by two TAs is compared. A group of 24 students with rather similar backgrounds was randomly divided into two labs and each group was taught by a different TA.Their test scores at the end of the semester show the following characteristics:

$$n_1 = 13, \bar{x} = 74.5, s_x^2 = 82.6$$

and

$$n_2 = 11, \bar{y} = 71.8, s_y^2 = 112.6.$$

Assuming underlying normal distributions with $\sigma_1^2 = \sigma_2^2$, find a 95 percent confidence interval for $\mu_1 - \mu_2$. Are the two labs different? Summarize the assumptions you used for your analysis.

Problem 8.4 *Two machines (called A and B in this problem) are compared. Machine A cost \$ 3000 and machine B cost \$ 4500. One machine of each type was operated during 30 days and the daily outputs were recorded. The results are summarized below:*
Machine A*:* $\overline{x}_A = 200kg \quad s_A = 5.1kg.$

Machine B*:* $\overline{x}_B = 270kg \quad s_B = 4.9kg.$

Is there statistical evidence indicating that any one of these machines has better output/cost performance than the other? Use $\alpha = 0.5$.

Problem 8.5 The average biological oxygen demand (BOD) at a certain experimental station has to be estimated. From measurements at other similar stations we know that the variance of BOD samples is about 8.0 $(mg/liter)^2$. How many observations should we sample

if we want to be 90 percent confident that the true mean is within 1 $mg/liter$ of our sample average? (Hint: Using CLT, we may assume the sample average has approximately normal distribution).

Problem 8.6 An automobile manufacturer recommends that any purchaser of one of its new cars bring it in to a dealer for a 3000-mile checkup. The company wishes to know whether the true average mileage for initial servicing differs from 3000. A random sample of 50 recent purchasers resulted in a sample average mileage of 3208 and a sample standard deviation of 273 miles. Does the data strongly suggest that true average mileage for this checkup is something other than the recommended value?

Problem 8.7 The following data were obtained on mercury residues on birds' breast muscles:
Mallard ducks: $m = 16, \bar{x} = 6.13, s_1 = 2.40$
Blue-winged teals: $n = 17, \bar{y} = 6.46, s_2 = 1.73$
Construct a 95% confidence interval for the difference between true average mercury residues μ_1, μ_2 in these two types of birds in the region of interest. Does your confidence interval indicate that $\mu_1 = \mu_2$ at a 95% confidence level?

Problem 8.8 A manufacturer of a certain type of glue claims that his glue can withstand 230 units of pressure. To test this claim, a sample of size 24 is taken. The sample mean is 191.2 units and the sample standard deviation is 21.3 units.
(a) Propose a statistical model to test this claim and test the manufacturer's claim.
(b) What is the highest claim that the manufacturer can make without rejection of this claim?

8.4.2 Exercise Set B

Problem 8.9 Suppose that $Y_1, \ldots, Y_n$ are a sample, that is, they are independent, identically distributed, with common mean μ and common variance σ^2. Recall that the sample variance is equal to

$$S^2 = \frac{\sum_{i=1}^n (Y_i - \overline{Y})^2}{n-1}$$

(a) Show that

$$\sum_{i=1}^n (Y_i - \overline{Y})^2 = \sum_{i=1}^n (Y_i - \mu)^2 - n(\overline{Y} - \mu)^2.$$

(b) Show that S^2 is an unbiased estimate of σ^2, that is

$$\mathrm{E}(S^2) = \sigma^2$$

Problem 8.10 (a) The president of a cable company claims that its 0.3–inch cable will support an average load of 4200 pounds. Twenty four of these cables are tested to failure, yielding the following data:

4201.3 4262.4 3983.0 3943.0 4141.3 4168.5 4050.0 4142.7

4270.0 4002.9 4393.9 3868.0 4123.5 4192.5 3986.6 4276.7
4253.9 4303.4 4099.2 4136.1 4492.7 4292.7 3820.9 3621.4

Propose a statistical model for the given data and test the president's claim. Check that your model's assumptions are consistent with the data.

(b) A different supplier has provided a sample of thirty six 0.3–inch cables which, after tested to failure, yielded the following data:

4047.3 4302.6 4069.4 3914.8 4133.2 3658.6 4221.9 3913.1 4129.9 4068.7
4389.9 3943.9 4446.6 3796.3 4117.4 3816.9 4353.4 4009.5 4432.9 4072.1
3862.0 3939.3 3875.2 3989.0 4203.2 4334.9 4358.6 4189.9 4219.7 4238.0
4033.2 4005.2 4428.8 3938.0 4171.6 3974.7

Propose a statistical model for (all) the given data and test the hypothesis that the cable from the two companies have the same average strength. Check that your model's assumptions are consistent with the data.

Problem 8.11 A politician must decide whether or not to run in the next local election. He would be inclined to do so if at least 30% of the voters would favor his candidacy. The results of a poll of 20 local citizens yielded the following results:

30% favor the politician 35% favor other candidates 35% are still undecided

Should the candidate decide to run based on the results of this survey? Do you think that the sample size is appropriate? If not, suggest an appropriate sample size.

Problem 8.12 The number of hours needed by twenty employees to complete a certain task have been measured before and after they participated of a special training program. The data is displayed on Table 7.1.

How would you model these data in order to answer the question: Was the training program successful? Was it? Also check that your model's assumptions are consistent with the data.

Table 7.1:

Problem 8.13 In order to process a certain chemical product, a company is considering the convenience of acquiring (for approximately the same price) either 100 large machines or 200 small ones. One important consideration is the average daily processing capacity (in hundred of pounds).

One machine of each type was tested for a period of 10 days, yielding the following results:

Large Machine: $\overline{x}_1 = 120 \quad s_1 = 1.5$
Small Machine: $\overline{x}_2 = 65 \quad s_2 = 1.6$

Model the data and identify the parameter of main interest. Construct a 95% confidence interval for this parameter. What is your recommendation to management?

Problem 8.14 A study is made to see if increasing the substrate concentration has appreciable effect on the velocity of a chemical reaction. With the substrate concentration of 1.5 moles per liter, the reaction was run 15 times with an average velocity of 7.5 micromoles per 30 minutes and a standard deviation of 1.5. With a substrate concentration of 2.0 moles per

Employee	Before Training	After Training	Difference
1	14.6	10.6	4.0
2	17.5	15.4	2.1
3	13.5	13.2	0.3
4	13.9	12.2	1.7
5	15.0	11.7	3.3
6	20.5	18.6	1.9
7	14.4	10.3	4.1
8	14.6	10.3	4.3
9	17.9	10.4	7.5
10	16.7	16.8	-0.1
11	14.7	14.6	0.1
12	17.3	14.6	2.7
13	11.7	10.5	1.2
14	13.7	10.9	2.8
15	16.8	11.8	5.0
16	15.7	13.4	2.3
17	15.7	13.6	2.1
18	16.7	16.7	0.0
19	15.5	16.7	-1.2
20	17.2	13.8	3.4

liter, 12 runs were made yielding an average velocity of 8.8 micromoles per 30 minutes and a sample standard deviation of 1.2. Would you say that the increase in substrate concentration increases the mean velocity by as much as 0.5 micromoles per 30 minutes? Use a 0.01 level of significance and assume the populations to be approximately normally distributed with equal variances.

Problem 8.15 *(Hypothetical)* A study was made to estimate the difference in annual salaries of professors in University of British Columbia (UBC) and University of Toronto (UT). A random sample of 100 professors in UBC showed an average salary of \$46,000 with a standard deviation \$12,000. A random sample of 200 professors in UT showed an average salary of \$51,000 with a standard deviation of \$14,000. Test the hypothesis that the average salary for professors teaching in UBC differs from the average salary for professors teaching in UT by \$5,000.

Problem 8.16 A UBC student will spend, on the average, \$8.00 for a Saturday evening gathering in pub. A random sample of 12 students attending a homecoming party showed an average expenditure of \$8.9 with standard deviation of \$1.75. Could you say that attending a homecoming party costs students more than gathering in pub?

Problem 8.17 The following data represent the running times of films produced by two different motion-picture companies.

	Times (minutes)						
Company I	103	94	110	87	98		
Company II	97	82	123	92	175	88	118

Compute a 90% confidence interval for the difference between the average running times of films produced by the two companies. Do the films produced by Company II run longer than those by Company I?

Problem 8.18 It is required to compare the effect of two dyes on cotton fibers. A random sample of 10 pieces of yarn were chosen; 5 pieces were treated with dye A, and 5 with dye B. The results were

Dye A	4	5	8	8	10
Dye B	6	2	9	4	5

(a) Test the significance of the difference between the two dyes. (Assume normality, common variance, and significance level $\alpha = 0.05$.)
(b) How big a sample do you estimate would be needed to detect a difference equal to 0.5 with probability 99%.

Chapter 9

Simulation Studies

9.1 Monte Carlo Simulation

Consider the integral

$$I = \int_0^1 g(t)dt.$$

Suppose that g is such that this integral cannot be easily integrated and we need to approximate it by numerical means. For simplicity suppose that $0 \leq g(t) \leq 1$ for all $0 \leq t \leq 1$.

If we are dealing with a function $h(t)$ which is not between 0 and 1 but we know that

$$a \leq h(t) \leq b, \quad \text{for all } \ 0 \leq t \leq 1,$$

then the function

$$g(t) = \frac{h(t) - a}{b - a}$$

does take values between 0 and 1 and

$$\int_0^1 h(t)dt = (b - a) \int_0^1 g(t)dt + a$$

Suppose that we want to estimate I with an error smaller than $\delta = 0.01$, with probability equal to 0.99. In other words, if $\hat{I}$ is the estimate of I, we require that

$$P\{|\hat{I} - I| < 0.01\} = 0.99.$$

First of all, we notice that

$$I = \int_0^1 g(t)dt = \mathrm{E}\{g(U)\},$$

where U is a random variable with uniform distribution on the interval $(0, 1)$. If we generate n independent random variables

$$U_1, U_2, \ldots, U_n$$

with uniform distribution on $(0, 1)$, then by the Central Limit Theorem

$$\hat{I} = \frac{1}{n}\sum_{i=1}^{n} g(U_i) = \overline{g(U)}$$

is approximately normal with mean $I = \mathrm{E}\{g(U)\}$ and variance σ^2/n, where

$$\sigma^2 \;=\; \int_0^1 g^2(t)dt - I^2 \;\leq\; \int_0^1 g(t)dt - I^2 \;=\; I(1-I).$$

Now,

$$\begin{aligned} P\{|\hat{I} - I| < 0.01\} &= P\{\sqrt{n}|\hat{I} - I|/\sigma < \sqrt{n}0.01/\sigma\} \\ &\approx P\{|Z| < \sqrt{n}0.01/\sigma\} \\ &= 2\Phi[\sqrt{n}0.01/\sigma] - 1. \end{aligned}$$

But,

$$\begin{aligned} 2\Phi[\sqrt{n}0.01/\sigma] - 1 = 0.99 \quad &\Rightarrow \quad \Phi[\sqrt{n}0.01/\sigma] = 0.995 \\ &\Rightarrow \quad \frac{\sqrt{n}0.01}{\sigma} = 2.58 \\ &\Rightarrow \quad n = \frac{\sigma^2(2.58)^2}{(0.01)^2}. \end{aligned}$$

Finally, since $I(1-I)$ reaches its maximum at $I = 0.5$ it follows that $I(1-I) \leq 0.25$ for all I, and so, a conservative estimate for n is

$$n = \frac{\sigma^2(2.58)^2}{(0.01)^2} \leq \frac{(0.25)(2.58)^2}{(0.01)^2} = 16,641$$

Therefore, an estimate of I based on $n = 16,641$ independent uniform random variables, U_i, will include an error smaller than 0.01, with probability 0.99.

The Monte Carlo method can also be used to estimate an integral of the form

$$J = \int_a^b f(t)dt, \tag{9.1}$$

where $f(t)$ takes values between c and d. That is, the domain of integration can be any given bounded interval, $[a, b]$, and the function can take values on any given bounded interval $[c, d]$. For example, we may wish to estimate the integral

$$J = \int_1^3 \exp\{t^2\}dt.$$

In this case the domain of integration is $[1, 3]$ and the function ranges over the interval

$$[2.7183, 20.086]$$

.

In order to estimate J, first we must make the change of variables

$$u = \frac{t-a}{b-a},$$

to obtain

$$J = (b-a)\int_0^1 f[(b-a)u + a]du = \int_0^1 g(u)du,$$

where

$$g(u) = (b-a)f[(b-a)u + a].$$

In the case of our numerical example we have

$$J = (3-1)\int_0^1 \exp\{[(3-1)u+1]^2\}du == 2\int_0^1 \exp\{2u+1]^2\}du,$$

and

$$g(u) = 2\exp\{[2u+1]^2\}$$

The second step is to linearly modify the function $g(u)$ so that the resulting function, $h(u)$, takes values between 0 and 1. That is,

$$h(u) = \frac{g(u)-(b-a)c}{(b-a)(d-c)} \quad \Rightarrow \quad g(u) = (b-a)(d-c)h(u) + (b-a)c$$

Notice that, since

$$(b-a)c \leq g(u) \leq (b-a)d,$$

then

$$0 \leq h(u) \leq 1.$$

In the case of our numerical example,

$$h(u) \;=\; \frac{2\exp\{[2u+1]^2\} - 2.7183}{2(20.086-2.7183)} \;=\; \frac{2\exp\{2u+1]^2\} - 2.7183}{34.7354}.$$

Finally,

$$\begin{aligned} J &= \int_0^1 g(u)du = (b-a)(d-c)\int_0^1 h(u)du + \frac{c}{d-c} \\ &= (b-a)(d-c)I + \frac{c}{d-c}, \end{aligned}$$

where I is of the desired form (that is, the integral between 0 and 1 of a function that takes values between 0 and 1).

9.2 Exercises

Problem 9.1 Use the Monte Carlo integration method with $n = 1500$ to approximate the following integrals.
(a)

$$I = \int_0^1 \exp\{-x^2\}dx.$$

What is the (approximated) probability that the approximation error is less than $d = 0.05$? Less than $d = 0.01$?
(b)

$$I = \int_{-1}^2 \exp\{x^2\}dx.$$

Problem 9.2 Let

$$I = \int_0^{\pi/2} \exp\{-\cos^2(x)\}\cos(x)\sin(x)dx$$

(a) Use the Monte Carlo method, with $n = 100$, to estimate I.
(b) Construct a 95% confidence interval for I based on the Monte Carlo data.
(c) Is the true value of I included in your confidence interval?
Hint: use the change of variables $y = \cos^2(x)$ to exactly evaluate the integral.
(d) Repeat (a)–(c) with $n = 500$ and $n = 1000$.
(e) What is the needed sample size if the 95% confidence interval must have total length equal to 0.02?

Problem 9.3 (a) Generate 100 samples of size $n = 10$ from the following distributions: (1) Uniform on the interval $(0, 1)$; (2) exponential with mean 1; (3) discrete with $f(1) = 1/3, f(2) = 1/3$ and $f(9) = 1/3$; (4) discrete with $f(1) = 1/8, f(3) = 1/8$ and $f(9) = 3/4$ and (5) $f(1) = 1/3, f(5) = 1/3$ and $f(9) = 1/3$.
(b) For each distribution calculate the corresponding sample means and discuss the merits of the CLT approximation to the distribution of the sample mean in each case. You can use histograms, Q-Q plots, box plots, etc. for your analysis.
(c) Repeat (a) and (b) with $n = 20$ and $n = 50$.
(d) Concisely state your conclusions.

Chapter 10

Comparison of several means

10.1 An example

The main ideas will be illustrated by the following example.

Example 10.1 A construction company wants to compare several different methods of drying concrete block cylinders. To that effect, the engineer in charge of acquisition and testing of materials sets up an experiment to compare five different drying methods referred to as drying methods A, B, C, D and E. One important feature of the concrete block cylinders is their compressive strength (in hundreds of kilograms per square centimeter), which can be determined by means of a destructive strength test. After selecting a carefully designed experiment (we will discuss this important step later on) the engineer collected the data displayed in Table 9.1.

Table 10.1: Concrete Blocks Compressive Strength

Type	Compressive Strength (100 pounds per square inch)										Mean	SD
A	47.90	47.95	49.39	48.80	53.15	49.06	50.62	46.80	34.66	47.48	45.05	7.5
	44.56	50.41	35.99	45.15	57.53	50.05	40.79	30.38	29.13	41.21		
B	37.69	37.79	62.75	51.62	39.73	65.68	64.62	46.64	52.01	61.38	52.29	8.70
	52.58	40.47	53.85	55.06	49.14	49.71	57.68	50.54	62.18	54.60		
C	61.93	63.39	52.87	47.26	50.97	58.45	48.87	66.48	57.79	48.51	54.29	6.32
	58.91	42.85	48.40	53.28	55.00	49.97	49.47	54.21	51.37	60.26		
D	82.31	51.82	64.11	61.06	47.72	53.08	56.99	55.49	52.72	64.97	56.83	10.40
	44.81	46.36	44.76	68.43	76.49	48.58	61.41	55.97	46.83	52.76		
E	39.72	40.98	44.74	29.94	47.18	32.84	35.39	43.54	50.21	42.12	41.15	8.16
	26.72	44.68	34.48	46.54	54.80	56.89	34.46	42.88	44.30	30.64		

Propose a statistical model and answer the following questions:

(a) Are the model's assumptions consistent with the data?
(b) Propose unbiased estimates for the unknown parameters in the model.
(c) Are the (population) mean compressive strengths for the five methods different?
(d) If the answer to question (b) is positive, what method is the best? The worst?

Solution to Example 10.1:

We propose the following model. Each measurement will be represented as the sum of two terms, an unknown constant, μ_i, and a random variable, ε_{ij}.

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, \ldots, k \quad \text{and} \quad j = 1, \ldots, n_i.$$

The first subscript, i, ranges from 1 to k, where k is the number of populations being compared, usually called **treatments**. In our example, we are comparing five types of drying methods, therefore $k = 5$. The second subscript, j, ranges from 1 to n_i, where n_i is the number of measurements for each treatment. In our example, we have $n_1 = n_2 = \ldots = n_5 = 20$.

The unknown parameters μ_i represent the treatment averages. Differences among the μ_i's account for the part of the variability observed in the data that is due to differences among the treatments being compared in the experiment.

The random variables ε_{ij} account for the additional variability that is caused by other factors not explicitly considered in the experiment (different batches of raw material, different mixing times, measurement errors, etc.). The best we can hope regarding the global effect of these uncontrolled factors is that it will average out. In this way these factors will not unduly enhance or worsen the performance of any treatment.

An important technique that can be used to achieve this (averaging out) is called **randomization**. The experimental units available for the experiment (the 100 concrete blocks cylinders in the case of our example) must be randomly assigned to the different treatments, so that each experimental unit has, in principle, the same chance of being assigned to any treatment. One practical way for doing this in the case of our example is to number the blocks from 1 to 100 and then to draw (without replacement) groups of 20 numbers. The units with numbers in the first group are assigned to treatment A, the units with numbers in the second group are assigned to treatment B, and so on. The actual labeling of the treatments as A, B, etc. can also be randomly decided.

The model assumptions are:

(1) **Independence**. The random variables ε_{ij} are independent.
(2) **Constant Treatment Means**. $\mathrm{E}(\varepsilon_{ij}) = 0$ for all i and j.
(3) **Constant Variance**. $\mathrm{Var}(\varepsilon_{ij}) = \sigma^2$ for all i and j.
(4) **Normality**. The variables ε_{ij} are normal.

These assumptions can be summarized by saying that the variables ε_{ij}'s are iid $N(0, \sigma^2)$.

(a) The Q–Q plots of Figure 10.1 (a)-(e) suggest that assumption (4) is consistent with the data. Figure 10.1 (f) displays the box–plots for the combined data (first from the left) and for each drying method. The variability within the samples seem roughly constant (the boxes are of approximately equal size). This suggests that assumption (3) is also consistent with the data.

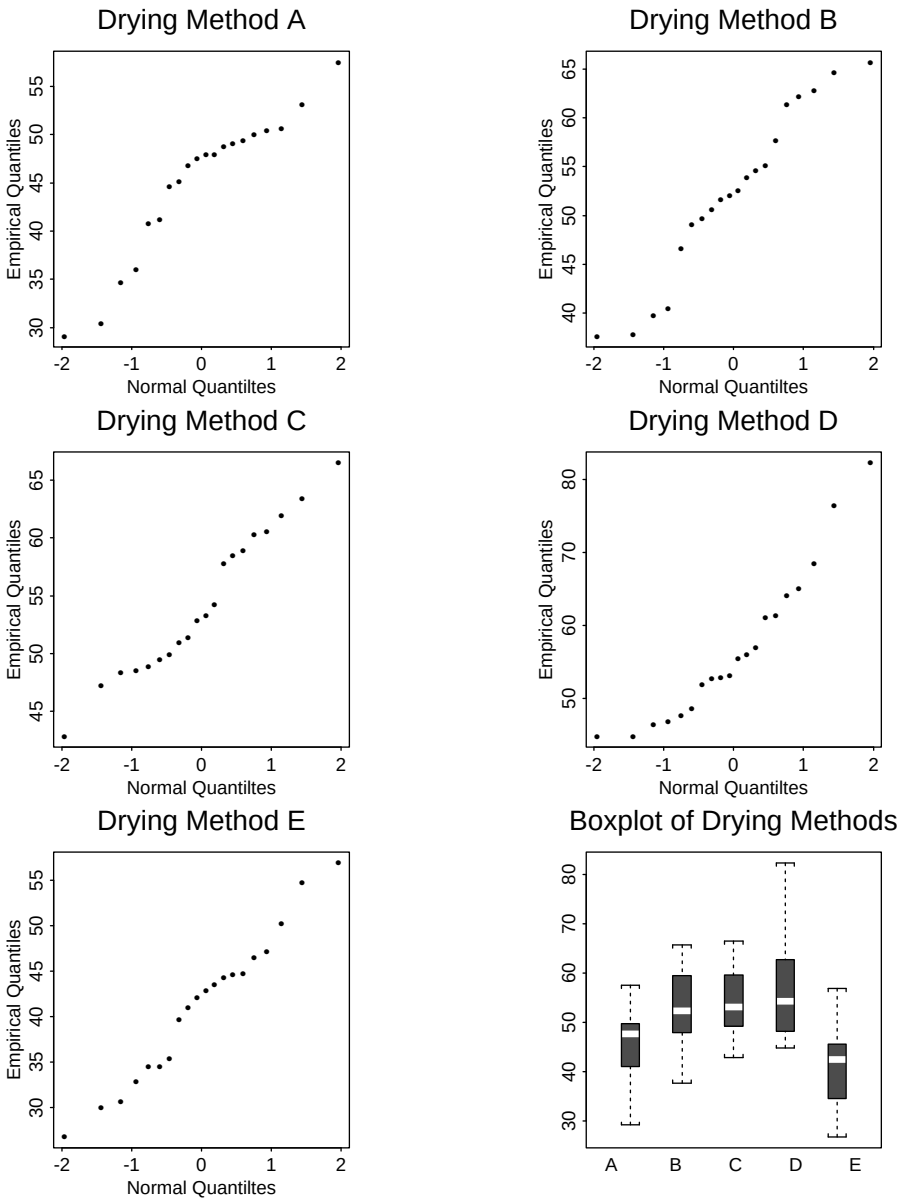


Figure 10.1: Q-Q plots and boxplot

(b) The unknown parameters in the model are μ_i $(i = 1, \ldots, k)$ and σ^2.
In what follows we will use the following notations.

$$n = n_1 + \ldots + n_k, \qquad \text{the total sample size,}$$

$$\begin{aligned} y_{i.} &= y_{i1} + \ldots + y_{in_i} \\ &= \sum_{j=1}^{n_i} y_{ij}, \qquad \text{the } i^{th} \text{ treatment's total,} \end{aligned}$$

$$\begin{aligned} \overline{y}_{i.} &= \frac{y_{i1} + \ldots + y_{in_i}}{n_i} = \frac{\sum_{j=1}^{n_i} y_{ij}}{n_i} \\ &= \frac{y_{i.}}{n_i}, \qquad \text{the } i^{th} \text{ treatment's mean,} \end{aligned}$$

$$\begin{aligned} s_i &= \sqrt{\frac{(y_{i1} - \overline{y}_{i.})^2 + \ldots + (y_{in_i} - \overline{y}_{i.})^2}{n_i - 1}} \\ &= \sqrt{\frac{\sum_{j=1}^{n_i}(y_{ij} - \overline{y}_{i.})^2}{n_i - 1}} \qquad \text{the } i^{th} \text{ treatment's standard deviation,} \end{aligned}$$

$$\begin{aligned} y_{..} &= y_{1.} + \ldots + y_{k.} = \sum_{i=1}^{k} y_{i.} \\ &= \sum_{i=1}^{k} \sum_{j=1}^{n_i} y_{ij}, \qquad \text{the overall total,} \end{aligned}$$

and

$$\begin{aligned} \overline{y}_{..} &= \frac{\sum_{i=1}^{k} y_{i.}}{n} \\ &= \frac{\sum_{i=1}^{k} \sum_{j=1}^{n_i} y_{ij}}{n}, \qquad \text{the overall mean.} \end{aligned}$$

In the case of our example

$$\overline{y}_{1.} = 45.05, \quad \overline{y}_{2.} = 52.29, \quad \overline{y}_{3.} = 54.29, \quad \overline{y}_{4.} = 56.83, \quad \overline{y}_{5.} = 41.15,$$

and

$$s_1 = 7.50, \quad s_2 = 8.70, \quad s_3 = 6.32, \quad s_4 = 10.40, \quad s_5 = 8.16.$$

(See columns 3 and 4 of Table 9.1). In addition,

$$y_{\cdot\cdot} = \sum_{i=1}^{k} n_i \overline{y}_{i\cdot} = 20[45.05 + 52.29 + 54.29 + 56.83 + 41.15] = 4992.2$$

and

$$\overline{y}_{\cdot\cdot} = 4992/100 = 49.92.$$

It is not difficult to show that the $\overline{y}_{i\cdot}$ are unbiased estimates for the unknown parameters μ_i. In fact, the reader can easily verify that

$$\mathrm{E}(\overline{Y}_{i\cdot}) = \mu_i, \quad \text{and} \quad \mathrm{Var}(\overline{Y}_{i\cdot}) = \frac{\sigma^2}{n_i}, \quad \text{for} \;\; i = 1, \ldots, k.$$

Analogously, it is not difficult to verify (see Problem 8.9) that $S_1^2, S_2^2, \ldots, S_k^2$ are k different unbiased estimates for the common variance σ^2:

$$\mathrm{E}(S_i^2) = \sigma^2, \quad \text{for} \;\; i = 1, \ldots, k.$$

These k estimates can be combined to obtain an unbiased estimate for σ^2. The reader is encouraged to verify that the combined estimate

$$S^2 = \frac{\sum_{i=1}^{k}(n_i - 1)S_i^2}{n - k}$$

is also unbiased and has a variance smaller than that of the individual S_i^2's.

(c) Roughly speaking one can answer this question positively if there is evidence that a substantial part of the variability in the data is due to differences among the treatments.

The total variability observed in the data is represented by the **total sum of squares**,

$$\begin{aligned} SST &= \sum_{i=1}^{k}\sum_{j=1}^{n_i}[y_{ij} - \overline{y}_{\cdot\cdot}]^2 \\ &= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{[\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}]^2}{\sum_{i=1}^{k} n_i} \\ &= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{n}. \end{aligned}$$

We will now show that the total sum of squares, SST, can be expressed as the sum of two terms, the **error sum of squares**, SSe, and the **treatment sum of squares**, SSt. That is,

$$SST = SSe + SSt, \tag{10.1}$$

where

$$\begin{aligned} SSe &= \sum_{i=1}^{k}\sum_{j=1}^{n_i}[y_{ij} - \overline{y}_{i.}]^2 \\ &= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \sum_{i=1}^{k}\frac{y_{i.}^2}{n_i}. \end{aligned}$$

and

$$SSt = \sum_{i=1}^{k} n_i[\overline{y}_{i.} - \overline{y}_{..}]^2.$$

The first term on the right–hand side of equation (1), SSe, represents the differences between items in the same treatment or **within–treatment variability** (this source of variability is also called **intra–group–variability**). The second term, SSt, represents the differences between items from different treatments or **between–groups variability** (this source of variability is also called **inter–group–variability**).

To prove equation (1) we add and subtract $\overline{y}_{i.}$ and expand the square to obtain

$$\begin{aligned} \sum_{i=1}^{k}\sum_{j=1}^{n_i}[y_{ij} - \overline{y}_{..}]^2 &= \sum_{i=1}^{k}\sum_{j=1}^{n_i}[(y_{ij} - \overline{y}_{i.}) + (\overline{y}_{i.} - \overline{y}_{..})]^2 \\ &= \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \overline{y}_{i.})^2 \;+\; \sum_{i=1}^{k}\sum_{j=1}^{n_i}(\overline{y}_{i.} - \overline{y}_{..})^2 \;+\; 2\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \overline{y}_{i.})(\overline{y}_{i.} - \overline{y}_{..}) \\ &= SSe \;+\; SSt \;+\; 2\sum_{i=1}^{k}(\overline{y}_{i.} - \overline{y}_{..})\sum_{j=1}^{n_i}(y_{ij} - \overline{y}_{i.}) \\ &= SSe \;+\; SSt \;+\; 2\sum_{i=1}^{k}(\overline{y}_{i.} - \overline{y}_{..})[n_i\overline{y}_{i.} - n_i\overline{y}_{i.}] \\ &= SSe + SSt. \end{aligned}$$

In the case of our example we have

$$SST = 259273.7 - \frac{(4992.2)^2}{100} = 10049.11,$$

$$SSe = 259273.7 - \frac{(901.01)^2 + (1045.72)^2 + (1085.79)^2 + (1136.67)^2 + (823.05)^2}{20} = 6587.75$$

and

$$SSt = SST - SSe = 3461.36.$$

Degrees of Freedom

The sums of squares cannot be compared directly. They must first be divided by their respective degrees of freedoms.

Since we use n squares and only one estimated parameter in the calculation of SST, we conclude that

$$df(SST) = n - 1.$$

Since there are n squares and k estimated parameters (the k treatment means) in the calculation of SSe, we conclude that

$$df(SSe) = n - k.$$

The degrees of freedom for SSt are obtained by the difference

$$df(SSt) = df(SST) - df(SSe) = (n-1) - (n-k) = k - 1.$$

ANALYSIS OF VARIANCE

All the calculations made so far can be summarized on a table called the **analysis of variance** (ANOVA) table.

Table 10.2: ANOVA TABLE

Source	Sum of Squares	*df*	Mean Squares	F
Drying Methods	3461.36	4	865.25	12.45
Error	6587.75	95	69.34	
Total	10049.11	99		

(c) To answer question (c) we must compare the variability due to the treatments with the variability due to other sources. In other words, we must find out if the "treatment effect" is strong enough to stand out above the "noise" caused by other sources of variability.

To do so, the ratio

$$F = \frac{MSt}{MSe}$$

is compared with the value $F[df(MSt), df(MSe)]$ from the F–Table, attached at the end of these notes. In our case

$$F = \frac{865.25}{69.34} = 12.45$$

and

$$F(4, 95) \approx F(4, 60) = 2.53.$$

Since $F > F(4, 95)$ we conclude that there are statistically significant differences among the drying methods.

(d) To answer question (d) we must perform **multiple comparisons** of the treatment means. It is intuitively clear that if the number of treatments is large and therefore the total number of comparisons of pairs of means

$$K = \binom{k}{2} = k(k-1)/2$$

is very large, there will be a greater chance that some of the 95% confidence intervals will fail to include the value zero, even if all the μ_i were the same. For example, $K = 3$ when $k = 3$, $K = 6$ when $k = 4$ and $K = 10$ when $k = 5$.

To compensate for the fact that the probability of declaring two means different when they are not is larger than the significance level $\alpha = 0.05$ used for each comparison, we must use the smaller significance level, δ, given by

$$\delta = 0.05/K$$

Each individual confidence interval is constructed so that it has probability $1 - \delta$ of including the true treatment mean difference. It can be shown that this procedure (called Bonferroni multiple comparisons) is conservative: If all the treatment means are equal,

$$\mu_1 = \mu_2 = \ldots = \mu_k,$$

then the probability that one or more of these intervals do not include the true difference, 0, is at most α.

The procedure to compute the simultaneous confidence intervals is as follows. In the first place, we must find the appropriate value, $t_{(n-k)}(\delta) = t_{(n-k)}(\alpha/K)$, from the Student's t table (see Table 7.1). As before, the number of degrees of freedom corresponds to those of the MSe, that is, $df = n - k$.

The second step is to determine the standard deviation of the difference of treatments means, $\overline{Y}_{i.} - \overline{Y}_{m.}$. It is easy to see that

$$\mathrm{Var}(\overline{Y}_{i.} - \overline{Y}_{m.}) = \sigma^2 \left[\frac{1}{n_i} + \frac{1}{n_m}\right].$$

Therefore,

$$\text{estimated SD}(\overline{Y}_{i.} - \overline{Y}_{m.}) \approx \sqrt{MSe}\sqrt{\left[\frac{1}{n_i} + \frac{1}{n_m}\right]}.$$

In the case of our example $k = 5$ and therefore $K = 10$. The observed differences between the 10 pairs of treatments (sample) means are given in Table 9.3.

Table 10.3: MULTIPLE COMPARISONS

Treatments	Observed Difference	$\hat{d}_{i,m}$	Significance
A–B	-7.24	7.56	
A–C	-9.24	7.56	*
A–D	-11.78	7.56	*
A–E	3.9	7.56	
B–C	-2.0	7.56	
B–D	-4.54	7.56	
B–E	11.14	7.56	*
C–D	-2.54	7.56	
C–E	13.14	7.56	*
D–E	15.68	7.56	*

As explained before, the (precision) number $\hat{d}_{i,m}$ is calculated by the formula

$$\hat{d}_{i,m} \;=\; t_{(n-k)}\sqrt{MSe}\sqrt{\frac{1}{n_i} + \frac{1}{n_m}}.$$

In the case of our example, since

$$n_1 = n_2 = n_3 = n_4 = n_5 = 20,$$

all the $\hat{d}_{i,m}$ are equal to

$$\hat{d} \;=\; t_{(95)}(0.05/10)\ \sqrt{69.34}\ \sqrt{\frac{2}{20}} \;=\; 7.56.$$

The differences marked with an star, *, on Table 9.3 are statistically significant. For example, the * on the line A–C together with the fact that the sign of the difference is negative, is

interpreted as evidence that method A is worse (less strong) than method C. The conclusions from Table 9.3 are: the methods A and E are not significantly different and appear to be significantly worse than than the others. Observe that, although method A is not significantly worse than method B (at the current level $\alpha = 0.05$) their difference, 7.24, is almost significant (fairly close to 7.56). □

10.2 Exercises

10.2.1 Exercise Set A

Problem 10.1 Three different methods are used to transport milk from a farm to a dairy plant. Their daily costs (in \$100) are given in the following:
Method 1: 8.10 4.40 6.00 7.00
Method 2: 6.60 8.60 7.35
Method 3: 12.00 11.20 13.30 10.55 11.50
(1) Calculate the sample mean and sample variance for the cost of each method.
(2) Calculate the grand mean and the pooled variance for the costs of the three methods.
(3) Test the difference of the costs of the three methods.

Problem 10.2 Six samples of each of four types of cereal grain grown in a certain region were analyzed to determine thiamin content, resulting in the following data (micrograms/grams):
Wheat: 5.2 4.5 6.0 6.1 6.7 5.8
Barley: 6.5 8.0 6.1 7.5 5.9 5.6
Maize: 5.8 4.7 6.4 4.9 6.0 5.2
Oats: 8.3 6.1 7.8 7.0 5.5 7.2
Carry out the analysis of variance for the given data. Do the data suggest that at least two of the four different grains differ with respect to true average thiamin content? Use $\alpha = 0.5$.

Problem 10.3 A psychologist is studying the effectiveness of three methods of reducing smoking. He wants to determine whether the mean reduction in the number of cigarettes smoked daily differs from one method to another among male patients. Twelve men are included in the experiment. Each smoked 60 cigarettes per day before the treatment. Four randomly chosen members of the group pursue method I; four pursue method II; and so on. The results are as follows (Table 9.4):

(a) Use a one-way analysis of variance to test whether the mean reduction in the number

Table 10.4:

Method I	Method II	Method III
52	41	49
51	40	47
51	39	45
52	40	47

of cigarettes smoked daily is equal for three methods. (Let the significance level equal 0.05).
(b) Use confidence intervals to determine which method results in a larger reduction in smoking.

10.2.2 Exercise Set B

Problem 10.4 For best production of certain molds, the furnaces need to heat quickly up to a temperature of 1500°F. Four furnaces were tested several times to determine the times (in

minutes) they took to reach 1500°F, starting from room temperature, yielding the following results

Are the furnaces average heating times different? If so, which is the fastest? The slowest?

Table 10.5:

Furnace	n_i	$\overline{x}_i$	s_i
1	15	14.21	0.52
2	15	13.11	0.47
3	10	15.17	0.60
4	10	12.42	0.43

Problem 10.5 Three specific brands of alkaline batteries are tested under heavy loading conditions. Given here are the times, in hours, that 10 batteries of each brand functioned before running out of power. Use analysis of variance to determine whether the battery brands take significantly different times to completely discharge. If the discharge times are significantly different (at the 0.05 level of confidence), determine which battery brands differ from one another. Specify and check the model assumptions.

Table 10.6:

Battery Type		
1	2	3
5.60	5.38	6.40
5.43	6.63	5.91
4.83	4.60	6.56
4.22	2.31	6.64
5.78	4.55	5.59
5.22	2.93	4.93
4.35	3.90	6.30
3.63	3.47	6.77
5.02	4.25	5.29
5.17	7.35	5.18

Problem 10.6 Five different copper-silver alloys are being considered for the conducting material in large coaxial cables, for which conductivity is a very important material characteristic. Because of differing availabilities of the five kinds, it was impossible to make as many samples from alloys 2 and 3 as from other alloys. Given next are the coded conductivity measurements from samples of wire made from each of the alloys. Determine whether the alloys have significantly different conductivities. If the conductivities are significantly different (at $\alpha = 0.05$), determine which alloys differ from one another. Specify and check the model assumptions.

Problem 10.7 Show that

$$\mathrm{E}(\overline{Y}_i) = \mu_i, \quad i = 1, \ldots, k,$$
$$\mathrm{E}(S_i^2) = \sigma^2, \quad i = 1, \ldots, k$$

Table 10.7:

Alloy				
1	2	3	4	5
60.60	58.88	62.90	60.72	57.93
58.93	59.43	63.63	60.41	59.85
58.40	59.30	62.33	59.60	61.06
58.63	56.97	63.27	59.27	57.31
60.64	59.02	61.25	59.79	61.28
59.05	58.59	62.67	62.35	59.68
59.93	60.19	61.29	60.26	57.82
60.82	57.99	60.77	60.53	59.29
58.77	59.24		58.91	58.65
59.11	57.38		58.55	61.96
61.40			61.20	57.96
59.00			59.73	59.42
			60.12	59.40
			60.49	60.30
				60.15

and

$$\mathrm{E}(\mathrm{MSe}) = \sigma^2.$$

Is the variance of MSe smaller than the variance of S_i^2? Why?

Problem 10.8 To study the correlation between the solar insulation and wind speed in the United States, 26 National Weather Service stations used three different types of solar collectors–2D Tracking, NS Tracking and EW Tracking to collect the solar insulation and wind speed data. An engineer wishes to compare whether these three collectors give significantly different measurements of wind speed. The values of windspeed corresponding to attainment of 95% integrated insulation are reported in the following Table 9.8.

Are there statistically significant differences in measurement among the three different apertures? Specify and check the model assumptions.

Table 10.8:

Station No.	Site	Latitude	2D Tracking	NS Tracking	EW Tracking
1	Brownsville, Texas	25.900	11.0	11.0	11.0
2	Apalachicola, Fla.	29.733	7.9	7.9	8.0
3	Miami, Fla.	25.800	8.7	8.6	8.7
4	Santa Maia, Calif.	34.900	9.6	9.7	9.5
5	Ft. Worth, Texas	32.833	10.8	10.7	10.9
6	Lake Charles, La.	30.217	8.5	8.4	8.6
7	Phoenix, Ariz.	33.433	6.6	6.6	6.5
8	El Paso, Taxes	31.800	10.3	10.3	10.3
9	Charleston, S.C.	32.900	9.2	9.1	9.2
10	Fresno, Calif.	36.767	6.2	6.3	6.1
11	Albuquerque, N.M.	35.050	9.0	9.0	8.9
12	Nashville, Tenn.	36.117	7.7	7.6	7.7
13	Cape Hatteras, N.C	35.267	9.2	9.2	9.3
14	Ely, Nev.	39.283	10.0	10.1	10.1
15	Dodge City, Kan.	37.767	12.0	11.9	12.0
16	Columbia, Mo.	38.967	9.0	8.9	9.1
17	Washington, D.C.	38.833	9.3	9.1	9.5
18	Medford, Ore.	42.367	6.8	6.9	6.5
19	Omaha, Neb.	41.367	10.4	10.3	10.5
20	Madison, Wis.	43.133	9.5	9.5	9.6
21	New York, N.Y.	40.783	10.4	10.3	10.4
22	Boston, Mass.	42.350	11.4	11.2	11.4
23	Seattle, Wash.	47.450	9.0	9.0	9.1
24	Great Falls, Mont.	47.483	12.9	12.6	13.0
25	Bismarck, N.D.	46.767	10.8	10.7	10.8
26	Caribou, Me.	46.867	11.4	11.3	11.5

Chapter 11

The Simple Linear Regression Model

11.1 An example

Consider the following example:

Example 11.1 Due to differences in the cooling rates when rolled, the average elastic limit and the ultimate strength of reinforcing metal bars is determined by the bar size. The measurements in Table 10.1 (in hundreds of pounds per square inch) were obtained from a sample of bars.

The experimental units (metal bars) are numbered from 1 to 35. Notice that each experimental unit, i, gave rise to three different measurements:

The diameter of the ith metal bar	x_i
The elastic limit of the ith metal bar	y_i
The ultimate strength of the ith metal bar	z_i

We will investigate the relationship between the variables x_i and y_i. Likewise, the relationship between the variables x_i and z_i can be investigated in an analogous way (see Problem 10.2).

First of all we notice that the roles of y_i and x_i are different. Reasonably, one must assume that the elastic limit, y_i, of the ith metal bar is somehow determined (or influenced) by the diameter, x_i, of the bar. Consequently, the variable y_i can be considered as a **dependent** or **response** variable and the variable x_i can be considered as an **independent** or **explanatory** variable.

A quick look at Figure 11.1 (a) will show that there is not an exact (deterministic) relationship between x_i and y_i. For example, bars with the same diameter (3, say) have different elastic limits (436.82, 449.40 and 412.63). However, the plot of y_i versus x_i shows that in general, larger values of x_i are associated with smaller values of y_i.

In cases like this we say that the variables are statistically related, in the sense that the average elastic limit is a decreasing function, $f(x_i)$, of the diameter.

Table 11.1: Elastic Limit and Ultimate Strength of Metal Bars

Unit	Bar Diameter ($\frac{1}{8}$ of an inch)	Elastic Limit (100 psi)	Ultimate Strength (100 psi)	Unit	Bar Diameter ($\frac{1}{8}$ of an inch)	Elastic Limit (100 psi)	Ultimate Strength (100 psi)
1	3	436.82	683.65	19	7	361.14	605.12
2	3	449.40	678.48	20	7	356.06	604.17
3	3	412.63	681.41	21	8	328.59	568.11
4	4	425.00	672.29	22	8	321.64	576.69
5	4	419.71	673.26	23	8	321.14	570.47
6	4	415.74	671.31	24	9	297.28	538.99
7	4	422.94	674.42	25	9	286.04	537.11
8	5	407.76	646.44	26	9	291.99	537.44
9	5	416.84	654.32	27	10	231.15	502.76
10	5	388.39	649.31	28	10	249.13	498.88
11	5	416.25	654.24	29	10	249.81	495.17
12	5	384.35	644.20	30	10	251.22	499.21
13	5	412.91	640.15	31	11	200.76	455.28
14	6	379.64	627.52	32	11	216.99	460.75
15	6	371.11	621.45	33	11	210.26	460.96
16	6	369.34	626.11	34	12	162.30	411.13
17	6	384.91	632.73	35	12	167.63	410.74
18	7	362.89	601.73				

Each elastic limit measurement, y_i, can be viewed as a particular value of the random variable, Y_i, which in turn can be expressed as the sum of two terms, $f(x_i)$ and ε_i. That is,

$$Y_i = f(x_i) + \varepsilon_i, \quad i = 1, \ldots, 35. \tag{11.1}$$

It is usually assumed that the random variables ε_i satisfy the following assumptions:

(1) **Independence**. The random variables ε_i are independent.
(2) **Constant Mean**. $\mathrm{E}(\varepsilon_i) = 0$ for all i.
(3) **Constant Variance**. $\mathrm{Var}(\varepsilon_i) = \sigma^2$ for all i.
(4) **Normality**. The variables ε_i are normal.

These assumptions can be summarized by simply saying that the variables Y_i's are independent normal random variables with

$$\mathrm{E}(Y_i) = f(x_i) \quad \text{and} \quad \mathrm{Var}(Y_i) = \sigma^2.$$

The model (10.1) above is called **linear** if the function $f(x_i)$ can be expressed in the form

$$f(x_i) = \beta_0 + \beta_1 g(x_i),$$

where the function $g(x)$ is completely specified, and β_0 and β_1 are (usually unknown) parameters.

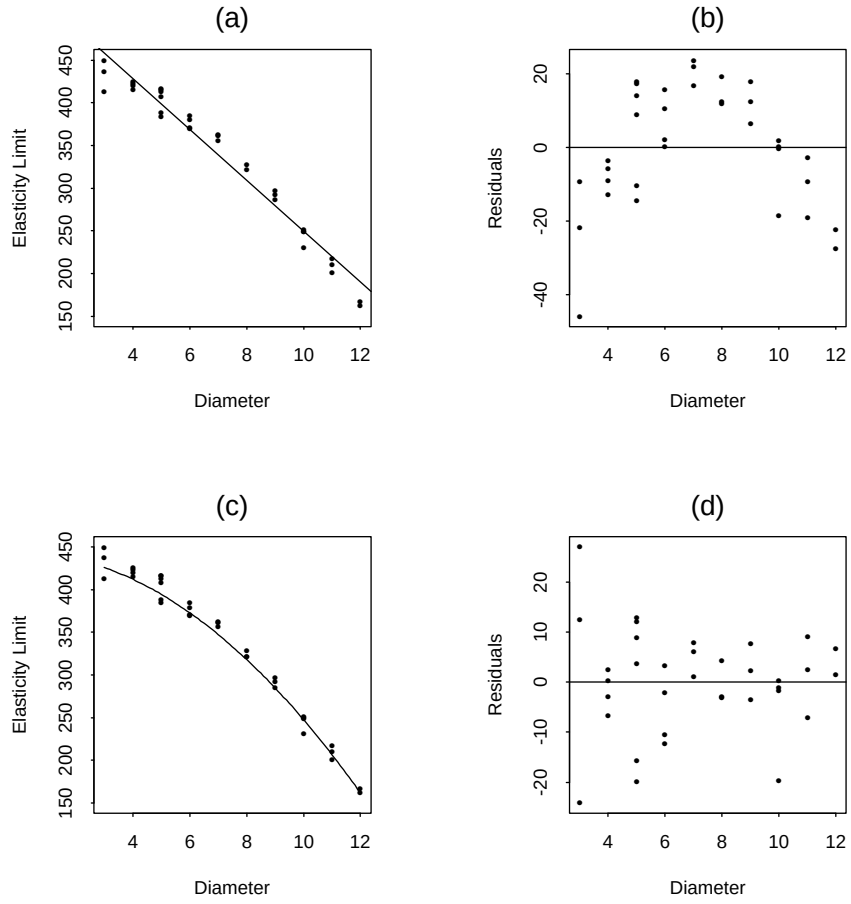


Figure 11.1

The linear model,

$$Y_i = \beta_0 + \beta_1 g(x_i) + \varepsilon_i, \qquad i = 1, \ldots, 35, \tag{11.2}$$

is very flexible as many possible **mean response** functions, $f(x_i)$, satisfy the **linear** form given above. For example, the functions

$$f(x_i) = 5.0 + 4.2x_i \quad \text{and} \quad f(x_i) = \beta_0 + 3\sin(2x_i),$$

are linear, in the sense explained above, with

$$g(x_i) = x_i, \quad \beta_0 = 5 \quad \text{and} \quad \beta_1 = 4.2$$

in the first case, and

$$g(x_i) = \sin(2x_i), \quad \beta_0 = \text{unspecified} \quad \text{and} \quad \beta_1 = 3,$$

in the second case.

On the other hand, there are some functions that can not be expressed in this linear form. One example is the function

$$f(x_i) = \frac{\exp\{\beta_1 x_i\}}{1 + \exp\{\beta_1 x_i\}}.$$

The shape assumed for $f(x_i)$ is sometimes suggested by scientific or physical considerations. In other cases, as in the present example, the shape of $f(x_i)$ is suggested by the data itself. The plot of y_i versus x_i (see Figure 11.1) indicates that, at least in principle, the simple linear mean response function

$$f(x_i) = \beta_0 + \beta_1 x_i, \quad \text{that is} \quad g(x_i) = x_i.$$

may be appropriate. In other words, to begin our investigation we will use the **tentative** working assumption that, on the average, the elastic limit of the metal bars is a linear function $(\beta_0 + \beta_1 x_i)$ of their diameters.

Of course, the values of β_0 and β_1 are unknown and must be empirically determined, that is, estimated from the data. One popular method for estimating these parameters is the **method of least squares**. Given the tentative values b_0 and b_1 for β_0 and β_1, respectively, the regression residuals

$$r_i(b_0, b_1) = y_i - b_0 - b_1\ x_i, \quad i = 1, \ldots, n,$$

measure the vertical distances between the observed value, y_i, and the tentatively estimated mean response function, $b_0 + b_1\ x_i$.

The method of least squares consists of finding the values $\hat{\beta}_0$ and $\hat{\beta}_1$ of b_0 and b_1, respectively, which minimize the sum of the squares of the residuals. It is expected that, because of this minimization property, the corresponding mean response function,

$$\hat{f}(x_i) = \hat{\beta}_0 + \hat{\beta}_1\ x_i$$

will be "close to " or will **fit** the data points. In other words, the least square estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are the solution to the minimization problem:

$$\min_{b_0,b_1} \sum_{i=1}^{n} r_i^2(b_0,b_1) \quad = \quad \min_{b_0,b_1} \sum_{i=1}^{n} [y_i - b_0 - b_1 x_i]^2.$$

To find the actual values of $\hat{\beta}_0$ and $\hat{\beta}_1$, we differentiate the function

$$S(b_0,b_1) = \sum_{i=1}^{n} r_i^2(b_0,b_1)$$

with respect to b_0 and b_1, and set these derivatives equal to zero to obtain the so called **LS equations**:

$$\begin{aligned} \frac{\partial}{\partial b_0} S(b_0,b_1) &= -2\sum_{i=1}^{n}[y_i - b_0 - b_1 x_i] = 0 \\ \frac{\partial}{\partial b_1} S(b_0,b_1) &= -2\sum_{i=1}^{n}[y_i - b_0 - b_1 x_i]x_i = 0, \end{aligned}$$

The LS equations can be rewritten as

$$\begin{aligned} \sum_{i=1}^{n} y_i - nb_0 - b_1 \sum_{i=1}^{n} x_i &= 0 \\ \sum_{i=1}^{n} y_i x_i - b_0 \sum_{i=1}^{n} x_i - b_1 \sum_{i=1}^{n} x_i^2 &= 0, \end{aligned}$$

or equivalently,

$$\overline{y} - b_0 - b_1 \overline{x} = 0 \tag{11.3}$$

$$\overline{xy} - b_0 \overline{x} - b_1 \overline{xx} = 0, \tag{11.4}$$

where

$$\overline{xy} = \frac{1}{n}\sum_{i=1}^{n} y_i x_i \quad \text{and} \quad \overline{xx} = \frac{1}{n}\sum_{i=1}^{n} x_i^2.$$

From (10.3) we have

$$b_0 = \overline{y} - b_1\overline{x}. \tag{11.5}$$

From this and (10.4) we have

$$\begin{aligned} b_1\overline{xx} &= \overline{xy} - b_0\overline{x} \\ &= \overline{xy} - \left[\overline{y} - b_1\overline{x}\right]\overline{x} \\ &= \overline{xy} - \overline{y}\ \overline{x} + b_1\ \overline{x}\ \overline{x} \\ \Rightarrow\ b_1 &= \frac{\overline{xy} - \overline{x}\ \overline{y}}{\overline{xx} - \overline{x}\ \overline{x}}. \end{aligned}$$

Therefore (see (10.5)),

$$\begin{aligned} \hat{\beta}_1 &= \frac{\overline{xy} - \overline{y}\ \overline{x}}{\overline{xx} - \overline{x}\ \overline{x}}, \quad \text{and} \\ \hat{\beta}_0 &= \overline{y} - \hat{\beta}\ \overline{x}. \end{aligned}$$

In the case of our numerical example we have

$$\overline{x} = 7.086, \quad \overline{y} = 336.565, \quad \overline{xy} = 2162.353 \quad \text{and} \quad \overline{xx} = 57.657,$$

Therefore,

$$\hat{\beta}_1 = \frac{2162.353 - (7.086)(336.565)}{57.657 - (7.086^2)} = -29.86,$$

$$\hat{\beta}_0 = 336.565 - (-29.86)(7.086) = 548.16.$$

and

$$\hat{f}(x) = 548.16 - (29.86)x.$$

The plot of $\hat{f}(x)$ versus x (solid line in Figure 11.1 (a)) and the plot of the regression residuals,

$$e_i = y_i - \hat{f}(x_i) = y_i - [\ 548.16\ -\ (29.86)x_i],$$

versus x_i (Figure 11.1 (b)) shows that the current fit may not be appropriate. The residuals show a clear *negative–positive–negative* pattern. Since the residuals, e_i, estimate the unobservable model errors,

$$\varepsilon_i = y_i - \beta_0 - \beta_1 x_i,$$

one would expect the plot of the e_i versus the x_i will not show any particular pattern. In other words, if the specified mean response function is correct, the estimated mean response function $\hat{f}(x_i)$ should "extract" most of the signal (systematic behavior) contained in the data and the residuals, e_i, should behave as patternless random noise.

Now that the tentatively specified simple transformation

$$g(x) = x$$

for the explanatory variable, x, is considered to be incorrect, the next step in the analysis is to specify a new transformation. We will try the mean response function

$$f(x) = \beta_0 + \beta_1 x^2, \quad \text{that is} \quad g(x) = x^2.$$

Notice that if the mean elastic limit of the bars is a function of the bar surface

$$\frac{(\text{diameter})^2 * \pi}{4}\ \ (\frac{1}{8}\ \text{inch})^2 = x^2(\pi/4)\ \ (\frac{1}{8}\ \text{inch})^2,$$

then the newly proposed mean response function will be appropriate. To simplify the notation we will write

$$w_i = x_i^2,$$

to represent the squared diameter of the ith metal bar.
The new estimates for β_0 and β_1 and $f(x)$ are

$$\hat{\beta}_1 \ = \ \frac{\overline{wy} - \overline{y}\ \overline{w}}{\overline{ww} - \overline{w}\ \overline{w}} = -2.022, \quad \text{and}$$

$$\hat{\beta}_0 \ = \ \overline{y} - \hat{\beta}_1\ \overline{w} = 336.56 + (2.022)(57.65) = 453.128.$$

and

$$\hat{f}(x) = 453.128 - (2.022)\ x^2.$$

The plot for this new fit (solid line on Figure 11.1 (c)) and the residuals plot (Figure 11.1 (d)) indicate that this second fit is appropriate.

Inference in the Linear Regression Model

It is not difficult to show that, if the model is correct, the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased:

$$\mathrm{E}(\hat{\beta}_0) = \beta_0 \qquad \text{and} \qquad \mathrm{E}(\hat{\beta}_1) = \beta_1.$$

Also, it can be shown that

$$\mathrm{Var}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\overline{w}^2}{\sum_{i=1}^n [w_i - \overline{w}]^2} \right]$$

and

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n [w_i - \overline{w}]^2}.$$

Finally, it can be shown that under the model,

$$\mathrm{E}\left(\sum_{i=1}^n [Y_i - \hat{\beta}_0 - \hat{\beta}_1 w_i]^2 \right) = (n-2)\sigma^2$$

a model–based unbiased estimate for σ^2 is given by

$$\begin{aligned} s^2 &= \frac{\sum_{i=1}^n [Y_i - \hat{\beta}_0 - \hat{\beta}_1 w_i]^2}{n-2} \\ &= \frac{n[(\overline{yy} - \overline{y}\ \overline{y}) - \hat{\beta}_1^2 (\overline{ww} - \overline{w}\ \overline{w})]}{n-2}. \end{aligned}$$

In the case of our example,

$$s^2 = \frac{35[(120178.7 - 336.5646^2) - (2.022^2)(4993.429 - 57.657^2)]}{35-2} = 86.53.$$

In summary, the empirically estimated standard deviations of $\hat{\beta}_0$ and $\hat{\beta}_1$ are

$$\mathrm{SD}(\hat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\overline{w}^2}{\sum_{i=1}^n [w_i - \overline{w}]^2}} = s\sqrt{\frac{1}{n} + \frac{\overline{w}^2}{n[\overline{ww} - \overline{w}\ \overline{w}]}}$$

and

$$\mathrm{SD}(\hat{\beta}_1) = \frac{s}{\sqrt{\sum_{i=1}^n [w_i - \overline{w}]^2}} = \frac{s}{\sqrt{n[\overline{ww} - \overline{w}\ \overline{w}]}}$$

In the case of our example,

$$\mathrm{SD}(\hat{\beta}_0) = \sqrt{86.53}\sqrt{\frac{1}{35} + \frac{57.657^2}{35[4993.429 - 57.657^2]}} = 1.60$$

and

$$\mathrm{SD}(\hat{\beta}_1) = \sqrt{\frac{86.53}{35(4993.429 - 57.657^2)}} = 0.0385$$

Confidence Intervals

95% confidence intervals for the model parameters, β_0 and β_1, and also for the mean response, $f(x)$, can now be easily obtained. First we derive the 95% confidence intervals for β_0 and β_1. As before, the intervals are of the form

$$\hat{\beta}_0 \pm \hat{d}_{\beta_0} \qquad \text{and} \qquad \hat{\beta}_1 \pm \hat{d}_{\beta_1}$$

where

$$\hat{d}_{\beta_0} = t_{(n-2)}(\alpha)\ \mathrm{SD}(\hat{\beta}_0) \qquad \text{and} \qquad \hat{d}_{\beta_1} = t_{(n-2)}(\alpha)\ \mathrm{SD}(\hat{\beta}_1).$$

In the case of our example, $n - 2 = 35 - 2 = 33$, $t_{(33)}(0.05) \approx t_{(30)}(0.05) = 2.04$ and so

$$\hat{d}_{\beta_0} = (2.04)(1.60) = 3.26 \qquad \text{and} \qquad \hat{d}_{\beta_1} = (2.04)(0.0385) = 0.0785$$

Therefore, the 95% confidence intervals for β_0 and β_1 are

$$453.125 \pm 3.26 \qquad \text{and} \qquad -2.021 \pm 0.0785,$$

respectively.

Notice that, since the confidence interval for β doesn't include the value zero, we conclude that there is a linear decreasing relationship between the square of the bar diameter and its elastic limit. When the bar surface increases by one unit ($\frac{1}{64}$ inch^2) the average elastic limit decreases by two hundred psi.

Finally, we can also construct a 95% confidence interval for the average response, $f(x)$, at any given value of x. It can be shown that the variance of $\hat{f}(w)$ is

$$\text{Var}(\hat{f}(x)) = \sigma^2 \left[\frac{1}{n} + \frac{(w - \overline{w})^2}{n[\overline{ww} - \overline{w}\ \overline{w}]} \right]$$

where $w = x^2$. Therefore, the empirically estimated standard deviation of $\hat{f}(x)$ is

$$\text{SD}(\hat{f}(x)) = s \sqrt{\frac{1}{n} + \frac{(w - \overline{w})^2}{n[\overline{ww} - \overline{w}\ \overline{w}]}}.$$

In the case of our example we have,

$$\text{SD}(\hat{f}(x)) = \sqrt{86.53} \sqrt{\frac{1}{35} + \frac{(w - 57.657)^2}{35[4993.429 - 57.657^2]}}.$$

For instance, if the value of interest is $x = 8.0$,

$$\text{SD}(\hat{f}(8.0)) = \sqrt{86.53} \sqrt{\frac{1}{35} + \frac{(16.0 - 57.657)^2}{35[4993.429 - 57.657^2]}} = 1.59.$$

The corresponding 95% confidence interval for $f(8.0)$ is then,

$$\hat{f}(8.0)) \pm \hat{d}$$

where

$$\hat{d} = (2.04)(1.59) \approx (2.04)(1.59) = 3.24.$$

Since

$$\hat{f}(8.0) = 453.125 - (2.022)(8.0^2) = 323.72,$$

the 95% confidence interval for $f(8.0)$ is equal to

$$323.72 \pm 3.24.$$

11.2 Exercises

Problem 11.1 The number of hours needed by twenty employees to complete a certain task have been measured before and after they participated of a special training program. The data is displayed on Table 7.2. Notice that these data have already been partially studied in Problem 7.12. Investigate the relationship between the before training and the after training times using linear regression. State your conclusions.

Problem 11.2 Investigate the relationship between the diameter bar and the ultimate strength shown in Table 10.1. State your conclusions.

Problem 11.3 Table 10.2 reports the yearly worldwide frequency of earthquakes with magnitude 6 or greater from January, 1953 to December, 1965.

(a) Make scatter-plots of the frequencies against magnitudes and the log-frequencies against the magnitudes.
(b) Propose your regression model and estimate the coefficients of your model.
(c) Test the null hypothesis that the slope is equal to zero.

Table 11.2:

Magnitude	Frequency	Magnitude	Frequency
6.0	2750	7.4	57
6.1	1929	7.5	45
6.2	1755	7.6	31
6.3	1405	7.7	23
6.4	1154	7.8	18
6.5	920	7.9	13
6.6	634	8.0	9
6.7	487	8.1	7
6.8	376	8.2	7
6.9	276	8.3	4
7.0	213	8.4	2
7.1	141	8.5	2
7.2	110	8.6	1
7.3	85	8.7	1

Problem 11.4 In a certain type of test specimen, the normal stress on a specimen is known to be functionally related to the shear resistance. The following is a set of experimental data on the variables.

x, normal stress	y, shear resistance
26.8	26.5
25.4	27.3
28.9	24.2
23.6	27.1
27.7	23.6
23.9	25.9
24.7	26.3
28.1	22.5
26.9	21.7
27.4	21.4
22.6	25.8
25.6	24.9

(a) Write the regression equation.
(b) Estimate the shear resistance for a normal stress of 24.5 pounds per square inch.
(c) Construct 95% confidence intervals for regression coefficients β_0 and β_1.
(d) Check the normality assumption through the residuals.

Problem 11.5 The amounts of a chemical compound y, which dissolved in 100 grams of water at various temperatures, x, were recorded as follows:

$x(^oC)$	y (grams)		
0	8	6	8
15	12	10	14
30	25	21	24
45	31	33	28
60	44	39	42
75	48	51	44

(a) Find the equation of the regression line.
(b) Estimate the amount of chemical that will dissolve in 100 grams of water at 50^oC.
(c) Test the hypothesis that $\beta_0 = 6$, using a 0.01 level of significance, against the alternative that $\beta_0 \neq 6$.
(d) Is the linear model adequate?

Chapter 12

Appendix

12.1 Appendix A: tables

This appendix includes five tables: normal table, t-distribution table, F-distribution table, cumulative Poisson distribution table and cumulative binomial distribution table.